Exact Dyadic Extraction of Rvachev's Up-Function

from Finite Sinc-Product Splines

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Abstract

Let $\Phi(t) = \prod_{k\geq 0} \operatorname{sinc}_{\mathrm{rad}}(\pi t/2^k)$ be the Fourier transform of Rvachev's up-function under the 2π convention, and let up_n be the inverse transform of the prefix product with factors $k = 0, \dots, n$: a compactly supported piecewise-polynomial box spline of degree at most n whose values at dyadic rationals are rational. Fix a dyadic rational $x \in [-1, 1]$, let $s(x)$ be the least $s \geq 0$ with $2^s x \in \mathbb{Z}$, and put $d = \lfloor s(x)/2 \rfloor$.

The volume proves that the convergence $\mathrm{up}_n(x) \to \mathrm{up}(x)$ is not merely asymptotic: for every $n \geq s(x)$,

$$\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{d} C_r(x)\, 4^{-rn}, \qquad C_r(x) = (-1)^r a_r 4^{-r}\, \mathrm{up}^{(2r)}(x),$$

exactly, with no remainder, where the positive rationals a_r are the coefficients of $1/\Phi$. Three finite mechanisms drive the proof: the omitted Fourier tail is a scaled copy of up supported in a single polynomial cell of the finite spline; on that cell the all-orders deconvolution series is a polynomial in a nilpotent operator and terminates; and every derivative $\mathrm{up}^{(r)}(x)$ with $r > s(x)$ vanishes. At the last level before the onset, where x is a knot of the spline, the defect against the law is computed in closed form through Rvachev's functional-differential equation: it is $-\varepsilon_k\, 2^{-\binom{s}{2}}\, B_s/s!$ with $s = s(x)$, ε_k the Thue–Morse sign of the knot and B_s the Bernoulli number, so the onset $n \geq s(x)$ is exact at every even depth and improves by exactly one level at odd depth; all d modes are nonzero when $s(x) \geq 2$, so d is the exact number of modes.

Geometric Lagrange interpolation at the nodes $1, Q, \dots, Q^d$, $Q = 1/4$, turns the law into a single extraction row: for any $N \geq s(x)$,

$$\mathrm{up}(x) = \frac{1}{(Q;Q)_d} \sum_{j=0}^{d} (-1)^{d-j} Q^{\binom{d-j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_Q \mathrm{up}_{N+j}(x),$$

recovering the exact value from $d+1$ consecutive rational samples. Every mode coefficient, and hence every even derivative of up at x, is recovered by the same interpolation; the row's ℓ^∞ amplification is bounded by $(-Q;Q)_\infty/(Q;Q)_\infty = 1.96926\ldots$. Against the corpus's Exponents volume, whose signed quarter-base Richardson combination cancels these modes asymptotically for general x, the contribution here is exactness at dyadic points after the depth. Worked examples, an exact-arithmetic algorithm with its verification specification, and a formalization register — the row's algebra and the cell identity at $x = 1/4$ are kernel-verified in Lean; the general theorem is not — complete the volume.

Contents

1 The result in one page

Rvachev's up-function up [12, 1] is the even, nonnegative C^∞ probability density with support $[-1,1]$ whose 2π-frequency Fourier transform is the dyadic sinc product

$$\Phi(\xi) := \widehat{\mathrm{up}}_{2\pi}(\xi) = \prod_{k=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\left(\frac{\pi\xi}{2^k}\right), \qquad \xi \in \mathbb{R} \tag{1}$$

(catalogue entries `rvachev-up` and `up-fourier-product`; both facts are Lean-proved in the repository [10]). Keeping only the factors $k = 0, \ldots, n$ produces the finite product Φ_n and its inverse transform up_n, a compactly supported piecewise polynomial of degree n with knots on a dyadic lattice (Section 2). At a rational point every value $\mathrm{up}_n(x)$ is a rational number that can be written down exactly, and $\mathrm{up}_n(x) \to \mathrm{up}(x)$ as $n \to \infty$.

The problem. Fix a dyadic rational $x \in [-1,1]$. Recover $\mathrm{up}(x)$ *exactly*, as an element of $\mathbb{Q} = \mathbb{Q}$, from finitely many of the finite-spline values $q_n := \mathrm{up}_n(x)$, by one closed formula whose weights do not depend on x.

The two integers and the two constants. Throughout, $s(x)$ is the *dyadic depth* of x, the least $s \geq 0$ with $2^s x \in \mathbb{Z}$ (Theorem 3.1), and

$$d := \lfloor s(x)/2 \rfloor, \qquad Q := \frac{1}{4}. \tag{2}$$

The *reciprocal-product coefficients* $a_m \in \mathbb{Q}$ are defined by the Taylor expansion at the origin

$$\frac{1}{\Phi(z)} = \sum_{m=0}^{\infty} a_m (2\pi z)^{2m}, \qquad a_0 = 1,\ a_1 = \frac{1}{18},\ a_2 = \frac{31}{16200},\ a_3 = \frac{2347}{42865200}; \tag{3}$$

they are positive rationals, and the section that introduces them proves this, identifies them with the catalogue's Rvachev–Appell coefficients, and gives the Bernoulli–Bell formula behind the listed values.

Theorem 1.1 (Exact eventual geometric law at a dyadic point)**.** *Let $x \in [-1,1]$ be a dyadic rational with depth $s = s(x)$ and let $d = \lfloor s/2 \rfloor$. Then for every integer $n \geq s$,*

$$\boxed{\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{r=1}^{d} C_r(x)\, 4^{-rn}, \qquad C_r(x) = (-1)^r\, a_r\, 4^{-r}\, \mathrm{up}^{(2r)}(x),} \tag{4}$$

equivalently

$$\mathrm{up}_n(x) = \sum_{r=0}^{d} (-1)^r\, a_r\, \mathrm{up}^{(2r)}(x)\, 4^{-r(n+1)} \qquad (n \geq s). \tag{5}$$

Every quantity in Equation (4) lies in $\mathbb{Q}$. The identity is exact: there is no remainder term once $n \geq s$.

The two displays are the same statement, because $a_r 4^{-r(n+1)} = (a_r 4^{-r})\, 4^{-rn}$; the sources split on which of the two factors 4^{-r} is absorbed into the coefficient, and this volume always prints $C_r(x)$ with the factor 4^{-r} absorbed, so that the modes of the *sequence* $(q_n)_n$ are literally $4^{-n}, 4^{-2n}, \ldots, 4^{-dn}$ (Theorem 1.4). The proof of Theorem 1.1 occupies the body of the volume. Its three ingredients are geometric and finite: the omitted Fourier tail is an exactly rescaled copy of up (Theorem 2.4); for $n \geq s$ that copy is supported inside one polynomial cell of up_n centred at x (Theorems 3.4 and 3.6), so that deconvolution becomes finite linear algebra on a polynomial jet; and the Taylor jet of up at a dyadic point terminates at order s, which is why only $d = \lfloor s/2 \rfloor$ even orders survive. The reciprocal-product coefficients and finite local deconvolution (Section 4), the termination of the dyadic jet (Section 5), and the eventual geometric law itself (Section 6) assemble these into Equation (4); its sharpness — the exact onset and the exact number of modes — is Section 7.

The Exponents volume [9] proves, for its prefix product p_n (defined in its equation `p3:eq:prefix-product`), the sharp C^r error law

$$\left\| p_n^{(r)} - \mathrm{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n} \qquad \text{(its } \texttt{p3:thm:sharp-prefix-law}\text{)},$$

and a signed geometric Richardson combination with quarter-base Lagrange weights whose leading term is only *asymptotic*, with an $\mathcal{O}_{n\to\infty}\left(\delta_n^{2M+2}\right)$ remainder at a general point (its equations `p3:eq:richardson-combination` and, with the same stem, `-weights`, `-moments`, `-leading`, `-stability`). The contribution of the present volume is that *at a dyadic point of depth s the expansion is exact for every $n \geq s$*, so the very same Richardson row returns the exact value rather than an accelerated approximation. That is the content of the next theorem.

Theorem 1.2 (One-row extraction). *Let x, s, d be as in Theorem 1.1 and let $N \geq s$ be any integer. Then*

$$\boxed{\mathrm{up}(x) = \frac{1}{(Q;Q)_d} \sum_{j=0}^{d} (-1)^{d-j}\, Q^{\binom{d-j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_Q \mathrm{up}_{N+j}(x), \qquad Q = \frac{1}{4}.} \tag{6}$$

Reindexing by $i = d - j$ gives the same row in the order in which S6 prints it,

$$\mathrm{up}(x) = \frac{1}{(Q;Q)_d} \sum_{i=0}^{d} (-1)^{i}\, Q^{\binom{i+1}{2}} \begin{bmatrix} d \\ i \end{bmatrix}_Q q_{N+d-i}, \tag{7}$$

since $\left[\begin{smallmatrix} d \\ d-i \end{smallmatrix}\right]_Q = \left[\begin{smallmatrix} d \\ i \end{smallmatrix}\right]_Q$. The $d+1$ weights are rational, sum to 1, and depend on neither x nor N; they depend on x only through $d = \lfloor s(x)/2 \rfloor$.

Proof. Deferred to Theorem 8.7: the row is geometric Lagrange interpolation at the nodes $Q^N, \ldots, Q^{N+d}$ extrapolated to 0, applied to the degree-$\leq d$ polynomial that Theorem 1.1 supplies; the closed form of the weights is Theorem 8.3, and the reversed order is the substitution $i = d - j$ together with the symmetry $\left[\begin{smallmatrix} d \\ d-i \end{smallmatrix}\right]_Q = \left[\begin{smallmatrix} d \\ i \end{smallmatrix}\right]_Q$. □

Remark 1.3 (Empty products at $d = 0$). For $s \in \{0, 1\}$ one has $d = 0$; the sum in Equation (4) is empty, $(Q;Q)_0 = 1$ and $\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right]_Q = 1$ by the catalogue's empty-product conventions, and the row Equation (6) reads $\mathrm{up}(x) = \mathrm{up}_N(x)$ for every $N \geq s$. This is not vacuous: it asserts that the finite splines are already exact at the five points $0, \pm\frac{1}{2}, \pm 1$ from level s on. Theorem 3.5 verifies it directly, with a convention-dependent exception at the single level $n = 0$ for $x = \pm\frac{1}{2}$.

Remark 1.4 (The two exponent conventions, reconciled). S1, S2 and S6 write the eventual law with modes $4^{-r(n+1)}$ and coefficients $(-1)^r a_r\, \mathrm{up}^{(2r)}(x)$ (S1 in the equivalent form $\gamma_{2r}\, \mathrm{up}^{(2r)}(x)/(2r)!$); S3, S4 and S5 write modes 4^{-rn} with coefficients $C_r(x)$. Both are correct and differ by the factor 4^{-r}, as Equations (4) and (5) show. The factor has a geometric meaning: $4^{-(n+1)} = \delta_n^2$ is the square of the half-width $\delta_n = 2^{-(n+1)}$ of the omitted tail (Theorem 2.4), so the shifted form is the one that deconvolution produces, while the unshifted form is the one that names the ratios of the sequence. A reader comparing a coefficient table across sources must first decide which of the two is tabulated: the mode coefficient $4/9$ at $x = \frac{1}{4}$ in S1 and S2 is the coefficient $C_1\left(\frac{1}{4}\right) = \frac{1}{9}$ of this volume. The same shift appears once more, disguised, in the indexing bridge to the Exponents volume (Section 2): there $\delta_N = 2^{-N}$ and the modes are 4^{-rN} with $N = n + 1$.

Remark 1.5 (Generality). Nothing in the proof of Theorem 1.2 used $Q = \frac{1}{4}$. For any base $q \in (0, 1)$, any polynomial P of degree at most d, and any N,

$$P(0) = \frac{1}{(q;q)_d} \sum_{j=0}^{d} (-1)^{d-j} q^{\binom{d-j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_q P(q^{N+j}).$$

What is special to up at a dyadic point is Theorem 1.1: that the sequence q_n *is* such a polynomial in q^n with $q = \frac{1}{4}$, exactly and from level s on.

Example 1.6 (The worked instance $x = \frac{1}{4}$). Here $2^2 \cdot \frac{1}{4} = 1$ and $2 \cdot \frac{1}{4} \notin \mathbb{Z}$, so $s = 2$ and $d = 1$: one geometric mode, two samples, onset $n = 2$. The truncated-power formula of Theorem 2.5 gives the samples at the levels $n = 2, 3$ in exact arithmetic,

$$q_2 = \mathrm{up}_2\big(\tfrac{1}{4}\big) = \frac{15}{16}, \qquad q_3 = \mathrm{up}_3\big(\tfrac{1}{4}\big) = \frac{179}{192}$$

(the level-2 value is the five-term sum $4\,[(\frac{9}{8})^2 - (\frac{7}{8})^2 - (\frac{5}{8})^2 + (\frac{3}{8})^2 - (\frac{1}{8})^2] = 4 \cdot \frac{15}{64}$; the level-3 value is the ten-term sum $\frac{1}{48}\sum_{k\le 9} \varepsilon_k(\frac{19}{2} - k)^3 = \frac{1}{48} \cdot \frac{179}{4}$ in the scaled form Equation (20)). With $d = 1$ the row Equation (6) has $(Q;Q)_1 = \frac{3}{4}$ and weights $(-Q,\,1)/\frac{3}{4} = \big(-\frac{1}{3},\,\frac{4}{3}\big)$, so

$$\mathrm{up}\big(\tfrac{1}{4}\big) = -\frac{1}{3}\cdot\frac{15}{16} + \frac{4}{3}\cdot\frac{179}{192} = -\frac{45}{144} + \frac{179}{144} = \frac{134}{144} = \frac{67}{72},$$

which is the classical value $F_{\mathrm{bd}}\big(\frac{3}{4}\big) = \frac{67}{72}$ [8, 2]. The single mode is $C_1\big(\frac{1}{4}\big) = q_2 \cdot 16 - \frac{67}{72}\cdot 16 = \frac{1}{9}$, that is

$$\mathrm{up}_n\big(\tfrac{1}{4}\big) = \frac{67}{72} + \frac{1}{9}\,4^{-n} = \frac{67}{72} + \frac{4}{9}\,4^{-(n+1)} \qquad (n \ge 2),$$

and the analytic formula for the coefficient agrees: the differential law $\mathrm{up}'(x) = 2\,\mathrm{up}(2x+1) - 2\,\mathrm{up}(2x-1)$ differentiated once more gives $\mathrm{up}''(\frac{1}{4}) = 8[\mathrm{up}(4) - \mathrm{up}(2)] - 8[\mathrm{up}(0) - \mathrm{up}(-2)] = -8$, and $C_1\big(\frac{1}{4}\big) = (-1)^1 a_1 4^{-1}\,\mathrm{up}''(\frac{1}{4}) = \frac{1}{18}\cdot\frac{1}{4}\cdot 8 = \frac{1}{9}$. Two independent checks: the level $n = 4$, which the row did not use, gives $\mathrm{up}_4(\frac{1}{4}) = \frac{715}{768} = \frac{67}{72} + \frac{1}{9}\cdot\frac{1}{256}$ exactly; and the level $n = 1 = s - 1$, at which $\frac{1}{4}$ is a knot of up_1 (Theorem 3.4), gives $\mathrm{up}_1(\frac{1}{4}) = 1$, whereas the law would predict $\frac{67}{72} + \frac{1}{36} = \frac{23}{24}$. The onset $n \ge s$ is therefore sharp at this point. All six packages' verifiers reproduce these rationals (S1, S2 and S5 print them; S3 and S4 include $x = \frac{1}{4}$ in their exhaustive sweeps).

A map of the volume. Section 2 fixes the Fourier normalization, defines Φ_n and up_n, proves the finite convolution model, the exact Thue–Morse truncated-power formula, rationality and strict positivity, and states the indexing bridge to the Exponents volume and to Lean once and for all. Section 3 defines the depth, proves that a dyadic point is the centre of one polynomial cell of up_n exactly when $n \ge s(x)$ and is a knot at $n = s(x) - 1$, and proves that the omitted tail sees a single polynomial piece. The sections that follow introduce the coefficients a_m (with the identification $a_m = (-1)^m \gamma_{2m}/(2m)!$ of S1's moment-generating-function normalization and the Bernoulli–Bell formula), invert the tail convolution on a finite polynomial jet, prove that the dyadic Taylor jet of up terminates at order $s(x)$, and combine the three to obtain Theorem 1.1, including the statement that every one of the d modes is nonzero for $s \ge 2$ and the one-stage sharpening at odd depth. The extraction part proves Theorem 1.2 a second time as the first row of an inverse geometric Vandermonde system, recovers every $C_r(x)$ and every even derivative from the same samples, gives the integer base-4 form, the annihilating q-difference recurrence, the Romberg-style triangular table, the minimality of $d + 1$ samples and the uniform stability bound on the weights. The remaining parts give worked exact examples through depth seven, an exact algorithm with its verification suite, structural consequences, and the appendices with the finite q-binomial identities, the notation concordance, the repair log and the formalization register.

2 Fourier normalization and the finite splines

2.1 The 2π convention and the two sinc normalizations

All transforms in this volume are 2π-frequency transforms in the sense of the catalogue entry `fourier-two-pi`: for $f \in L^1(\mathbb{R})$,

$$\widehat{f}_{2\pi}(\xi) := \int_{\mathbb{R}} f(x)\,\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,\mathrm{d}x, \qquad f(x) = \int_{\mathbb{R}} \widehat{f}_{2\pi}(\xi)\,\mathrm{e}^{2\pi\mathrm{i}x\xi}\,\mathrm{d}\xi, \tag{8}$$

the inversion formula being asserted only where it holds classically (here: whenever $\widehat{f}_{2\pi} \in L^1$ and f is continuous). There is no scalar prefactor in either direction, convolution goes to the pointwise product, $\widehat{f * g}_{2\pi} = \widehat{f}_{2\pi}\,\widehat{g}_{2\pi}$, and dilation obeys $\widehat{\delta^{-1} f(\cdot/\delta)}_{2\pi}(\xi) = \widehat{f}_{2\pi}(\delta\xi)$ for $\delta > 0$. The sinc function is the *radian* sinc of the catalogue entry `sinc`,

$$\operatorname{sinc}_{\mathrm{rad}}(z) := \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases} \qquad \operatorname{sinc}_\pi(z) = \operatorname{sinc}_{\mathrm{rad}}(\pi z), \tag{9}$$

so that the product Equation (1) may equally be written $\prod_{k\geq 0} \operatorname{sinc}_\pi(\xi/2^k)$. Under Equation (8), $\Phi = \widehat{\mathrm{up}}_{2\pi}$ is the product Equation (1); this is the normalization of Rvachev and of Arias de Reyna [12, 1], and the catalogue records that the product converges locally uniformly on $\mathbb{C}$ to an entire function with $\Phi(0) = 1$.

2.2 Boxes, the finite product, and the finite spline

Definition 2.1 (Dyadic boxes, finite products, finite splines)**.** For $k \in \mathbb{N}_0$ let

$$b_k(x) := 2^k\,\mathbf{1}_{[-2^{-k-1},\,2^{-k-1}]}(x), \qquad x \in \mathbb{R}, \tag{10}$$

the centred uniform density of total mass one and half-width 2^{-k-1}. For $n \in \mathbb{N}_0$ define

$$\Phi_n(\xi) := \prod_{k=0}^{n} \operatorname{sinc}_{\mathrm{rad}}\left(\frac{\pi\xi}{2^k}\right), \qquad \mathrm{up}_n := b_0 * b_1 * \cdots * b_n, \qquad \delta_n := 2^{-(n+1)}. \tag{11}$$

Thus Φ_n has exactly $n+1$ factors, indexed $k = 0, \ldots, n$, and δ_n is the half-width of the last box b_n.

Lemma 2.2 (Transform of a box)**.** *$\widehat{b_k}_{2\pi}(\xi) = \operatorname{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$ for every $k \in \mathbb{N}_0$ and $\xi \in \mathbb{R}$.*

Proof. With $h = 2^{-k-1}$ and $\xi \neq 0$, $\widehat{b_k}_{2\pi}(\xi) = 2^k \int_{-h}^{h} \mathrm{e}^{-2\pi i x\xi}\,\mathrm{d}x = 2^k \dfrac{\sin(2\pi h\xi)}{\pi\xi} = \dfrac{\sin(\pi\xi/2^k)}{\pi\xi/2^k}$, since $2^k \cdot 2h = 1$; at $\xi = 0$ both sides equal 1. □

Proposition 2.3 (Finite convolution model)**.** *For every $n \in \mathbb{N}_0$:*

1. *$\widehat{\mathrm{up}_n}_{2\pi} = \Phi_n$. For $n \geq 1$ the function Φ_n is integrable, up_n is continuous, and $\mathrm{up}_n = \mathcal{F}_{2\pi}^{-1}[\Phi_n]$ holds pointwise; for $n = 0$, $\mathrm{up}_0 = b_0$ and inversion holds at every point except the two jumps $\pm\frac{1}{2}$, where the inversion integral returns the half-height $\frac{1}{2}$.*

2. *up_n is an even, nonnegative probability density with*

$$\operatorname{supp}(\mathrm{up}_n) = [-1 + \delta_n,\ 1 - \delta_n]. \tag{12}$$

3. *(Probabilistic model.) Let $V_0, V_1, \ldots$ be independent random variables, each uniform on $[-\frac{1}{2}, \frac{1}{2}]$, and put*

$$Z_n := \sum_{k=0}^{n} 2^{-k} V_k, \qquad Z := \sum_{k=0}^{\infty} 2^{-k} V_k. \tag{13}$$

The series converges absolutely and surely, $|Z| \leq 1$, up_n is the density of Z_n, and up is the density of Z. In particular up_n is the density of a sum of $n+1$ independent centred uniforms of half-widths $\frac{1}{2}, \frac{1}{4}, \ldots, 2^{-n-1}$.

Proof. (1) Each $b_k \in L^1$, so the convolution theorem and Theorem 2.2 give $\widehat{\mathrm{up}_n}_{2\pi} = \prod_{k=0}^{n} \widehat{b_k}_{2\pi} = \Phi_n$. Since $\left|\operatorname{sinc}_{\mathrm{rad}}(\pi\xi/2^k)\right| \leq 2^k/(\pi\,|\xi|)$ and also ≤ 1, one has $|\Phi_n(\xi)| \leq \min\{1,\, C_n\,|\xi|^{-(n+1)}\}$ with $C_n = 2^{\binom{n+1}{2}}\pi^{-(n+1)}$, which is integrable for $n \geq 1$. A convolution of a bounded function with an L^1

function is continuous, so $\mathrm{up}_n = b_0 * (b_1 * \cdots * b_n)$ is continuous for $n \geq 1$, and the classical inversion theorem applies. For $n = 0$ the statement is the Dirichlet–Jordan theorem for the box.

(2) Convolution of even, nonnegative functions of integral one is even, nonnegative, and of integral one (Fubini). The support of a convolution of compactly supported nonnegative functions is the Minkowski sum of the supports, so the support radius is $\sum_{k=0}^{n} 2^{-k-1} = 1 - 2^{-(n+1)}$.

(3) $\left|2^{-k}V_k\right| \leq 2^{-k-1}$, so the series converges absolutely for every sample point and $|Z| \leq \sum_k 2^{-k-1} = 1$. The density of $2^{-k}V_k$ is b_k, and the density of a sum of independent variables is the convolution of the densities, so Z_n has density up_n. The characteristic function of Z at $-2\pi\xi$, that is the 2π-transform of its law, is $\mathbb{E}\,\mathrm{e}^{-2\pi\mathrm{i}\xi Z} = \lim_n \mathbb{E}\,\mathrm{e}^{-2\pi\mathrm{i}\xi Z_n} = \lim_n \Phi_n(\xi) = \Phi(\xi)$ by dominated convergence and the locally uniform convergence of the product. The finite measure $\mathrm{up}(x)\,\mathrm{d}x$ has the same transform Equation (1), and a finite Borel measure is determined by its Fourier transform; hence Z has density up. □

Proposition 2.4 (Head–tail factorization). *For $\delta > 0$ write $T_\delta(y) := \delta^{-1}\,\mathrm{up}(y/\delta)$ for the copy of* up *rescaled to the support $[-\delta, \delta]$; it is an even probability density with $\widehat{T_\delta}_{2\pi}(\xi) = \Phi(\delta\xi)$. For every $n \in \mathbb{N}_0$,*

$$\Phi(\xi) = \Phi_n(\xi)\,\Phi(\delta_n\xi), \qquad \boxed{\mathrm{up} = \mathrm{up}_n * T_{\delta_n},} \qquad Z \overset{\text{law}}{=} Z_n + \delta_n Z', \tag{14}$$

where Z' is a copy of Z independent of Z_n. Equivalently, $\Phi_n(\xi) = \Phi(\xi)/\Phi(\delta_n\xi)$ wherever the denominator is nonzero.

Proof. Reindex the tail of the product by $k = n + 1 + j$:

$$\prod_{k=n+1}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\left(\frac{\pi\xi}{2^k}\right) = \prod_{j=0}^{\infty} \mathrm{sinc}_{\mathrm{rad}}\left(\frac{\pi(2^{-(n+1)}\xi)}{2^j}\right) = \Phi(\delta_n\xi),$$

which is the first identity. The dilation rule gives $\widehat{T_{\delta_n}}_{2\pi}(\xi) = \Phi(\delta_n\xi)$, so the first identity says $\widehat{\mathrm{up}}_{2\pi} = \widehat{\mathrm{up}_n}_{2\pi}\,\widehat{T_{\delta_n}}_{2\pi} = \widehat{\mathrm{up}_n * T_{\delta_n}}_{2\pi}$. Both up and $\mathrm{up}_n * T_{\delta_n}$ are in L^1, and an L^1 function is determined almost everywhere by its transform; both sides are continuous (the right side as a convolution of a bounded function with an L^1 function), so they agree everywhere. For the random-variable form, the omitted sum is $\sum_{k\geq n+1} 2^{-k}V_k = 2^{-(n+1)}\sum_{j\geq 0} 2^{-j}V_{n+1+j}$, and $Z' := \sum_{j\geq 0} 2^{-j}V_{n+1+j}$ is built from the variables $V_{n+1}, V_{n+2}, \ldots$, hence is independent of Z_n and has the law of Z. □

The factorization Equation (14) is exact for every n; no limit is taken. It is the source of the base $Q = \frac{1}{4}$: the omitted tail lives at the spatial scale δ_n, its law is even so only even moments enter, and $\delta_{n+1}^2/\delta_n^2 = \frac{1}{4}$.

2.3 The exact truncated-power formula

Write $\varepsilon_k = (-1)^{\mathrm{wt}_2(k)}$ for the signed Thue–Morse sequence, where $\mathrm{wt}_2(k)$ is the number of ones in the binary expansion of k, so that $(\varepsilon_k)_{k\geq 0} = 1, -1, -1, 1, -1, 1, 1, -1, \ldots$; and write $(y)_+^n = y^n$ for $y > 0$ and 0 for $y \leq 0$, with the convention $(y)_+^0 = \mathbf{1}_{(0,\infty)}(y)$ at $n = 0$.

Theorem 2.5 (Exact finite spline with Thue–Morse knots). *For every $n \geq 1$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathrm{up}_n(x) = \frac{2^{\binom{n+1}{2}}}{n!} \sum_{k=0}^{2^{n+1}-1} \varepsilon_k \left(x + 1 - \delta_n - \frac{k}{2^n}\right)_+^n.} \tag{15}$$

Consequently:

1. *The knots are the* 2^{n+1} *points*

$$t_{n,k} \coloneqq -1+\delta_n+\frac{k}{2^n} = -1+(2k+1)\,\delta_n = \big(2k+1-2^{n+1}\big)\delta_n, \qquad 0 \le k < 2^{n+1}, \tag{16}$$

the odd multiples of δ_n *in* $[-1+\delta_n, 1-\delta_n]$*; consecutive knots are* $2\delta_n$ *apart, the first and last knots are the endpoints of the support.*

2. $\mathrm{up}_n \in C^{n-1}(\mathbb{R})$*, and on each open interval free of knots* up_n *coincides with a polynomial of degree at most* n*.*

3. *Off the knots the* n*-th derivative is the step function*

$$\mathrm{up}_n{}^{(n)}(x) = 2^{\binom{n+1}{2}} \sum_{k=0}^{2^{n+1}-1} \varepsilon_k\, \mathbf{1}_{(t_{n,k},\infty)}(x), \tag{17}$$

which jumps by $2^{\binom{n+1}{2}}\varepsilon_k \neq 0$ *at* $t_{n,k}$*. Hence every* $t_{n,k}$ *is a genuine knot:* up_n *agrees with no polynomial on any neighbourhood of* $t_{n,k}$*.*

For $n=0$ *the right-hand side of Equation (15) equals* $\mathbf{1}_{[-\frac{1}{2},\frac{1}{2})}$*, which is* $b_0 = \mathrm{up}_0$ *except at the jump* $x=\frac{1}{2}$*.*

Proof. Step 1: the top derivative is atomic. With $h_k = 2^{-k-1}$ one has $b_k = 2^k\big(\mathbf{1}_{(-h_k,\infty)} - \mathbf{1}_{(h_k,\infty)}\big)$ almost everywhere, hence in the sense of distributions $Db_k = 2^k(\delta_{-h_k} - \delta_{h_k})$, a difference of two unit point masses. Convolution of compactly supported distributions is associative and commutative and satisfies $D(u*v) = (Du)*v$, so

$$D^{n+1}\,\mathrm{up}_n = (Db_0)*(Db_1)*\cdots*(Db_n) = \prod_{k=0}^{n} 2^k \sum_{\beta\in\{0,1\}^{n+1}} (-1)^{\beta_0+\cdots+\beta_n}\,\delta_{c(\beta)},$$
$$c(\beta) = \sum_{k=0}^{n}(-1)^{1-\beta_k}h_k = -(1-\delta_n) + \sum_{k=0}^{n}\beta_k 2^{-k},$$

where $\beta_k = 1$ selects the right endpoint $+h_k$ of the k-th box (sign -1) and $\beta_k = 0$ the left endpoint $-h_k$ (sign $+1$), and the second form of $c(\beta)$ uses $\sum_k h_k = 1-\delta_n$ and $2h_k = 2^{-k}$. Put

$$k(\beta) \coloneqq \sum_{j=0}^{n}\beta_j 2^{n-j}.$$

As β runs over the binary cube, $k(\beta)$ runs exactly once over $0,1,\dots,2^{n+1}-1$ (binary expansion with the digits read in reverse order), $\sum_j \beta_j 2^{-j} = k(\beta)/2^n$, and $(-1)^{\sum_j\beta_j} = \varepsilon_{k(\beta)}$ because reversing the binary digits does not change their sum. With $\prod_{k=0}^{n}2^k = 2^{\binom{n+1}{2}}$ this gives

$$D^{n+1}\,\mathrm{up}_n = 2^{\binom{n+1}{2}}\sum_{k=0}^{2^{n+1}-1}\varepsilon_k\,\delta_{t_{n,k}}, \qquad t_{n,k} = -1+\delta_n+\frac{k}{2^n}. \tag{18}$$

Step 2: integrate back. Let $R(x)$ denote the right-hand side of Equation (15). For $n \ge 1$ the function $y \mapsto (y)_+^n/n!$ is of class C^{n-1}, its n-th derivative is $\mathbf{1}_{(0,\infty)}$ off the origin, and its $(n+1)$-st distributional derivative is δ_0. Hence $D^{n+1}R$ equals the right-hand side of Equation (18), so $D^{n+1}(\mathrm{up}_n - R) = 0$ and $\mathrm{up}_n - R$ is a polynomial of degree at most n (the only distributions annihilated by D^{n+1}). For $x < -1+\delta_n = t_{n,0}$ every truncated power in R vanishes, and up_n vanishes by Equation (12); a polynomial vanishing on a half-line is identically zero. This proves Equation (15).

Consequences. (1) is read off the formula: $t_{n,k} = -1+(2k+1)2^{-(n+1)}$, and $2k+1-2^{n+1}$ runs over the odd integers from $1-2^{n+1}$ to $2^{n+1}-1$. (2) Each summand is C^{n-1} and polynomial of degree at most n off its own knot. (3) Differentiate Equation (15) n times off the knots, using $D^n\,(y)_+^n = n!\,\mathbf{1}_{(0,\infty)}(y)$; the knots are pairwise distinct, so exactly one indicator switches at each $t_{n,k}$, and the jump is $2^{\binom{n+1}{2}}\varepsilon_k$. A polynomial has a continuous n-th derivative, so no polynomial agrees with up_n on a neighbourhood of a knot. For $n = 0$, $\left(x+\frac{1}{2}\right)_+^0 - \left(x-\frac{1}{2}\right)_+^0 = \mathbf{1}_{[-\frac{1}{2},\frac{1}{2})}(x)$ with the stated convention. □

Corollary 2.6 (Scaled cell formula and rationality). *Let $n \geq 1$ and $x \in [-1, 1]$, and put*

$$y := 2^n(x + 1 - \delta_n) = 2^n(x + 1) - \tfrac{1}{2}, \qquad 0 \leq \lfloor y \rfloor \leq 2^{n+1} - 1. \tag{19}$$

Then

$$\mathrm{up}_n(x) = \frac{2^{-\binom{n}{2}}}{n!} \sum_{0 \leq k \leq \lfloor y \rfloor} \varepsilon_k\,(y-k)^n, \tag{20}$$

where the sum is empty if $y < 0$. In particular:

1. *if $x \in \mathbb{Q}$ then $\mathrm{up}_n(x) \in \mathbb{Q}$;*
2. *if $x = m/2^s$ with $m \in \mathbb{Z}$ and $n \geq s$, then $y \in \mathbb{Z} + \frac{1}{2}$ and $2^{\binom{n+1}{2}}\,n!\,\mathrm{up}_n(x) \in \mathbb{Z}$.*

Proof. $x + 1 - \delta_n - k/2^n = 2^{-n}(y - k)$, and the term is active exactly when $k < y$, that is $k \leq \lfloor y \rfloor$ when $y \notin \mathbb{Z}$ and $k \leq y - 1$ when $y \in \mathbb{Z}$; in the latter case the term $k = y$ contributes 0 anyway. Factor 2^{-n^2} out of the n-th powers: $2^{\binom{n+1}{2}-n^2} = 2^{-\binom{n}{2}}$. For $x \leq 1$ one has $y \leq 2^{n+1} - \frac{1}{2}$, so no index beyond $2^{n+1} - 1$ is ever active, and for $x \geq -1$ one has $y \geq -\frac{1}{2}$. Rationality is now immediate. If $2^s x \in \mathbb{Z}$ and $n \geq s$, then $2^n(x+1) \in \mathbb{Z}$, so $y - k \in \mathbb{Z} + \frac{1}{2}$, $(y-k)^n \in 2^{-n}\mathbb{Z}$, and the sum lies in $2^{-n}\mathbb{Z}$; multiplying by $2^{\binom{n}{2}}n! \cdot 2^n = 2^{\binom{n+1}{2}}n!$ gives an integer. □

The scaled formula Equation (20) is the form in which the sources compute (S3, S4, S6) and is, term for term, the definition of `fabiusUniformSpline` in the repository (Section 2.5). The denominator bound in (2) is sharp in the examples: $\mathrm{up}_2(\frac{1}{4}) = \frac{15}{16}$ with $2^3 \cdot 2! = 16$, and $\mathrm{up}_3(\frac{1}{4}) = \frac{179}{192}$ with $2^6 \cdot 3! = 384$.

Remark 2.7 (Prouhet cancellation, in advance). Summing Equation (18) over all k gives $\sum_{k<2^{n+1}} \varepsilon_k = 0$ for $n \geq 0$, which is the reason the step function Equation (17) returns to 0 beyond the last knot. More is true: $\sum_{k<2^m} \varepsilon_k\,k^j = 0$ for all $0 \leq j < m$ (the generating polynomial $\sum_{k<2^m} \varepsilon_k z^k = \prod_{h<m}(1 - z^{2^h})$ has a zero of order m at $z = 1$; catalogue entry `thue-morse-block-sum`). This is the mechanism by which, at a dyadic point, the local degree of up_n collapses from n to $s(x)$; the section on local degree collapse develops it.

2.4 Basic analytic properties and strict positivity

Proposition 2.8 (Basic properties of up and up_n).

1. up *is an even, nonnegative C^∞ probability density with topological support $[-1, 1]$, $\mathrm{up}(0) = 1$, $\mathrm{up}(\pm 1) = 0$, all derivatives vanishing at ± 1, and*

$$\mathrm{up}'(x) = 2\,\mathrm{up}(2x + 1) - 2\,\mathrm{up}(2x - 1), \qquad x \in \mathbb{R}. \tag{21}$$

2. $\mathrm{up}_n(x) > 0$ *for $|x| < 1 - \delta_n$, for every $n \geq 0$.*
3. $\mathrm{up}(x) > 0$ *for $|x| < 1$.*
4. $\mathrm{up}_n(0) = 1$ *for every $n \geq 0$.*

Proof. (1) The regularity, support, normalization and the differential law are the content of the catalogue entry `rvachev-up`, Lean-proved in the repository [10]; we recall the one-line derivations that are used again below. Smoothness: by Theorem 2.3(1), $|\Phi(\xi)| \leq |\Phi_K(\xi)| \leq C_K\,|\xi|^{-(K+1)}$ for every K, so $\xi^m\Phi(\xi) \in L^1$ for every m and the inversion integral may be differentiated under the sign arbitrarily often. The value at 0: by Equation (14) with $n = 0$, $\mathrm{up} = b_0 * T_{1/2}$, and $T_{1/2}$ is a probability density supported in $[-\frac{1}{2}, \frac{1}{2}]$, on which $b_0 \equiv 1$; hence $\mathrm{up}(0) = \int b_0(-y)T_{1/2}(y)\,\mathrm{d}y = 1$. The same identity reads $\mathrm{up}(x) = \int_{x-1/2}^{x+1/2} 2\,\mathrm{up}(2y)\,\mathrm{d}y = \int_{2x-1}^{2x+1} \mathrm{up}(z)\,\mathrm{d}z$, and differentiating gives Equation (21).

(2) Let $f, g \geq 0$ be integrable with $f > 0$ on $(-\alpha, \alpha)$ and $g > 0$ on $(-\beta, \beta)$, f vanishing outside $[-\alpha, \alpha]$. For $|x| < \alpha + \beta$ the set $\{y : |y| < \beta,\ |x - y| < \alpha\} = (-\beta, \beta) \cap (x - \alpha, x + \alpha)$ is a nonempty open interval (its endpoints $\max(-\beta, x - \alpha) < \min(\beta, x + \alpha)$ precisely because $|x| < \alpha + \beta$), and on it the integrand of $(f * g)(x) = \int f(x - y)g(y)\,\mathrm{d}y$ is strictly positive; hence $(f * g)(x) > 0$. Apply this inductively to $\mathrm{up}_n = \mathrm{up}_{n-1} * b_n$ with $\alpha = 1 - \delta_{n-1}$ and $\beta = \delta_n$, noting $\alpha + \beta = 1 - \delta_n$, starting from $\mathrm{up}_0 = b_0 > 0$ on $(-\frac{1}{2}, \frac{1}{2})$.

(3) Fix $|x| < 1$ and choose n with $2\delta_n < 1-|x|$. For $|y| \leq \delta_n$ one has $|x - y| \leq |x|+\delta_n < 1-\delta_n$, so by (2) and continuity ($n \geq 1$) the function $y \mapsto \mathrm{up}_n(x-y)$ is bounded below by a positive constant c on $[-\delta_n, \delta_n]$, which carries the whole mass of T_{δ_n}. By Equation (14), $\mathrm{up}(x) = \int \mathrm{up}_n(x - y)T_{\delta_n}(y)\,\mathrm{d}y \geq c > 0$.

(4) $\mathrm{up}_n = b_0 * g_n$ with $g_n = b_1 * \cdots * b_n$ a probability density supported in $[-\frac{1}{2}+\delta_n, \frac{1}{2}-\delta_n] \subset [-\frac{1}{2}, \frac{1}{2}]$ ($g_0 = \delta_0$), on which $b_0 \equiv 1$; hence $\mathrm{up}_n(0) = \int b_0(-y)g_n(y)\,\mathrm{d}y = 1$. □

2.5 The indexing bridge

The finite spline of this volume has appeared under three indexings in the corpus. The box below fixes the dictionary once; every later use of an outside result is to be read through it.

One spline, three indexings

All six sources, and this volume, index the finite spline by the *largest factor index*: $\Phi_n = \prod_{k=0}^{n} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$ has $n + 1$ factors, up_n has degree n, and the omitted tail has half-width $\delta_n = 2^{-(n+1)}$. Two other conventions are in use in the corpus, and every formula transported from elsewhere must be shifted once.

- **The Exponents volume** [9] indexes by the *number of factors*: its prefix product (equation `p3:eq:prefix-product`) is $\Phi_n(\xi) = \prod_{j=0}^{n-1} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^j)$ with n factors, $\widehat{p_n}_{2\pi} = \Phi_n$, $p_n = u_{2^{-1}} * \cdots * u_{2^{-n}}$ with $u_a = (2a)^{-1}\mathbf{1}_{[-a,a]}$, and its tail scale is $\delta_N = 2^{-N}$. The bridge is

$$\boxed{\mathrm{up}_n = p_{n+1}, \qquad \Phi_n = \Phi_{n+1}, \qquad \delta_n^{(\text{here})} = 2^{-(n+1)} = \delta_{n+1}^{(\text{there})},} \tag{22}$$

  so the sharp error law $2^{\binom{r+3}{2}-1}/(9 \cdot 4^N)$ of `p3:thm:sharp-prefix-law` and the exact CDF law $1/(9 \cdot 4^N)$ of `p3:cor:kolmogorov` read, for up_n, with $4^N = 4^{n+1}$; and a mode 4^{-rN} there is a mode $4^{-r(n+1)} = 4^{-r} \cdot 4^{-rn}$ here (Theorem 1.4).

- **The Lean development** [10] works on the Fabius interval. Its centred spline `fabiusUniformSpline p y` is, for $y \in [0, 1]$, the sum $\left(2^{\binom{p}{2}}p!\right)^{-1} \sum_{r<\lfloor 2^p y+1/2\rfloor} \varepsilon_r \left(2^p y - \frac{1}{2} - r\right)^p$, that is Equation (20) with $y = 2^p(x + 1) - \frac{1}{2}$ replaced by $2^p y - \frac{1}{2}$. Hence, for $n \geq 1$ and $x \in [-1, 1]$,

$$\boxed{\mathrm{up}_n(x) = \texttt{fabiusUniformSpline}\ n\ (1 + x) = \texttt{fabiusUniformSpline}\ n\ (1 - |x|),} \tag{23}$$

  the second form by evenness of up_n. The degree matches the index: up_n *is* `fabiusUniformSpline n`, and the Lean theorem `fabiusUniformSpline_eq_centeredPartialCDF` identifies it on $[0, 1]$ with the distribution function of $\sum_{j=1}^{n} 2^{-j}U_j + 2^{-n-1}$, U_j uniform on $[0, 1]$ (the finite-level form of the fold $\mathrm{up}(x) = F_{\mathrm{bd}}(1 - |x|)$ of the catalogue). The module `QuarterSplineLocalPolynomial.lean` writes P_n for the density of the first n centred uniforms, the report convention of the inverse frontier, and records

$P_n =$ `fabiusUniformSpline` $(n-1)$: thus

$$P_n = \mathrm{up}_{n-1} = p_n, \tag{24}$$

and its kernel-checked local formula $P_n(\frac{1}{4}+z) = \frac{5}{72}+z+4z^2-\frac{4}{9}4^{-n}$ for $n \geq 3$, $|z| \leq 2^{-n}$, is in this volume's terms $\mathrm{up}_m(\frac{3}{4}-z) = \frac{5}{72}+z+4z^2-\frac{1}{9}4^{-m}$ for $m = n-1 \geq 2$, the depth-two instance $x = \frac{3}{4}$ of Theorem 1.1 with $C_1(\frac{3}{4}) = -\frac{1}{9}$ (compare $C_1(\frac{1}{4}) = +\frac{1}{9}$ in Theorem 1.6; the sign flips with up'').

The rule of this volume: *every* formula keeps the sources' n; p_n, Φ_n (Exponents), P_n (report) and `fabiusUniformSpline` are mentioned only in this box and in the formalization register, always through Equations (22) to (24). The most common error the shift produces is the confusion between 4^{-n} and $4^{-(n+1)}$ in the coefficient of a mode; the second most common is an onset stated as $N \geq s+1$ (factor count) being copied as $n \geq s+1$ (largest index). The onset in this volume's indexing is $n \geq s$, and Theorem 1.6 shows it cannot be lowered.

Setting, in five lines

2π-frequency transforms, radian sinc; $\Phi = \widehat{\mathrm{up}}_{2\pi} = \prod_{k\geq 0} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$; Φ_n keeps $k = 0, \dots, n$ ($n+1$ factors); $\mathrm{up}_n = b_0 * \cdots * b_n$ is its inverse transform, the density of $Z_n = \sum_{k\leq n} 2^{-k} V_k$, an even C^{n-1} box spline of degree n on $[-1+\delta_n, 1-\delta_n]$, $\delta_n = 2^{-(n+1)}$, with the exact Thue–Morse truncated-power formula Equation (15) and knots at the odd multiples of δ_n; $\mathrm{up} = \mathrm{up}_n * T_{\delta_n}$ exactly, for every n; $\mathrm{up}_n(x) \in \mathbb{Q}$ at rational x, with denominator dividing $2^{\binom{n+1}{2}} n!$ once $n \geq s(x)$; $\mathrm{up}_n = p_{n+1}$ in the Exponents volume and $\mathrm{up}_n =$ `fabiusUniformSpline` n in Lean.

3 Dyadic depth and cell geometry

3.1 The depth

Definition 3.1 (Dyadic depth). A dyadic rational is a number $x = m/2^n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}_0$ (catalogue entry `fabius-dyadic-cells`). Its *depth* is

$$s(x) := \min\{s \in \mathbb{N}_0 : \ 2^s x \in \mathbb{Z}\}. \tag{25}$$

Thus $s(x) = 0$ exactly when $x \in \mathbb{Z}$, and for $s = s(x) \geq 1$ the integer $a := 2^s x$ is odd, so that $x = a/2^s$ is the unique representation of x with an odd numerator; for $0 < x < 1$ this is the catalogue's "reduced at level s". The four letters r (S1), d (S2), m (S3) and s (S4, S5, S6) of the sources all denote $s(x)$.

If $a = 2^s x$ were even for $s \geq 1$, then $2^{s-1}x = a/2$ would be an integer, contradicting minimality; this is the whole content of "odd".

Lemma 3.2 (Representation independence and invariances). *Let $x = m/2^n$ with $m \in \mathbb{Z}$, $n \in \mathbb{N}_0$, not necessarily reduced, and let $\nu_2(m)$ be the 2-adic valuation of m, with $\nu_2(0) = +\infty$. Then*

$$s(x) = \max\{0, \ n - \nu_2(m)\}, \tag{26}$$

which is invariant under $(m, n) \mapsto (2m, n+1)$. Moreover, for every dyadic x and every $c \in \mathbb{Z}$,

$$s(-x) = s(x), \qquad s(x+c) = s(x), \qquad s(1-|x|) = s(x). \tag{27}$$

Proof. If $m = 0$ then $x = 0$, $s(x) = 0$, and the right-hand side of Equation (26) is $\max\{0, -\infty\} = 0$. Otherwise write $m = 2^v u$ with $v = \nu_2(m)$ and u odd. Then $2^s x = 2^{s+v-n}u$ is an integer if and only if $s+v-n \geq 0$, because u is odd; the least such $s \geq 0$ is $\max\{0, n-v\}$. Replacing (m,n) by $(2m, n+1)$ increases both v and n by one. The invariances: $2^s x$ is an integer if and only if $-2^s x$ is; if and only if $2^s x + 2^s c$ is; and $1-|x|$ is $1-x$ or $1+x$, both covered by the first two. □

The third invariance is what allows S6 to work with $a = 1 - |x| \in [0, 1]$ throughout: the depth, and hence d, is the same for x and for its fold $1 - |x|$, and Equation (23) transports the samples.

3.2 Knots and cells

Lemma 3.3 (The knot lattice of up_n). *Let $n \geq 1$. The knot set of up_n is*

$$K_n := \{t_{n,k} : 0 \leq k < 2^{n+1}\} = \{(2j+1)\delta_n : \ j \in \mathbb{Z},\ |2j+1| \leq 2^{n+1} - 1\}, \tag{28}$$

the odd multiples of δ_n of absolute value at most $1 - \delta_n$. The complement of K_n in $(-1, 1)$ is the disjoint union of the $2^{n+1} - 1$ open cells

$$\mathcal{C}_{n,j} := (2j\delta_n - \delta_n,\ 2j\delta_n + \delta_n), \qquad -2^n < j < 2^n, \tag{29}$$

each of width $2\delta_n$ and centred at the even multiple $2j\delta_n$ of δ_n; on each cell up_n is a polynomial of degree at most n, and on no neighbourhood of a knot is it a polynomial. Outside $[-1 + \delta_n, 1 - \delta_n]$ the function vanishes identically. The cell centres are exactly the even multiples of δ_n in $(-1, 1)$; 0 is always a cell centre.

Proof. By Theorem 2.5(1), $t_{n,k} = (2k+1-2^{n+1})\delta_n$ with $0 \leq k < 2^{n+1}$, and $2k+1-2^{n+1}$ runs over the odd integers of absolute value at most $2^{n+1}-1$. Two consecutive odd multiples $(2j-1)\delta_n < (2j+1)\delta_n$ of δ_n bound the open interval $\mathcal{C}_{n,j}$, whose midpoint is $2j\delta_n$; both bounding points are knots exactly when $|2j \pm 1| \leq 2^{n+1} - 1$, that is $|j| \leq 2^n - 1$. The polynomial and genuine-knot statements are Theorem 2.5(2),(3), and the vanishing outside the support is Equation (12). Since $1 = 2^{n+1}\delta_n$ is an even multiple of δ_n, the even multiples strictly between -1 and 1 are $2j\delta_n$ with $|j| < 2^n$, and $j = 0$ gives 0. □

3.3 A dyadic point becomes a cell midpoint

For $x \in \mathbb{R}$ and $n \in \mathbb{N}_0$ call

$$W_n(x) := (x - \delta_n,\ x + \delta_n) \tag{30}$$

the *tail window* at x: it is the open interval of half-width δ_n around x, which is exactly the set of points $x - y$ with y in the interior of the support $[-\delta_n, \delta_n]$ of the tail kernel T_{δ_n} of Theorem 2.4. Its width $2\delta_n$ equals the width of a cell of up_n. Whether the window meets a knot is therefore decided by the position of x relative to the lattice $\delta_n\mathbb{Z}$, and that position is governed by the depth.

Lemma 3.4 (A dyadic point is a cell midpoint exactly from level $s(x)$ on). *Let $x \in [-1, 1]$ be dyadic with depth $s = s(x)$, and let $n \geq 1$. Write $\theta_n := x/\delta_n = 2^{n+1}x$.*

- *(i) If $n \geq s$, then θ_n is an even integer. If moreover $|x| < 1$, the point x is the centre of the cell $\mathcal{C}_{n,\theta_n/2}$, the two knots adjacent to x are $x \pm \delta_n$, and the tail window $W_n(x)$ is that cell: it contains no knot. If $x = \pm 1$, then $\mathrm{up}_n \equiv 0$ on $W_n(x)$. In all cases there is a polynomial $P_{n,x}$ of degree at most n, the* cell polynomial, *with*

$$\mathrm{up}_n(x + h) = P_{n,x}(h) \qquad \textit{for } |h| < \delta_n, \tag{31}$$

  and $P_{n,\pm 1} = 0$.
- *(ii) If $n = s - 1$ (so $s \geq 2$), then $\theta_n = 2^s x$ is an odd integer of absolute value at most $2^s - 1$: the point x is a knot of up_n, the only knot in $W_n(x)$, and up_n agrees with no polynomial on $W_n(x)$.*
- *(iii) If $1 \leq n \leq s - 2$, then $\theta_n \notin \mathbb{Z}$: the point x lies in the interior of a cell but is not its centre, and $W_n(x)$ contains exactly one knot, which is different from x; again up_n agrees with no polynomial on $W_n(x)$.*

Consequently $W_n(x)$ is free of knots if and only if $n \geq s$. For the degree-zero spline the same trichotomy holds with $\delta_0 = \frac{1}{2}$ and the two jumps $\pm\frac{1}{2}$ of b_0 as knots.

Proof. For $s = 0$, $x \in \{-1, 0, 1\}$ and $\theta_n = 2^{n+1}x$ is an even integer for every $n \geq 0$; cases (ii) and (iii) are empty. For $s \geq 1$ write $x = a/2^s$ with a odd; then $\theta_n = a\,2^{n+1-s}$. This is an integer if and only if $n + 1 \geq s$, and it is then even if and only if $n + 1 - s \geq 1$, that is $n \geq s$; at $n = s - 1$ it equals the odd integer a; and for $n \leq s - 2$ it is a rational with denominator $2^{s-1-n} \geq 2$.

(i) For $n \geq s$ and $|x| < 1$, $x = 2j\delta_n$ with $j = \theta_n/2$ and $|j| < 2^n$, so by Theorem 3.3 x is the centre of $\mathcal{C}_{n,j} = W_n(x)$, whose endpoints $(2j \pm 1)\delta_n = x \pm \delta_n$ are the adjacent knots and whose interior contains none. The cell polynomial is the polynomial that up_n equals on $\mathcal{C}_{n,j}$, shifted by x. For $x = 1$ the window is $(1 - \delta_n, 1 + \delta_n)$, on which $\mathrm{up}_n = 0$ by Equation (12); the case $x = -1$ is symmetric; so $P_{n,\pm 1} = 0$.

(ii) At $n = s - 1$, $\theta_n = a$ is odd with $|a| = 2^s\,|x| < 2^s$ (note $|x| < 1$ because $s \geq 1$), hence $|a| \leq 2^s - 1 = 2^{n+1} - 1$ and $x = a\delta_n \in K_n$ by Equation (28). The neighbouring knots $x \pm 2\delta_n$ lie outside the open window $W_n(x)$, so x is the only knot in it, and Theorem 2.5(3) says up_n is not a polynomial near x.

(iii) In units of δ_n the window is the open interval $(\theta_n - 1, \theta_n + 1)$ of length 2 about a non-integer, which contains exactly the two integers $\lfloor \theta_n \rfloor$ and $\lfloor \theta_n \rfloor + 1$, exactly one of which is odd; call it $2j + 1$. Both integers are strictly inside the window, and $|2j + 1| \leq 2^{n+1} - 1$ because $|\theta_n| < 2^{n+1}$ and $\theta_n \notin \mathbb{Z}$ force $|\lfloor \theta_n \rfloor|, |\lfloor \theta_n \rfloor + 1| \leq 2^{n+1}$ with the value $\pm 2^{n+1}$ even. So $(2j + 1)\delta_n$ is a knot of up_n lying in $W_n(x)$, different from x since θ_n is not an integer, and it is the only one since the next odd integers are outside. The point x itself is not a knot and lies in the open cell between the odd integers $2j \pm 1$, at distance $|\theta_n - 2j|\,\delta_n \neq 0$ from its centre. Non-polynomiality on $W_n(x)$ follows again from Theorem 2.5(3).

The "if and only if" collects (i)–(iii). For $n = 0$, $\theta_0 = 2x$, $\delta_0 = \frac{1}{2}$, and the knots of b_0 are $\pm\frac{1}{2} = \pm\delta_0$, which are the odd multiples of δ_0 in $[-\frac{1}{2}, \frac{1}{2}]$; the same case analysis applies verbatim. □

Theorem 3.4 is the knot-geometry explanation of the onset $n \geq s$ in Theorem 1.1. The main theorem is obtained by deconvolving $\mathrm{up} = \mathrm{up}_n * T_{\delta_n}$ on the tail window; that inversion is finite linear algebra only when the window carries a single polynomial, which by the lemma happens for all $n \geq s$ and for no $n < s$. At the levels $n < s$ the window straddles a knot, and the convolution mixes two polynomial pieces; the resulting values may still coincide with the eventual law by accident (the section on the odd-level sharpening isolates the one systematic instance, $n = s - 1$ with s odd), but no such coincidence is needed, and Theorem 1.6 shows that at $n = s - 1$ with s even it fails.

Example 3.5 (Depth zero and depth one, exactly)**.** The five dyadic points with $d = 0$ are $0, \pm 1$ (depth 0) and $\pm\frac{1}{2}$ (depth 1), and Theorem 1.1 asserts $\mathrm{up}_n(x) = \mathrm{up}(x)$ from level $n = s(x)$ on. This holds, and can be seen directly:

- $\mathrm{up}_n(0) = 1 = \mathrm{up}(0)$ for all $n \geq 0$ and $\mathrm{up}_n(\pm 1) = 0 = \mathrm{up}(\pm 1)$ for all $n \geq 0$, by Theorem 2.8(1),(4) and Equation (12).
- $\mathrm{up}_n(\pm\frac{1}{2}) = \frac{1}{2} = \mathrm{up}(\pm\frac{1}{2})$ for all $n \geq 1$. Indeed $\mathrm{up}_n = b_0 * g_n$ with $g_n = b_1 * \cdots * b_n$ an even probability density supported in $[-\frac{1}{2} + \delta_n, \frac{1}{2} - \delta_n]$, and $b_0(\frac{1}{2} - y) = 1$ exactly for $y \in [0, 1]$, so $\mathrm{up}_n(\frac{1}{2}) = \int_0^1 g_n = \frac{1}{2}$ by evenness (the point 0 carries no mass). The same computation with the tail $T_{1/2}$ in place of g_n gives $\mathrm{up}(\frac{1}{2}) = \frac{1}{2}$.

At $n = 0 = s(\frac{1}{2}) - 1$ the point $\frac{1}{2}$ is a jump of b_0, in accordance with Theorem 3.4(ii): $\mathrm{up}_0(\frac{1}{2})$ is 0 with the right-continuous convention of Theorem 2.5, 1 with the closed box Equation (10), and $\frac{1}{2}$ with the half-height inversion convention of Theorem 2.3(1). Only the last convention makes the law hold one level early; S3 calls it the symmetric-rectangle convention at $s = 1$. This is the single convention-dependent value in the volume, and the onset $n \geq s$ avoids it.

3.4 The tail sees only one polynomial piece

Lemma 3.6 (Jet transfer through the tail). *Let $x \in [-1,1]$ be dyadic with depth s, let $n \geq \max\{s,1\}$, and let $P_{n,x}$ be the cell polynomial of Equation (31). Then, with Z' a random variable of law* up*,*

$$\boxed{\mathrm{up}^{(r)}(x) = \int_{-\delta_n}^{\delta_n} P_{n,x}^{(r)}(-y)\, T_{\delta_n}(y)\; \mathrm{d}y = \mathbb{E}\big[P_{n,x}^{(r)}(-\delta_n Z')\big], \qquad 0 \leq r \leq n.} \tag{32}$$

In particular $\mathrm{up}(x) = \mathbb{E}[P_{n,x}(-\delta_n Z')]$*: the value of* up *at x is a moment functional of Z' applied to a single polynomial of degree at most n, and the omitted tail never sees a second polynomial piece.*

Proof. Start from $\mathrm{up}(x+h) = \int \mathrm{up}_n(x+h-y)T_{\delta_n}(y)\,\mathrm{d}y$, valid for all h by Equation (14).

Orders $r \leq n-1$. By Theorem 2.5(2) the derivatives ${\mathrm{up}_n}^{(r)}$, $r \leq n-1$, exist everywhere, are continuous and bounded (compact support). Differentiating under the integral sign r times is justified by dominated convergence, the difference quotients of ${\mathrm{up}_n}^{(r-1)}$ being bounded by $\sup|{\mathrm{up}_n}^{(r)}|$ and T_{δ_n} being integrable; hence $\mathrm{up}^{(r)}(x) = \int {\mathrm{up}_n}^{(r)}(x-y)T_{\delta_n}(y)\,\mathrm{d}y$.

Order $r = n$. The function ${\mathrm{up}_n}^{(n-1)}$ is continuous and piecewise polynomial, hence Lipschitz, hence absolutely continuous with almost-everywhere derivative $g := {\mathrm{up}_n}^{(n)}$, the bounded step function Equation (17) defined off the finite knot set. Then

$$\frac{\mathrm{up}^{(n-1)}(x+h) - \mathrm{up}^{(n-1)}(x)}{h} = \int \frac{1}{h}\int_0^h g(x+t-y)\,\mathrm{d}t\; T_{\delta_n}(y)\,\mathrm{d}y.$$

For every y outside the finite set $x - K_n$ the inner average tends to $g(x-y)$ as $h \to 0$, because g is continuous at $x-y$; the inner average is bounded by $\sup|g|$; and T_{δ_n} is integrable. Dominated convergence gives $\mathrm{up}^{(n)}(x) = \int g(x-y)T_{\delta_n}(y)\,\mathrm{d}y$.

Localization. T_{δ_n} vanishes outside $[-\delta_n, \delta_n]$, and for $|y| < \delta_n$ the point $x-y$ lies in the tail window $W_n(x)$, on which $\mathrm{up}_n(x-y) = P_{n,x}(-y)$ by Theorem 3.4(i) (this is where $n \geq s$ enters), so that ${\mathrm{up}_n}^{(r)}(x-y) = P_{n,x}^{(r)}(-y)$ for $r \leq n-1$ and $g(x-y) = P_{n,x}^{(n)}(-y)$ for y off the two endpoints. The endpoints $y = \pm\delta_n$ are a null set. Substituting yields the integral in Equation (32), and the substitution $y = \delta_n z$ with $T_{\delta_n}(\delta_n z)\,\mathrm{d}y = \mathrm{up}(z)\,\mathrm{d}z$ turns it into the expectation. □

Remark 3.7 (What the lemma says and what it does not). Equation (32) holds for the n-jet only. For $r > n$ the right-hand side would be 0, and $\mathrm{up}^{(r)}(x)$ is not in general 0; the correct statement for higher orders is the termination theorem for the dyadic jet, proved later, which says $\mathrm{up}^{(r)}(x) = 0$ for $r > s(x)$, a bound independent of n. The two facts combine in the deconvolution section: since up is even, the law of Z' is even and only the even moments $\mathbb{E}[(Z')^{2m}]$ enter; the finite Taylor expansion of $P_{n,x}^{(r)}$ then expresses the n-jet of up at x through the n-jet of $P_{n,x}$ by a triangular system, whose inverse is the reciprocal moment series truncated at degree n, with the coefficients a_m of Equation (3). For $n < s$ the lemma is false as stated, because the window contains a knot and $\mathrm{up}_n(x-y)$ is given by two different polynomials on the two sides of it; for irrational x no level n makes x a cell centre, which is why the eventual law is a genuinely dyadic phenomenon.

Depth and cells, in four lines

$s(x) = \min\{s : 2^s x \in \mathbb{Z}\}$, computed from any representation $m/2^n$ as $\max\{0, n - \nu_2(m)\}$, invariant under $x \mapsto -x$, $x + c$, $1 - |x|$; the knots of up_n are the odd multiples of $\delta_n = 2^{-(n+1)}$, all genuine, and the cell centres are the even multiples; a dyadic x is a cell centre of up_n if and only if $n \geq s(x)$, is a knot at $n = s(x) - 1$, and sits off centre with one knot inside its tail window at every lower level; for $n \geq s(x)$ the whole n-jet of up at x is the expectation of the jet of one cell polynomial, $\mathrm{up}^{(r)}(x) = \mathbb{E}[P_{n,x}^{(r)}(-\delta_n Z')]$, $0 \leq r \leq n$.

4 The reciprocal-product coefficients and finite deconvolution

Two finite mechanisms turn the eventual law of Section 6 from an asymptotic expansion into an identity. The first, proved in this section, is algebraic: on the one polynomial cell of up_n centred at a dyadic point, division by the omitted Fourier factor is a polynomial in a nilpotent operator, so the all-orders deconvolution series is a finite sum with rational coefficients. The second, proved in Section 5, is arithmetic: the derivative jet of up at a dyadic point of depth s stops at order s, and its last even entry is not zero. This section introduces the coefficients the first mechanism produces, fixes their normalization against the three normalizations used by the sources and by the catalogue, proves that every one of them is a positive rational, and proves the deconvolution identity.

Standing conventions. $\Phi(z) = \widehat{\mathrm{up}}_{2\pi}(z) = \prod_{k\geq 0} \mathrm{sinc}_{\mathrm{rad}}(\pi z/2^k)$ is the 2π-convention transform of the up-function (catalogue entries `rvachev-up`, `up-fourier-product`, `sinc`); it is entire, even, and $\Phi(0) = 1$. Φ_n is its prefix with the $n+1$ factors $k = 0, \ldots, n$, $\mathrm{up}_n = \mathcal{F}_{2\pi}^{-1}[\Phi_n]$, and

$$\delta_n := 2^{-(n+1)}.$$

From Section 2 we use two facts. First, up_n is the density of $\sum_{k=0}^{n} 2^{-k-1} U_k$ with $U_0, U_1, \ldots$ independent and uniform on $[-1, 1]$; equivalently it is the convolution of the box densities $b_k = 2^k \mathbf{1}_{[-2^{-k-1}, 2^{-k-1}]}$, $k = 0, \ldots, n$. Hence it is even, nonnegative, supported on $[-(1-\delta_n), 1-\delta_n]$, of class C^{n-1} when $n \geq 1$, and on each open interval between consecutive knots it is one polynomial of degree at most n; its knots are the odd multiples of δ_n (the sums $\sum_{k=0}^{n} \pm 2^{-k-1}$). Second, the head–tail identity: with the dilated kernel

$$\rho_\delta(y) := \delta^{-1}\,\mathrm{up}(y/\delta) \qquad (\delta > 0), \tag{33}$$

which is a smooth even probability density supported on $[-\delta, \delta]$ with $\widehat{\rho_\delta}_{\,2\pi}(t) = \Phi(\delta t)$,

$$\mathrm{up} = \mathrm{up}_n * \rho_{\delta_n} \qquad (n \geq 0). \tag{34}$$

Indeed the factors of Φ omitted from Φ_n are $\prod_{k\geq n+1} \mathrm{sinc}_{\mathrm{rad}}(\pi t/2^k) = \prod_{j\geq 0} \mathrm{sinc}_{\mathrm{rad}}(\pi \delta_n t/2^j) = \Phi(\delta_n t)$, so $\Phi(t) = \Phi_n(t)\,\Phi(\delta_n t)$, and Equation (34) is Fourier inversion of this product. The scale of the omitted tail is therefore δ_n, and the ratio base is $Q = \delta_{n+1}^2/\delta_n^2 = 1/4$.

4.1 Definition, and the three normalizations of the same sequence

Lemma 4.1 (The reciprocal product near the origin). *Φ has no zero in the open unit disc $|z| < 1$, and ± 1 are simple zeros. Consequently $1/\Phi$ is holomorphic and even on $|z| < 1$, equals 1 at 0, and its Taylor series at the origin has radius of convergence exactly 1.*

Proof. The factor $\mathrm{sinc}_{\mathrm{rad}}(\pi z/2^k) = \sin(\pi z/2^k)/(\pi z/2^k)$ vanishes exactly at $z = 2^k j$, $j \in \mathbb{Z} \setminus \{0\}$, each zero being simple. Every such point has modulus at least 1, and $z = \pm 1$ is attained only by the pair $(k, j) = (0, \pm 1)$; since the product converges locally uniformly (catalogue entry `up-fourier-product`), its zero set is the union of the zero sets of the factors with multiplicities added. Evenness is inherited from the factors. The radius statement is Cauchy–Hadamard together with the simple pole of $1/\Phi$ at ± 1. ☐

Definition 4.2 (Reciprocal-product coefficients). The sequence $(a_m)_{m\geq 0}$ is defined by

$$\frac{1}{\Phi(z)} = \sum_{m=0}^{\infty} a_m\,(2\pi z)^{2m}, \qquad |z| < 1. \tag{35}$$

By Theorem 4.1 the series exists, has only even powers, and $a_0 = 1$.

The factor $(2\pi)^{2m}$ is part of the definition: it makes a_m the coefficient of the angular variable $\omega = 2\pi z$, in which the omitted tail acts on a polynomial cell as the differential operator $\sum_m (-1)^m a_m \delta_n^{2m} D^{2m}$ (Theorem 4.10). Three other descriptions of the same sequence occur in the sources and in the catalogue; the next proposition identifies them exactly.

Let X be a random variable with density up. Its moment-generating function and its raw moments are

$$M(z) := \mathbb{E}\,\mathrm{e}^{zX} = \int_{-1}^{1} \mathrm{e}^{zx}\,\mathrm{up}(x)\,\mathrm{d}x, \qquad m_n^{\mathrm{Rv}} := \mathbb{E}X^n = \int_{-1}^{1} x^n\,\mathrm{up}(x)\,\mathrm{d}x, \tag{36}$$

so that $m_{2j}^{\mathrm{Rv}} = c_j$ is the j-th even full moment of the catalogue entry `full-moments` and $m_{2j+1}^{\mathrm{Rv}} = 0$ (entry `rvachev-appell-deconvolution`). That entry defines the binomial-convolution reciprocal $(\gamma_n^{\mathrm{Rv}})_{n\ge 0}$ as the unique rational sequence with

$$\sum_{k=0}^{n} \binom{n}{k} m_k^{\mathrm{Rv}} \gamma_{n-k}^{\mathrm{Rv}} = \delta_{n,0} \qquad (n \in \mathbb{N}_0). \tag{37}$$

S1 expands the reciprocal moment-generating function $1/M(z)$ in the form $\sum_j \gamma_j\, z^j/j!$; S6 expands it in the form $\sum_k \gamma_k\, z^{2k}/(2k)!$, with the odd indices suppressed.

Proposition 4.3 (The three normalizations). (i) *M is entire and even, $M(z) = \sum_{n\ge 0} m_n^{\mathrm{Rv}} z^n/n!$, and*

$$M(z) = \Phi\!\left(\frac{\mathrm{i}z}{2\pi}\right) = \prod_{k=0}^{\infty} \frac{\sinh(z/2^{k+1})}{z/2^{k+1}}, \qquad \Phi(t) = M(2\pi\mathrm{i}t). \tag{38}$$

(ii) *For $|z| < 2\pi$,*

$$\frac{1}{M(z)} = \sum_{n=0}^{\infty} \gamma_n^{\mathrm{Rv}} \frac{z^n}{n!} = \sum_{m=0}^{\infty} (-1)^m a_m z^{2m}, \tag{39}$$

so that $\gamma_{2m+1}^{\mathrm{Rv}} = 0$ and

$$\boxed{a_m = (-1)^m \frac{\gamma_{2m}^{\mathrm{Rv}}}{(2m)!} \qquad (m \ge 0).} \tag{40}$$

In particular S1's γ_j and S6's γ_k are γ_j^{Rv} and $\gamma_{2k}^{\mathrm{Rv}}$, and S1's mode coefficient $\gamma_{2\ell}/(2\ell)!$ is $(-1)^\ell a_\ell$.

(iii) *The even moments and the coefficients are reciprocal under the two equivalent finite convolutions*

$$\sum_{j=0}^{m} (-1)^{m-j} a_{m-j} \frac{c_j}{(2j)!} = \delta_{m,0}, \qquad \sum_{k=0}^{m} \binom{2m}{2k} \gamma_{2k}^{\mathrm{Rv}} c_{m-k} = \delta_{m,0} \qquad (m \in \mathbb{N}_0). \tag{41}$$

Proof. (i) X is bounded, so M is entire with the displayed Taylor series, and even because up is even. Substituting $\xi = \mathrm{i}z/(2\pi)$ into $\Phi(\xi) = \int \mathrm{up}(x)\mathrm{e}^{-2\pi\mathrm{i}\xi x}\,\mathrm{d}x$ gives $\mathrm{e}^{-2\pi\mathrm{i}\xi x} = \mathrm{e}^{zx}$, hence $M(z) = \Phi(\mathrm{i}z/2\pi)$; inverting, $\Phi(t) = M(-2\pi\mathrm{i}t) = M(2\pi\mathrm{i}t)$. Termwise, $\mathrm{sinc}_{\mathrm{rad}}(\pi \cdot \mathrm{i}z/(2\pi 2^k)) = \mathrm{sinc}_{\mathrm{rad}}(\mathrm{i}z/2^{k+1}) = \sinh(z/2^{k+1})/(z/2^{k+1})$, which is the product.

(ii) By Theorem 4.1 and (i), M has no zero in $|z| < 2\pi$ (its zeros are $2\pi\mathrm{i}$ times the zeros of Φ), so $1/M$ is holomorphic there. Putting $z \mapsto \mathrm{i}z/(2\pi)$ in Equation (35) gives, for $|z| < 2\pi$,

$$\frac{1}{M(z)} = \sum_{m\ge 0} a_m (\mathrm{i}z)^{2m} = \sum_{m\ge 0} (-1)^m a_m z^{2m}.$$

Write the same function as an exponential generating series $\sum_n g_n z^n/n!$; then $g_{2m+1} = 0$ and $g_{2m}/(2m)! = (-1)^m a_m$. Multiplying the two convergent series $M(z)\cdot M(z)^{-1} = 1$ and comparing the coefficient of $z^n/n!$ gives $\sum_k \binom{n}{k} m_k^{\mathrm{Rv}} g_{n-k} = \delta_{n,0}$, which is Equation (37). Since $m_0^{\mathrm{Rv}} = 1$, the relation Equation (37) determines the sequence uniquely by triangular recursion, so $g_n = \gamma_n^{\mathrm{Rv}}$, which proves Equations (39) and (40).

(iii) Multiply the catalogue's expansion $\Phi(z) = \sum_j (-1)^j c_j (2\pi z)^{2j}/(2j)!$ (entry `full-moments`; it is Equation (38) read at $z = 2\pi i t$) by Equation (35) and compare the coefficients of $(2\pi z)^{2m}$: this is the first identity. The second is Equation (37) at $n = 2m$ after discarding the odd terms, which vanish. □

Formal versions. In module `RvachevMomentAppell`:

- `Fabius.rvachevRawMomentRat` is m_n^{Rv}, with even entries the executable rational moments `Fabius.moment` $j = c_j$ of `Arithmetic`;
- `Fabius.rvachevReciprocalMomentRat` is γ_n^{Rv}, and `Fabius.binomialConv_rvachevRawMomentRat_reciprocal` is Equation (37);
- `Fabius.rvachevReciprocalMomentRat_eq_completeBellPolynomial` is the Bell-polynomial form of that reciprocal.

Thus Equation (40) expresses every a_m as a Lean-computable rational, $(-1)^m/(2m)!$ times the $2m$-th entry of `rvachevReciprocalMomentRat`. The analytic identification Equation (39) of the reciprocal sequence with the Taylor coefficients of $1/M$ is not formalized; the polynomial identities that use it are (see Theorem 4.11).

4.2 Rationality and positivity

Proposition 4.4 (Logarithm of the reciprocal product). *For $|z| < 1$, with the branch of the logarithm that vanishes at $z = 0$,*

$$\log \frac{1}{\Phi(z)} = \sum_{j=1}^{\infty} \alpha_j\, (2\pi z)^{2j}, \qquad \boxed{\alpha_j = \frac{\zeta(2j)}{j\,(2\pi)^{2j}\,(1-Q^j)} = \frac{|B_{2j}|}{2j\,(2j)!\,(1-Q^j)} \in \mathbb{Q}_{>0},} \tag{42}$$

where B_{2j} are the Bernoulli numbers and $Q = 1/4$. Equivalently, for $|z| < 2\pi$, the cumulants κ_j of X satisfy

$$\log M(z) = \sum_{j\ge 1} \kappa_{2j} \frac{z^{2j}}{(2j)!}, \qquad \kappa_{2j} = \frac{B_{2j}}{2j\,(1-Q^j)}, \qquad \alpha_j = (-1)^{j+1}\frac{\kappa_{2j}}{(2j)!}, \qquad \kappa_{2j+1} = 0. \tag{43}$$

Proof. For $|w| < \pi$, the product $\sin w/w = \prod_{h\ge1}(1 - w^2/(\pi^2 h^2))$ has all factors in the disc $|u-1| < 1$, so

$$\log \frac{\sin w}{w} = \sum_{h\ge1} \log\Big(1 - \frac{w^2}{\pi^2 h^2}\Big) = -\sum_{h\ge1}\sum_{j\ge1} \frac{w^{2j}}{j\,\pi^{2j} h^{2j}} = -\sum_{j\ge1} \frac{\zeta(2j)}{j\,\pi^{2j}}\, w^{2j},$$

the interchange being justified by absolute convergence ($\sum_h \sum_j |w|^{2j}/(j\pi^{2j}h^{2j}) = -\sum_h \log(1 - |w|^2/(\pi^2 h^2)) < \infty$). Put $w = \pi z/2^k$, $k \ge 0$, which satisfies $|w| < \pi$ when $|z| < 1$, and sum over k:

$$\sum_{k\ge0} \log \operatorname{sinc}_{\mathrm{rad}}\Big(\frac{\pi z}{2^k}\Big) = -\sum_{k\ge0}\sum_{j\ge1} \frac{\zeta(2j)}{j}\, z^{2j} Q^{jk} = -\sum_{j\ge1} \frac{\zeta(2j)}{j\,(1-Q^j)}\, z^{2j},$$

again by absolute convergence ($\zeta(2j) \le \zeta(2)$ and $\sum_k \sum_j |z|^{2j} Q^{jk}/j = -\sum_k \log(1 - |z|^2 Q^k) < \infty$). The left side is a holomorphic function on the disc whose exponential is $\Phi(z)$ and which vanishes at 0; by Theorem 4.1 the disc is a simply connected zero-free domain of Φ, so this function is the branch of $\log\Phi$ fixed in the statement. Negating and writing $z^{2j} = (2\pi z)^{2j}/(2\pi)^{2j}$ gives the first form of α_j; Euler's evaluation $\zeta(2j) = |B_{2j}|\,(2\pi)^{2j}/(2\,(2j)!)$ gives the second, which is visibly a positive rational. Finally $\log M(z) = \log\Phi(iz/2\pi) = -\sum_j \alpha_j (iz)^{2j} = \sum_j (-1)^{j+1}\alpha_j z^{2j}$ for $|z| < 2\pi$, and since $\operatorname{sign} B_{2j} = (-1)^{j+1}$, $(-1)^{j+1}(2j)!\,\alpha_j = B_{2j}/(2j(1-Q^j))$. Odd cumulants vanish because M is even. □

Remark 4.5 (The probabilistic reading). S1 obtains Equation (43) by writing $X = \sum_{k\geq 0} 2^{-k-1}U_k$: the $2j$-th cumulant of a uniform variable on $[-1,1]$ is $2^{2j-1}B_{2j}/j$ (from $\log(\sinh z/z)$), cumulants of independent summands add, and the $2j$-th cumulant of cU is c^{2j} times that of U; summing $\sum_{k\geq 0} 2^{-2j(k+1)} = Q^j/(1-Q^j)$ gives $\kappa_{2j} = 2^{2j-1}B_{2j}Q^j/(j(1-Q^j)) = B_{2j}/(2j(1-Q^j))$. The two derivations agree.

Theorem 4.6 (Rationality, positivity, recurrence, closed form, growth). (a) *$a_0 = 1$ and, for $m \geq 1$,*

$$\boxed{m\, a_m = \sum_{j=1}^{m} j\, \alpha_j\, a_{m-j}.} \tag{44}$$

(b) *In terms of the complete exponential Bell polynomials,*

$$a_m = \frac{1}{m!}\mathcal{B}_m^{\exp}(1!\,\alpha_1, 2!\,\alpha_2, \ldots, m!\,\alpha_m) = \sum_{k_1+2k_2+\cdots+mk_m=m} \prod_{j=1}^{m} \frac{\alpha_j^{k_j}}{k_j!}. \tag{45}$$

(c) *Every a_m is a positive rational number. Equivalently $(-1)^m\gamma_{2m}^{\mathrm{Rv}} > 0$ for every m; no even entry of the reciprocal moment sequence vanishes.*

(d) *$(2\pi)^{2m}a_m \to 2/\Phi(1/2) = 3.6115993860280564\ldots$ as $m \to \infty$, with $(2\pi)^{2m}a_m - 2/\Phi(1/2) = \mathcal{O}_{m\to\infty}(r^{-2m})$ for every $r < 2$; here $\Phi(1/2) = \prod_{k\geq 1}\mathrm{sinc}_{\mathrm{rad}}(\pi/2^k) = 0.5537712758888100\ldots$.*

Proof. Put $w = (2\pi z)^2$ and $A(w) = \sum_{j\geq 1}\alpha_j w^j$, so that Equations (35) and (42) read $\sum_m a_m w^m = \exp A(w)$ for $|w| < 4\pi^2$. Since $A(0) = 0$, the composition $\exp\circ A$ is also a well-defined formal power series and the analytic and formal coefficients agree.

(a) Differentiating $F = \exp A$ gives $F' = A'F$; the coefficient of w^{m-1} is Equation (44).

(b) The exponential formula $\exp\bigl(\sum_{j\geq 1} x_j w^j/j!\bigr) = \sum_{m\geq 0}\mathcal{B}_m^{\exp}(x_1,\ldots,x_m)\, w^m/m!$ with $x_j = j!\,\alpha_j$ gives the first form; expanding $\exp A = \prod_j \exp(\alpha_j w^j) = \prod_j \sum_{k_j} \alpha_j^{k_j} w^{jk_j}/k_j!$ and collecting w^m gives the second (a finite sum, because $k_j = 0$ for $j > m$).

(c) By Theorem 4.4 every α_j is a positive rational. Induction on m in Equation (44): if $a_0,\ldots,a_{m-1}$ are positive rationals, the right side is a positive rational and so is a_m. The reformulation is Equation (40).

(d) By Theorem 4.1, $1/\Phi$ is meromorphic on $|z| < 2$ with simple poles at ± 1 and no other pole there (the zeros $2^k j$ of Φ other than ± 1 have modulus at least 2). Near $z = 1$, $\mathrm{sinc}_{\mathrm{rad}}(\pi z) = \sin(\pi z)/(\pi z) = (1-z) + \mathcal{O}_{z\to 1}\bigl((1-z)^2\bigr)$ and $\prod_{k\geq 1}\mathrm{sinc}_{\mathrm{rad}}(\pi z/2^k) = \Phi(z/2) \to \Phi(1/2)$, so the residue of $1/\Phi$ at 1 is $-1/\Phi(1/2)$; by evenness the residue at -1 is $1/\Phi(1/2)$. Hence

$$g(z) := \frac{1}{\Phi(z)} - \frac{1}{\Phi(1/2)}\left(\frac{1}{1-z} + \frac{1}{1+z}\right) = \frac{1}{\Phi(z)} - \frac{2}{\Phi(1/2)}\,\frac{1}{1-z^2}$$

is holomorphic on $|z| < 2$, and Cauchy's estimate on the circle $|z| = r < 2$ bounds its z^{2m}-coefficient by $\max_{|z|=r}|g|\cdot r^{-2m}$. The z^{2m}-coefficient of $1/\Phi$ is $(2\pi)^{2m}a_m$ and that of $2/(\Phi(1/2)(1-z^2))$ is $2/\Phi(1/2)$. Every factor $\mathrm{sinc}_{\mathrm{rad}}(\pi/2^k)$, $k \geq 1$, is positive, so $\Phi(1/2) > 0$; the numerical values are the convergent product evaluated to the digits shown. ☐

The values $(2\pi)^{2m}a_m$ for $m = 1,\ldots,5$ are 2.193, 2.982, 3.369, 3.527, 3.584; the next three are 3.603, 3.609, 3.611, approaching the limit 3.6116 roughly at the rate Q^m that the double pole of $1/\Phi$ at ± 2 (from the factors $k = 0$ and $k = 1$) dictates.

Proposition 4.7 (q-Pochhammer coefficient formula). *$\Phi(z) = \prod_{h\geq 1}\bigl(z^2/h^2; Q\bigr)_\infty$ for all $z \in \mathbb{C}$, and for every $m \geq 0$*

$$\boxed{(2\pi)^{2m}a_m = \sum_{\substack{\nu_1,\nu_2,\ldots\geq 0\\ \nu_1+\nu_2+\cdots=m}} \prod_{h=1}^{\infty} \frac{h^{-2\nu_h}}{(Q;Q)_{\nu_h}},} \tag{46}$$

the sum running over the finitely supported sequences $(\nu_h)_{h\geq 1}$ *of total weight* m*; each summand is positive and the sum converges. In S1's normalization the same statement reads* $\gamma_{2m}/(2m)! = (-1)^m(4\pi^2)^{-m}\sum_\nu\prod_h h^{-2\nu_h}/\,(Q;Q)_{\nu_h}$.

Proof. Expanding each sinc factor by its Euler product,

$$\Phi(z) = \prod_{k\geq 0}\prod_{h\geq 1}\Big(1-\frac{z^2}{h^2 4^k}\Big) = \prod_{h\geq 1}\prod_{k\geq 0}\Big(1-\frac{z^2}{h^2}Q^k\Big) = \prod_{h\geq 1}\left(z^2/h^2;Q\right)_\infty;$$

the rearrangement is legitimate because $\sum_{k,h}|z|^2/(h^2 4^k) < \infty$ makes the double product absolutely convergent. For $|a|<1$ Euler's q-exponential identity ([7, (1.3.15)], [5])

$$\frac{1}{(a;Q)_\infty} = \sum_{\nu=0}^{\infty}\frac{a^\nu}{(Q;Q)_\nu}$$

applies to $a = z^2/h^2$ as soon as $|z|<1$. Fix $H\geq 1$ and let $P_H(z) = \prod_{h\leq H}\left(z^2/h^2;Q\right)_\infty^{-1}$. Multiplying H absolutely convergent power series,

$$P_H(z) = \sum_{m\geq 0} S_{H,m}\, z^{2m}, \qquad S_{H,m} = \sum_{\substack{\nu_1,\dots,\nu_H\geq 0\\ \nu_1+\cdots+\nu_H=m}}\prod_{h=1}^{H}\frac{h^{-2\nu_h}}{(Q;Q)_{\nu_h}},$$

a finite sum of positive terms that increases with H. As $H\to\infty$, $P_H\to 1/\Phi$ locally uniformly on $|z|<1$, because the tails $\prod_{h>H}\left(z^2/h^2;Q\right)_\infty \to 1$ locally uniformly and Φ is zero-free there; by Cauchy's formula the coefficients converge, $S_{H,m}\to(2\pi)^{2m}a_m$. A monotone sequence of finite sums of positive terms converges to the sum of the infinite series of those terms, which is the right side of Equation (46). The S1 form is Equation (40). □

For $m=1$ the formula reads $(2\pi)^2 a_1 = \zeta(2)/(1-Q) = \frac{4}{3}\cdot\frac{\pi^2}{6} = \frac{2\pi^2}{9}$, i.e. $a_1 = 1/18$; for $m=2$ it reads $(2\pi)^4 a_2 = \zeta(4)/\,(Q;Q)_2 + \frac{1}{2}(\zeta(2)^2-\zeta(4))/\,(Q;Q)_1^2 = \frac{32}{2025}\pi^4 + \frac{2}{135}\pi^4 = \frac{62}{2025}\pi^4$, i.e. $a_2 = 62/(2025\cdot 16) = 31/16200$, in agreement with Table 1. The same quarter-base q-calculus governs the extraction weights of the later sections; that is the conceptual content of the formula. For computation, Equation (44) is the tool.

Remark 4.8 (Which argument proves what). Three of the sources' arguments prove strict positivity and one does not.

- S1's proposition "Rational cumulants and reciprocal coefficients" proves *rationality only*: its recurrence $\gamma_n = -\sum_{j=1}^{n}\binom{n-1}{j-1}\kappa_j\gamma_{n-j}$ (the cumulant form of Equation (37)) has terms of both signs, so no sign information follows from it. Positivity in S1 comes from its q-Pochhammer formula, Theorem 4.7 here, whose summands are all positive: this is the second proof of Theorem 4.6(c).
- S2, S3 and S4 use the exponential recurrence Equation (44) with $\alpha_j>0$: the proof adopted above.
- S6's lemma "Rationality and nonvanishing of the reciprocal moments" proves, despite its title, the strict inequality $(-1)^k\gamma_k>0$ for its $\gamma_k = \gamma_{2k}^{\mathrm{Rv}}$, that is, $a_k>0$. Its argument: from Equation (38), $1/M(z) = \prod_{k\geq 0}(z/2^{k+1})/\sinh(z/2^{k+1})$, and $w/\sinh w = \sum_{r\geq 0}(2-2^{2r})B_{2r}w^{2r}/(2r)! = \sum_r(-1)^r b_r w^{2r}$ with $b_0=1$ and $b_r = 2(2^{2r-1}-1)\,|B_{2r}|\,/(2r)!>0$ for $r\geq 1$. The z^{2m}-coefficient of the product of the first K factors is $(-1)^m$ times a finite sum of positive terms containing the term $b_m 4^{-m}$ (all of $2m$ taken from the factor $k=0$); the partial products converge locally uniformly on $|z|<2\pi$ to $1/M$, so the coefficients converge and

  $$a_m = (-1)^m[z^{2m}]\frac{1}{M(z)} \;\geq\; \frac{b_m}{4^m} = \frac{2(2^{2m-1}-1)\,|B_{2m}|}{4^m\,(2m)!} \qquad (m\geq 1), \tag{47}$$

  a third proof, with an explicit lower bound ($1/24\leq 1/18$, $7/5760\leq 31/16200, \dots$).

The justification of coefficient extraction from an infinite product of series, which S1 and S6 pass over, is the locally uniform convergence used in Theorem 4.7 and in the third bullet.

Table 1: The first reciprocal-product coefficients, their logarithmic coefficients α_m and cumulants κ_{2m} (Theorem 4.4), and the catalogue's reciprocal moments $\gamma_{2m}^{\mathrm{Rv}} = (-1)^m (2m)!\, a_m$. The values $a_0, \ldots, a_5$ are the ground truth of the verification outputs of S4 and S5; the rest are computed by Equation (44) in exact arithmetic.

m	a_m	$\gamma_{2m}^{\mathrm{Rv}} = (-1)^m (2m)!\, a_m$
0	1	1
1	1/18	−1/9
2	31/16200	31/675
3	2347/42865200	−2347/59535
4	1904369/1311675120000	1904369/32531625
5	3309760579/88561680752160000	−3309760579/24405225075
6	1884491755011469/1980124051433319792000000	1884491755011469/4133856862760625

m	α_m	κ_{2m}
1	1/18	1/9
2	1/2700	−2/225
3	1/178605	16/3969
4	1/9639000	−16/3825
5	1/478533825	256/33759
6	691/15688261338750	−353792/16769025

4.3 Finite deconvolution on a polynomial jet

For $\delta > 0$ and a polynomial P define the tail-averaging operator

$$(\mathcal{A}_\delta P)(t) := \int_{-1}^{1} P(t - \delta y)\ \mathrm{up}(y)\ \mathrm{d}y = (P * \rho_\delta)(t), \tag{48}$$

the second form by the substitution $u = \delta y$ in Equation (33). Writing D for $\mathrm{d}/\mathrm{d}t$:

Lemma 4.9 (Symbol of the tail average)**.** *On the space of polynomials of degree at most n,*

$$\mathcal{A}_\delta = M(\delta D) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{c_j}{(2j)!}\, \delta^{2j} D^{2j}, \tag{49}$$

so $\mathcal{A}_\delta$ preserves degree and leading coefficient, commutes with D, and $(\mathcal{A}_\delta P)^{(r)} = \mathcal{A}_\delta(P^{(r)})$ for all $r \geq 0$.

Proof. Taylor's formula for a polynomial is finite: $P(t - \delta y) = \sum_{k=0}^{n} P^{(k)}(t)(-\delta y)^k/k!$. Integrating against $\mathrm{up}(y)\ \mathrm{d}y$ and using $m_k^{\mathrm{Rv}} = 0$ for odd k, $m_{2j}^{\mathrm{Rv}} = c_j$, gives Equation (49). The $j = 0$ term is the identity and every other term lowers degree. Differentiation under the integral sign in Equation (48) gives the last assertion. □

Lemma 4.10 (Finite polynomial deconvolution)**.** *Let P be a polynomial of degree at most n and $\delta > 0$. Then*

$$\boxed{P = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m a_m\, \delta^{2m} D^{2m} (\mathcal{A}_\delta P),} \tag{50}$$

and for every $r \geq 0$ and every t,

$$P^{(r)}(t) = \sum_{m=0}^{\lfloor (n-r)/2 \rfloor} (-1)^m a_m\, \delta^{2m} (\mathcal{A}_\delta P)^{(r+2m)}(t), \tag{51}$$

where an empty sum ($r > n$) is 0*. Both identities hold in the polynomial ring* $\mathbb{Q}[t]$ *with* $\mathbb{Q} = \mathbb{Q}$ *whenever* $P \in \mathbb{Q}[t]$ *and* $\delta \in \mathbb{Q}$*; no limiting process is involved.*

Proof. On the space V_n of polynomials of degree at most n, D is nilpotent, $D^{n+1} = 0$, so every power series in δD is a polynomial in D of degree at most n, and products of such series are computed by finite coefficient convolutions. Put $R := \sum_{m \le \lfloor n/2 \rfloor} (-1)^m a_m \delta^{2m} D^{2m}$. Composing with Equation (49),

$$R\,\mathcal{A}_\delta = \sum_{s \ge 0} \delta^{2s} D^{2s} \sum_{m+j=s} (-1)^m a_m \frac{c_j}{(2j)!} = \sum_{s \ge 0} \delta_{s,0}\, \delta^{2s} D^{2s} = \mathrm{id}_{V_n},$$

where the terms with $2s > n$ vanish on V_n regardless of their coefficients and the terms with $2s \le n$ are evaluated by the first identity of Equation (41). This is Equation (50). Apply D^r to it; since D commutes with R and with $\mathcal{A}_\delta$ and $D^{r+2m}\mathcal{A}_\delta P = 0$ once $r + 2m > n$, only $m \le \lfloor (n-r)/2 \rfloor$ survive, which is Equation (51). □

Remark 4.11 (The catalogue's Appell form of the same inverse). With $A_n^{\mathrm{Rv}}(x) = \sum_k \binom{n}{k} \gamma_{n-k}^{\mathrm{Rv}} x^k$ the Rvachev–Appell polynomials of the entry `rvachev-appell-deconvolution`, one has $A_n^{\mathrm{Rv}} = \sum_j \gamma_j^{\mathrm{Rv}} D^j(x^n)/j! = \sum_m (-1)^m a_m D^{2m}(x^n)$ by Equation (40), so the catalogue's deconvolution operator $\mathscr{D}_{\mathrm{up}} \colon x^n \mapsto A_n^{\mathrm{Rv}}$ is exactly the operator R of the proof at $\delta = 1$, and the catalogue's smoothing identity $\int (\mathscr{D}_{\mathrm{up}} P)(x-y)\,\mathrm{up}(y)\,\mathrm{d}y = P(x)$ is $\mathcal{A}_1 \mathscr{D}_{\mathrm{up}} = \mathrm{id}$, the other order of Equation (50). At scale δ, conjugating by the dilation $t \mapsto t/\delta$ gives $\mathcal{A}_\delta^{-1} P = \sum_k p_k \delta^k A_k^{\mathrm{Rv}}(t/\delta)$ for $P = \sum_k p_k t^k$. *Formal versions* (all in `RvachevMomentAppell`):

- `Fabius.rvachevAppellPolynomialRat`, the rational Appell family;
- `Fabius.rvachevDeconvolvedPolynomial`, the operator $\mathscr{D}_{\mathrm{up}}$;
- `Fabius.integral_eval_rvachevAppellPolynomial_sub_mul_rvachev`, the smoothing identity with $x - y$.

Consequently Theorem 4.10 at $\delta = 1$ is Lean-proved in the Appell normalization; the dilation to general δ is a change of variable.

The lemma is applied on the cell that the tail window $[x - \delta_n, x + \delta_n]$ occupies. We recall the geometric input from Section 2 in the precise form needed, with its short proof.

Lemma 4.12 (A dyadic point is the centre of one cell)**.** *Let* $x \in [-1, 1]$ *be dyadic with* $s(x) = s$ *and let* $n \ge s$*. Then there is one polynomial* $P_{n,x}$ *of degree at most* n *with*

$$\mathrm{up}_n(z) = P_{n,x}(z) \qquad \text{for all } z \in (x - \delta_n, x + \delta_n); \tag{52}$$

for $x = \pm 1$ *it is the zero polynomial. For* $n \ge 1$ *and* $0 \le r \le n-1$, $\mathrm{up}_n^{(r)}(x) = P_{n,x}^{(r)}(x)$*; for* $r = n$, $P_{n,x}^{(n)}$ *is the constant value of the piecewise-constant* n*-th derivative of* up_n *on the open cell. If* $x \in (-1, 1)$ *and* $n = s - 1 \ge 0$*, then* x *is a knot of* up_n*.*

Proof. Write $x = a/2^s$ with a odd if $s \ge 1$ and $a \in \{-1, 0, 1\}$ if $s = 0$. Since $\delta_n = 2^{-n-1}$, $x/\delta_n = a\,2^{n+1-s}$, an even integer when $n \ge s$ and the odd integer a when $n = s - 1$. The knots of up_n are the odd multiples of δ_n in $[-(1-\delta_n), 1-\delta_n]$. If $|x| < 1$, then $|a| \le 2^s - 1$, so $|x/\delta_n| \le 2^{n+1} - 2^{n+1-s} \le 2^{n+1} - 2$: x is an even multiple of δ_n at distance at least δ_n from the two extreme knots $\pm(2^{n+1} - 1)\delta_n$, hence the midpoint of the open cell between the two neighbouring odd multiples, on which up_n is one polynomial. If $x = \pm 1$, the interval $(x - \delta_n, x + \delta_n)$ lies outside the support $[-(1-\delta_n), 1-\delta_n]$ except for its endpoint, so $\mathrm{up}_n = 0$ there. The derivative statements follow from $\mathrm{up}_n \in C^{n-1}$ and from the polynomial form on the open cell. When $n = s-1$ and $|x| < 1$, x is an odd multiple of δ_n of modulus at most $(2^{n+1} - 1)\delta_n$, a knot. □

Theorem 4.13 (Exact local deconvolution on a centred cell). *Let $x \in [-1, 1]$ be dyadic with $s(x) = s$, let $n \geq s$, and let $P = P_{n,x}$ be the cell polynomial of Theorem 4.12. Then, for every $r \geq 0$,*

$$\mathrm{up}^{(r)}(x) = (\mathcal{A}_{\delta_n} P)^{(r)}(x) = \sum_{j=0}^{\lfloor (n-r)/2 \rfloor} \frac{c_j}{(2j)!}\, \delta_n^{2j}\, P^{(r+2j)}(x), \tag{53}$$

$$P^{(r)}(x) = \sum_{m=0}^{\lfloor (n-r)/2 \rfloor} (-1)^m a_m\, \delta_n^{2m}\, \mathrm{up}^{(r+2m)}(x). \tag{54}$$

In particular, at $r = 0$, since $P(x) = \mathrm{up}_n(x)$ and $\delta_n^{2m} = 4^{-m(n+1)} = Q^m\, 4^{-mn}$,

$$\boxed{\mathrm{up}_n(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m a_m\, 4^{-m(n+1)}\, \mathrm{up}^{(2m)}(x) = \mathrm{up}(x) + \sum_{m=1}^{\lfloor n/2 \rfloor} C_m(x)\, 4^{-mn},} \tag{55}$$

with $C_m(x) = (-1)^m a_m 4^{-m}\, \mathrm{up}^{(2m)}(x)$: an identity for each $n \geq s$, with no remainder.

Proof. By Equation (34), $\mathrm{up} = \mathrm{up}_n * \rho_{\delta_n}$ with up_n bounded and integrable and ρ_{δ_n} smooth with compact support. Differentiation under the integral sign therefore gives, for every $r \geq 0$,

$$\mathrm{up}^{(r)}(x) = \int_{\mathbb{R}} \mathrm{up}_n(z)\, \rho_{\delta_n}^{(r)}(x - z)\, \mathrm{d}z.$$

The integrand vanishes unless $|x - z| \leq \delta_n$; on the open interval $(x - \delta_n, x + \delta_n)$ we have $\mathrm{up}_n = P$ by Theorem 4.12, and the two endpoints have measure zero. Hence

$$\mathrm{up}^{(r)}(x) = \int_{\mathbb{R}} P(z)\, \rho_{\delta_n}^{(r)}(x - z)\, \mathrm{d}z = (P * \rho_{\delta_n})^{(r)}(x) = (\mathcal{A}_{\delta_n} P)^{(r)}(x),$$

the middle equality because $P * \rho_{\delta_n}$ is a polynomial whose r-th derivative is obtained by differentiating the kernel, and the last by Equation (48). Expanding with Equation (49) (applied to $P^{(r)}$, of degree at most $n - r$) gives Equation (53). Now Equation (51) with $\delta = \delta_n$, evaluated at $t = x$, reads $P^{(r)}(x) = \sum_m (-1)^m a_m \delta_n^{2m} (\mathcal{A}_{\delta_n} P)^{(r+2m)}(x)$, and each $(\mathcal{A}_{\delta_n} P)^{(r+2m)}(x)$ equals $\mathrm{up}^{(r+2m)}(x)$ by the identity just proved (at order $r + 2m$). This is Equation (54); $r = 0$ and $P(x) = \mathrm{up}_n(x)$ give Equation (55). □

An all-orders asymptotic expansion is not an identity

For an arbitrary x, dividing Φ by $\Phi(\delta_n t)$ on the Fourier side yields, for every truncation order K, $\mathrm{up}_n = \sum_{m \leq K} (-1)^m a_m \delta_n^{2m}\, \mathrm{up}^{(2m)} + \mathcal{O}_{n\to\infty}(\delta_n^{2K+2})$ in the uniform norm; this is the asymptotic Richardson calculus of the Exponents volume [9] (its `p3:eq:richardson-moments` and `p3:eq:richardson-leading`). At a dyadic point the displayed derivatives eventually vanish (Theorem 5.4), but that alone does not force the remainder to vanish: a remainder smaller than every power of 4^{-n} is not excluded by any finite number of vanishing terms. What excludes it is Theorem 4.13: on the centred cell the reciprocal series acts on a nilpotent operator, so the identity is a finite algebraic one and there is no remainder to estimate. The onset $n \geq s(x)$ is exactly the range in which the tail window $[x - \delta_n, x + \delta_n]$ crosses no knot.

Remark 4.14 (Six forms of one lemma). Theorems 4.10 and 4.13 absorb S1's "Finite jet inversion" and "Exact local deconvolution of a sinc prefix" (stated with $\gamma_k \delta^k / k!$ over all k, odd terms vanishing), S2's "Finite local deconvolution" (the a_m form with $0 \leq r \leq n$, inverting the triangular relation Equation (53)), S3's "Finite polynomial deconvolution" (the operator $M(\delta D)$ on polynomials, applied to a cell polynomial of degree at most s, see Theorem 5.6), S4's "Exact inverse on polynomials" and "The tail sees only one polynomial piece" (the kernel-differentiation route adopted here, in the p_N

indexing with $4^{-mN} = 4^{-m(n+1)}$), S5's "Finite polynomial inverse" and "Jet transfer at a cell centre" (the angular normalization $\Phi(-h^2D^2)$ with $h_N = 2^{-N} = \delta_n$), and S6's "Exact reciprocal moment deconvolution on a matched cell" (the form $\sum_k \gamma_k 4^{-k(n+1)} F^{(2k)}(a)/(2k)!$ for the centred CDF spline S_n at $a = 1 - |x|$, where $F^{(2k)}(a) = \mathrm{up}^{(2k)}(x)$ and $S_n(a) = \mathrm{up}_n(x)$). Under Theorem 4.3 and the index bridge of Section 2 all six are the identity Equation (55).

Proposition 4.15 (One stage earlier at odd depth). *Let $x \in (-1,1)$ be dyadic with $s(x) = s = 2d+1$ odd, $s \geq 3$, and put $n = s-1 = 2d$. Then Equation (55) also holds at this n:*

$$\mathrm{up}_{s-1}(x) = \sum_{m=0}^{d} (-1)^m a_m\, 4^{-ms}\ \mathrm{up}^{(2m)}(x). \tag{56}$$

For $s = 1$, $x = \pm 1/2$, the same formula at $n = 0$ reads $\mathrm{up}_0(\pm 1/2) = \mathrm{up}(\pm 1/2) = 1/2$ and holds if and only if the jump of the box $\mathrm{up}_0 = \mathbf{1}_{[-1/2,1/2]}$ is assigned its Fourier-inversion value $1/2$.

Proof. By Theorem 4.12, x is a knot of up_n; the tail window $[x-\delta_n, x+\delta_n]$ covers the right half of the cell $(x-2\delta_n, x)$ and the left half of the cell $(x, x+2\delta_n)$. Let P_- and P_+ be the cell polynomials of up_n on these two cells. Since $\mathrm{up}_n \in C^{n-1}$ ($n = 2d \geq 2$), the difference $P_+ - P_-$ vanishes to order n at x, so $P_+ - P_- = c\,(z-x)^n$ for a constant c. Put $P_* := \frac{1}{2}(P_- + P_+)$, so that $P_\pm = P_* \pm \frac{c}{2}(z-x)^n$ and $P_*(x) = P_\pm(x) = \mathrm{up}_n(x)$. As in the proof of Theorem 4.13, for every $r \geq 0$,

$$\begin{aligned}\mathrm{up}^{(r)}(x) &= \int_{x-\delta_n}^{x} P_-(z)\rho_{\delta_n}^{(r)}(x-z)\ \mathrm{d}z + \int_{x}^{x+\delta_n} P_+(z)\rho_{\delta_n}^{(r)}(x-z)\ \mathrm{d}z \\ &= (\mathcal{A}_{\delta_n} P_*)^{(r)}(x) + \frac{c}{2}\int_0^{\delta_n} u^n \left[\rho_{\delta_n}^{(r)}(-u) - (-1)^n \rho_{\delta_n}^{(r)}(u)\right] \mathrm{d}u,\end{aligned}$$

by the substitutions $z = x+u$ and $z = x-u$ in the two half-windows. The kernel ρ_{δ_n} is even, so $\rho_{\delta_n}^{(r)}$ has parity $(-1)^r$, and the bracket vanishes whenever $r \equiv n \pmod 2$. Since n is even, $\mathrm{up}^{(r)}(x) = (\mathcal{A}_{\delta_n} P_*)^{(r)}(x)$ for every even r. Now Equation (50) for P_* (degree at most $n = 2d$), evaluated at x, uses only the even-order derivatives $(\mathcal{A}_{\delta_n} P_*)^{(2m)}(x)$, $m \leq d$:

$$\mathrm{up}_n(x) = P_*(x) = \sum_{m=0}^{d} (-1)^m a_m \delta_n^{2m} (\mathcal{A}_{\delta_n} P_*)^{(2m)}(x) = \sum_{m=0}^{d} (-1)^m a_m \delta_n^{2m}\ \mathrm{up}^{(2m)}(x),$$

and $\delta_n^{2m} = 4^{-m(n+1)} = 4^{-ms}$. For $s = 1$, $n = 0$: the two cell polynomials of the box at $x = 1/2$ are the constants 1 and 0, the same computation gives $\mathrm{up}(1/2) = \int_0^{\delta_0} \rho_{\delta_0}(u)\ \mathrm{d}u = \frac{1}{2}$, and Equation (56) with $d = 0$ asserts $\mathrm{up}_0(1/2) = 1/2$, the mean of the two one-sided limits, which is the value the inverse Fourier integral of $\mathrm{sinc}_{\mathrm{rad}}(\pi t)$ takes at a jump. □

The parity computation also shows why the even-depth case admits no such sharpening in general: for $n = s-1$ odd the bracket does not vanish for even r, and the exact examples of Section 6 (for instance $x = 1/4$, where $\mathrm{up}_1(1/4) = 1$ while the law gives $67/72 + \frac{1}{9}\cdot\frac{1}{4} = 23/24$) show that Equation (55) fails at $n = s-1$ for even s.

What the first mechanism delivers

The coefficients a_m of $1/\Phi(z) = \sum_m a_m (2\pi z)^{2m}$ are positive rationals, computable by the Bernoulli recurrence Equation (44), equal to $(-1)^m \gamma_{2m}^{\mathrm{Rv}}/(2m)!$ in the catalogue's reciprocal-moment normalization, and given in closed q-Pochhammer form by Equation (46). At a dyadic point of depth s and every $n \geq s$ (and at $n = s-1$ when $s \geq 3$ is odd), $\mathrm{up}_n(x) = \sum_{m \leq \lfloor n/2 \rfloor} (-1)^m a_m 4^{-m(n+1)}\ \mathrm{up}^{(2m)}(x)$ exactly, because the deconvolution series is a polynomial in the nilpotent operator D on the cell polynomial. The sum still runs to $\lfloor n/2 \rfloor$; cutting it to

$\lfloor s/2 \rfloor$, independently of n, is the second mechanism.

5 Termination of the dyadic derivative jet

The second mechanism is arithmetic. Rvachev's functional-differential equation (catalogue entry `rvachev-up`, "Differential law")

$$\mathrm{up}'(x) = 2\,\mathrm{up}(2x+1) - 2\,\mathrm{up}(2x-1) \qquad (x \in \mathbb{R}) \tag{57}$$

expresses each derivative of up as a signed sum of values of up at dilated arguments; at a dyadic point the dilated arguments eventually all land on odd integers, where up vanishes.

5.1 The Thue–Morse derivative stencil

Recall $\varepsilon_k = (-1)^{s_2(k)}$, $s_2(k)$ the binary digit sum, so that $\varepsilon_{2k} = \varepsilon_k$ and $\varepsilon_{2k+1} = -\varepsilon_k$.

Theorem 5.1 (Thue–Morse derivative stencil). *For every $r \geq 0$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathrm{up}^{(r)}(x) = 2^{\binom{r+1}{2}} \sum_{k=0}^{2^r-1} \varepsilon_k\, \mathrm{up}\big(2^r x + 2^r - 1 - 2k\big).} \tag{58}$$

Proof. Induction on r. For $r = 0$ the sum has the single term $\mathrm{up}(x)$. Assume Equation (58) at order r and differentiate; the chain rule contributes 2^r, and Equation (57) applied to each term gives

$$\begin{aligned}\mathrm{up}^{(r+1)}(x) = 2^{\binom{r+1}{2}+r+1} \sum_{k<2^r} \varepsilon_k \Big[&\mathrm{up}\big(2^{r+1}x + 2^{r+1} - 1 - 2(2k)\big) \\ &- \mathrm{up}\big(2^{r+1}x + 2^{r+1} - 1 - 2(2k+1)\big)\Big].\end{aligned}$$

The first bracketed term carries the index $2k$ with sign $\varepsilon_k = \varepsilon_{2k}$, the second the index $2k+1$ with sign $-\varepsilon_k = \varepsilon_{2k+1}$; together they enumerate $0 \leq k' < 2^{r+1}$. Finally $\binom{r+1}{2} + r + 1 = \binom{r+2}{2}$. □

Corollary 5.2 (Exact derivative tiling; global form). *For $r \geq 0$, $0 \leq k < 2^r$ and $y \in [-1, 1]$,*

$$\boxed{\mathrm{up}^{(r)}\Big(-1 + \frac{2k+1+y}{2^r}\Big) = 2^{\binom{r+1}{2}}\, \varepsilon_k\, \mathrm{up}(y).} \tag{59}$$

Equivalently, with the signed global extension $F_{\mathrm{gl}}(t) = \sum_{b\geq 0} \varepsilon_b\, \mathrm{up}(t - 2b - 1)$ of the catalogue entry `fabius-global`,

$$\mathrm{up}^{(r)}(x) = 2^{\binom{r+1}{2}}\, F_{\mathrm{gl}}\big(2^r(x+1)\big) \qquad (x \in [-1, 1],\ r \geq 0). \tag{60}$$

Proof. At $x = -1 + (2k+1+y)/2^r$ the argument of the j-th term of Equation (58) is $2(k-j)+y$. For $j \neq k$ and $|y| \leq 1$ it has modulus at least 1, where up vanishes ($\mathrm{up}(\pm 1) = 0$ and the support is $[-1, 1]$); the term $j = k$ is $\varepsilon_k\, \mathrm{up}(y)$. For Equation (60): $t = 2^r(x+1) \in [0, 2^{r+1}]$ lies in a block $[2k, 2k+2]$ with $0 \leq k < 2^r$ (for $t = 2^{r+1}$ take $k = 2^r - 1$), say $t = 2k+1+y$ with $y \in [-1, 1]$; in the locally finite sum defining $F_{\mathrm{gl}}(t)$ only $b = k$ can contribute, the neighbouring translates being evaluated at ± 1 or beyond, so $F_{\mathrm{gl}}(t) = \varepsilon_k\, \mathrm{up}(y)$, and Equation (59) applies. □

Formal versions.

- `Fabius.iteratedDeriv_rvachevUp_eq_thueMorse_sum` (`Differential`) is Theorem 5.1, term for term;

- `Fabius.iteratedDeriv_rvachev` (`GlobalExtension`) is the global form Equation (60);
- `Fabius.iteratedDeriv_extendedFabius` (same module) is the scaling law $F_{\mathrm{gl}}^{(r)}(t) = 2^{\binom{r+1}{2}} F_{\mathrm{gl}}(2^r t)$ that the catalogue's `fabius-dyadic-cells` entry (field "Derivative scaling") invokes.

S5 and S6 derive the jet results from that scaling law; S1, S2, S3 and S4 from the stencil. The two routes are the same computation read through Theorem 5.2.

5.2 Values at the two shallowest points, and strict positivity

Lemma 5.3 (Values and positivity). $\mathrm{up}(0) = 1$, $\mathrm{up}(\pm 1/2) = 1/2$, *and* $\mathrm{up}(y) > 0$ *for every* $y \in (-1, 1)$.

Proof. The point 0 has depth 0 and $\pm 1/2$ depth 1. Equation (55) at $x = 0$, $n = 0$, has the single term $m = 0$ ($\lfloor 0/2 \rfloor = 0$): $\mathrm{up}(0) = \mathrm{up}_0(0) = 1$, the box b_0 at its centre. At $x = \pm 1/2$, $n = 1$, again only $m = 0$: $\mathrm{up}(\pm 1/2) = \mathrm{up}_1(\pm 1/2) = (b_0 * b_1)(\pm 1/2) = 2 \int_{1/4}^{3/4} \mathbf{1}_{[-1/2,1/2]}(y)\ \mathrm{d}y = \frac{1}{2}$. (Both agree with the catalogue: $\mathrm{up}(0) = 1$ is normative, and $\mathrm{up}(1/2) = F_{\mathrm{bd}}(1/2) = 1/2$ by the reflection $F_{\mathrm{bd}}(t) + F_{\mathrm{bd}}(1-t) = 1$.)

Positivity. First, $\mathrm{up}_n(y) > 0$ for $|y| < 1 - \delta_n$, by induction on n: $\mathrm{up}_0 = b_0$ is positive on $(-1/2, 1/2)$; and if up_{n-1} is nonnegative and positive on the open interval $(-(1-\delta_{n-1}), 1-\delta_{n-1})$, then for $|y| < 1 - \delta_n = (1 - \delta_{n-1}) + 2^{-n-1}$,

$$\mathrm{up}_n(y) = (\mathrm{up}_{n-1} * b_n)(y) = 2^n \int_{y-2^{-n-1}}^{y+2^{-n-1}} \mathrm{up}_{n-1}(u)\ \mathrm{d}u > 0,$$

because the interval of integration meets the positivity interval of up_{n-1} in a nonempty open interval. Now fix $y \in (-1, 1)$ and choose $n \geq 1$ with $2\delta_n < 1 - |y|$. For $|u| \leq \delta_n$, $|y - u| < 1 - \delta_n$, so the continuous function $u \mapsto \mathrm{up}_n(y - u)$ is positive on the compact support $[-\delta_n, \delta_n]$ of ρ_{δ_n}, and Equation (34) gives $\mathrm{up}(y) = \int \mathrm{up}_n(y - u) \rho_{\delta_n}(u)\ \mathrm{d}u \geq \min_{|u| \leq \delta_n} \mathrm{up}_n(y - u) > 0$. ☐

Formal versions.

- `Fabius.rvachevUp_zero` (`Basic`): $\mathrm{up}(0) = 1$;
- `Fabius.rvachevUp_pos_of_mem_Ioo` (`Monotonicity`): positivity on $(-1, 1)$.

5.3 The jet stops at the depth, and its last even entry is not zero

Theorem 5.4 (Dyadic jet termination). *Let* $x \in [-1, 1]$ *be dyadic with* $s(x) = s$, *and write* $x = a/2^s$ *(*a *odd if* $s \geq 1$*).*

(i) $\mathrm{up}^{(r)}(x) = 0$ *for every* $r > s$.

(ii) *If* $s \geq 1$, *the derivative of order exactly* s *is a signed power of two,*

$$\mathrm{up}^{(s)}(x) = 2^{\binom{s+1}{2}} \varepsilon_k, \qquad k = \frac{2^s(x+1) - 1}{2} = \frac{a + 2^s - 1}{2} \in \{0, \ldots, 2^s - 1\}. \tag{61}$$

(iii) *If* $s \geq 1$ *and* $0 \leq r < s$, *then* $\mathrm{up}^{(r)}(x) \neq 0$*; in particular, for odd* s,

$$\mathrm{up}^{(s-1)}(x) = 2^{\binom{s}{2} - 1} \varepsilon_{k'}, \qquad k' = \left\lfloor \frac{2^s(x+1) - 1}{4} \right\rfloor. \tag{62}$$

Thus, for $s \geq 1$, $s = \max\left\{r : \mathrm{up}^{(r)}(x) \neq 0\right\}$ *and the set of orders with a nonzero derivative is exactly* $\{0, 1, \ldots, s\}$*; for* $s = 0$ *(the points* $0, \pm 1$*) every derivative of positive order vanishes.*

Proof. Put $t_r := 2^r(x+1) = 2^{r-s}(a+2^s)$.

(i) For $r > s$, t_r is an even integer (for $s = 0$ directly, since $a+1 \in \{0,1,2\}$ and $r \geq 1$; for $s \geq 1$ because $r - s \geq 1$). Then every argument $2^r x + 2^r - 1 - 2j = t_r - 1 - 2j$ in Equation (58) is an odd integer, of modulus at least 1, where up vanishes. So the whole sum is zero. (Equivalently, t_r is a block endpoint in Equation (59), $y = \pm 1$.)

(ii) For $r = s \geq 1$, $t_s = a + 2^s$ is odd, and $0 < t_s < 2^{s+1}$ because $|a| < 2^s$; write $t_s = 2k+1$ with $0 \leq k < 2^s$. This is Equation (59) with $y = 0$, and $\mathrm{up}(0) = 1$.

(iii) For $0 \leq r < s$, $t_r = (a+2^s)/2^{s-r}$ is not an integer, so $t_r \in (2k, 2k+2)$ for a unique k, $0 \leq k < 2^r$ (as $0 < t_r < 2^{r+1}$), and $t_r = 2k+1+y$ with $y \in (-1,1)$. By Equation (59) and Theorem 5.3, $\mathrm{up}^{(r)}(x) = 2^{\binom{r+1}{2}}\varepsilon_k \,\mathrm{up}(y) \neq 0$. For odd s and $r = s-1$, $t_{s-1} = (a+2^s)/2$ is a half-integer: writing the odd number $A = a + 2^s = 2^s(x+1)$ as $4k'+1$ or $4k'+3$ gives $t_{s-1} = 2k'+1 \mp \frac{1}{2}$, so $y = \mp\frac{1}{2}$ with $k' = \lfloor (A-1)/4 \rfloor$, and $\mathrm{up}(\pm 1/2) = 1/2$ gives Equation (62). □

Formal versions.

- `Fabius.iteratedDeriv_rvachevUp_dyadic_eq_zero` (`DyadicDerivativeFiltration`) is (i);
- `Fabius.iteratedDeriv_rvachevUp_dyadic_critical` (same module) is (ii), in the form $\mathrm{up}^{(s)}((2c+1)/2^s) = -\varepsilon_c 2^{\binom{s+1}{2}}$ for $0 \leq c < 2^{s-1}$, which is Equation (61) with $k = c + 2^{s-1}$ and $\varepsilon_{c+2^{s-1}} = -\varepsilon_c$;
- `Fabius.dyadic_depth_eq_max_nonzero_iteratedDeriv` (same module) is their conjunction, "the depth is the largest order with a nonzero derivative";
- `Fabius.iteratedDeriv_rvachev_centeredDyadic_eq_zero` and `Fabius.iteratedDeriv_rvachev_centeredDyadic_ne_zero` (`OriginalPaperSupplement`) are the same two facts in the centred-cell coordinates of the original paper;
- `Fabius.iteratedDeriv_fabiusReal_dyadicGrid_eq_zero_iff` (`BoundedDerivatives`) is the "zero iff m even" form on the matched grid $m/2^k$, $m < 2^k$.

Part (iii) for $0 < r < s$ is not formalized as a separate statement; it is the positivity lemma composed with the tiling.

Corollary 5.5 (The even jet: half the depth). *Let $x \in (-1,1)$ be dyadic with $s(x) = s$ and $d = \lfloor s/2 \rfloor$. Then*

$$\mathrm{up}^{(2m)}(x) \neq 0 \iff m \leq d; \tag{63}$$

at the two endpoints $x = \pm 1$ (depth 0) the whole jet vanishes, $\mathrm{up}^{(r)}(\pm 1) = 0$ for all $r \geq 0$. For $s \geq 2$ the last nonzero even entry has magnitude

$$\left|\mathrm{up}^{(2d)}(x)\right| = \begin{cases} 2^{\binom{s+1}{2}}, & s = 2d, \\ 2^{\binom{s}{2}-1}, & s = 2d+1, \end{cases} \tag{64}$$

with the signs ε_k and $\varepsilon_{k'}$ of Theorem 5.4. In S6's terms, with $a = 1 - |x|$ and $F = F_{\mathrm{bd}}$, $F^{(2m)}(a) = \mathrm{up}^{(2m)}(x)$ vanishes for $m > d$ and is nonzero for $m \leq d$: "beyond half the dyadic depth" for the even derivatives is the same cutoff as "beyond the depth" for all derivatives, because $2m > s \iff m > \lfloor s/2 \rfloor$.

Proof. $2m \leq s \iff m \leq d$. For $2m \leq s$ the derivative is nonzero by Theorem 5.4(ii)–(iii) (and by Theorem 5.3 when $m = 0$); for $2m > s$ it vanishes by (i). At $x = \pm 1$, (i) with $s = 0$ gives the vanishing for $r \geq 1$, and $\mathrm{up}(\pm 1) = 0$. The magnitudes are Equation (61) at $2d = s$ and Equation (62) at $2d = s - 1$. For the CDF form: for $x \geq 0$, $\mathrm{up}(x) = F_{\mathrm{bd}}(1-x)$ (catalogue entry `fabius-relations`), so $\mathrm{up}^{(2m)}(x) = (-1)^{2m}F^{(2m)}(1-x) = F^{(2m)}(a)$; for $x < 0$ use evenness. □

Remark 5.6 (A second proof of termination, from the first mechanism; local degree collapse). Theorem 4.13 was proved for every $r \geq 0$, and its right side Equation (53) is empty when $r > n$. Taking $n = s$ gives, without any Thue–Morse computation, $\mathrm{up}^{(r)}(x) = (\mathcal{A}_{\delta_s} P_{s,x})^{(r)}(x) = 0$ for $r > s$, because $\mathcal{A}_{\delta_s} P_{s,x}$ is a polynomial of degree at most s: the jet of up at x terminates because up is, at x, the average of a degree-s polynomial over a window that crosses no knot. Conversely, for every $n \geq s$, Equation (54) at orders $r > s$ gives $P_{n,x}^{(r)}(x) = 0$, so

$$\deg P_{n,x} \leq s \qquad (n \geq s), \tag{65}$$

and at order $r = s$ it gives $P_{n,x}^{(s)}(x) = \mathrm{up}^{(s)}(x)$ (the terms $m \geq 1$ vanish by (i)). Thus the cell polynomial of *every* up_n, $n \geq s$, at a point of depth s has degree exactly s with the n-independent leading coefficient $\mathrm{up}^{(s)}(x)/s! = 2^{\binom{s+1}{2}} \varepsilon_k / s!$, although up_n has global degree n. This is S3's theorem "Local degree collapse", proved there by Prouhet cancellation in the truncated-power formula (blocks of 2^{n-s} consecutive Thue–Morse signs annihilate polynomials of degree below $n - s$). The two statements, jet termination and degree collapse, are equivalent through Theorem 4.13: each implies the other by one of Equations (53) and (54). By the catalogue's caveat in `fabius-dyadic-cells`, no local analyticity is inferred from this finite jet, and none is needed anywhere in this volume.

Formal versions.

- `Fabius.iteratedDeriv_fabiusUniformSpline_dyadic_center` (`PlateauLimit`) is the constancy of the leading coefficient, the "plateau" of the Lean corpus: in the centred-CDF picture at the anchor 2^{-r} (depth r), the r-th derivative of `fabiusUniformSpline` p equals $2^{\binom{r+1}{2}}$ for every $p \geq r$, the value of $\mathrm{up}^{(r)}$ there;
- `Fabius.rvachevDyadicTaylorPolynomial_natDegree_eq` (`OriginalPaperSupplement`): the Taylor polynomial of up at a centred dyadic of level s has degree exactly s.

5.4 The two mechanisms together: the exact number of modes

Corollary 5.7 (Finite geometric tail with all modes present). *Let $x \in [-1, 1]$ be dyadic with $s(x) = s$ and $d = \lfloor s/2 \rfloor$. For every $n \geq s$, and also for $n = s - 1$ when $s \geq 3$ is odd,*

$$\boxed{\mathrm{up}_n(x) = \sum_{m=0}^{d} (-1)^m a_m \, 4^{-m(n+1)} \, \mathrm{up}^{(2m)}(x) = \mathrm{up}(x) + \sum_{m=1}^{d} C_m(x) \, 4^{-mn},} \tag{66}$$

with $C_m(x) = (-1)^m a_m 4^{-m} \, \mathrm{up}^{(2m)}(x) \neq 0$ for every $1 \leq m \leq d$. In particular:

- *for $s \in \{0, 1\}$ (the points $0, \pm 1, \pm 1/2$) there is no mode at all, $\mathrm{up}_n(x) = \mathrm{up}(x)$ for every $n \geq s$;*
- *for $s \geq 2$ the tail has exactly $d \geq 1$ nonconstant geometric modes 4^{-mn}, $1 \leq m \leq d$, with the top coefficient*

$$|C_d(x)| = a_d \, 4^{-d} \left| \mathrm{up}^{(2d)}(x) \right| = \begin{cases} a_d \, 2^{\binom{s+1}{2} - 2d}, & s = 2d, \\ a_d \, 2^{\binom{s}{2} - 2d - 1}, & s = 2d + 1, \end{cases} \tag{67}$$

so the mode count d is sharp and the sample count $d + 1$ of the extraction in Section 6 cannot be lowered within the mode model.

All quantities in Equation (66) lie in $\mathbb{Q} = \mathbb{Q}$.

Proof. Equation (55) holds for $n \geq s$, and Equation (56) for $n = s - 1$ when $s \geq 3$ is odd; in both the sum runs to $\lfloor n/2 \rfloor \geq d$ (for $n = s - 1 = 2d$ it runs exactly to d). By Theorem 5.5 (and Theorem 5.4(i) at $x = \pm 1$), $\mathrm{up}^{(2m)}(x) = 0$ for $m > d$, so the sum stops at d; and for $1 \leq m \leq d$, $a_m > 0$ (Theorem 4.6) and $\mathrm{up}^{(2m)}(x) \neq 0$ give $C_m(x) \neq 0$. The rewriting $4^{-m(n+1)} = 4^{-m} 4^{-mn}$ produces $C_m(x)$, and Equation (64) gives Equation (67). For $s \leq 1$, $d = 0$.

Rationality. The samples $\mathrm{up}_n(x)$, $n = s, \dots, s+d$, are rational by the truncated-power formula of Section 2. Read at these $d+1$ values of n, Equation (66) is a square linear system for the $d+1$ unknowns $\mathrm{up}(x), C_1(x), \dots, C_d(x)$ whose matrix $\left(4^{-mn}\right)_{n,m}$ is the Vandermonde matrix of the distinct rational nodes $4^{-s}, \dots, 4^{-(s+d)}$; it is invertible over $\mathbb{Q}$, so the unknowns are rational, and then so is $\mathrm{up}^{(2m)}(x) = (-1)^m 4^m C_m(x)/a_m$. (This recovers the classical rationality of up at dyadic points from the finite splines alone.) The sample count $d+1$ cannot be lowered within the model: the kernel of the $d \times (d+1)$ system obtained from any d of the levels is spanned by the coefficient vector of $\prod_n (t - 4^{-n})$ over those levels, whose constant term $\pm \prod_n 4^{-n}$ is not zero, so d samples leave the constant term $\mathrm{up}(x)$ undetermined. □

The exact examples of the later sections confirm the top coefficient literally: at $x = 1/4$ ($s = 2$, $d = 1$) it is $a_1 2^{3-2} = 1/9$; at $x = 1/8$ ($s = 3$, $d = 1$) it is $a_1 2^{3-3} = 1/18$; at $x = 1/16$ ($s = 4$, $d = 2$) it is $a_2 2^{10-4} = 64 \cdot 31/16200 = 248/2025$; at $x = 1/32$ ($s = 5$, $d = 2$) it is $a_2 2^{10-5} = 124/2025$; the signs are ε_k or $\varepsilon_{k'}$ times $(-1)^d$.

The shape of the argument so far

1. *Cell geometry* (Section 2, recalled in Theorem 4.12): for $n \geq s(x)$ the tail window $[x - \delta_n, x + \delta_n]$ is one polynomial cell of up_n.
2. *First mechanism* (Theorem 4.13): on that cell the deconvolution identity Equation (55) holds exactly, a finite sum to $\lfloor n/2 \rfloor$ with the positive rational coefficients a_m of Theorem 4.6.
3. *Second mechanism* (Theorem 5.4, Theorem 5.5): $\mathrm{up}^{(2m)}(x) = 0$ for $m > \lfloor s/2 \rfloor$ and $\neq 0$ for $m \leq \lfloor s/2 \rfloor$.
4. *Together* (Theorem 5.7): an exact d-mode geometric law from $n = s(x)$ on, every mode present. The extraction of $\mathrm{up}(x)$ from $d+1$ consecutive samples by the quarter-base Gaussian-binomial row, and the recovery of every mode, are the subject of Section 6 and the sections that follow it.

6 The exact eventual geometric law

The three mechanisms are now in place: a dyadic point is the centre of one polynomial cell of up_n from level $n = s(x)$ on (Section 3); on such a cell the omitted tail is deconvolved exactly by the finite reciprocal-product operator with the rational coefficients a_m (Section 4); and the derivative jet of up at the point terminates at the order $s(x)$ (Section 5). This section assembles them into the proof of Theorem 1.1, deferred from Section 1, and records the immediate consequences that the six sources state in various forms: the derivative version of the law, the rationality of everything in sight, the rational tail generating function, and the eventually annihilating q-difference recurrence. Section 7 then shows that nothing in the statement can be improved.

Throughout, $x \in [-1, 1]$ is dyadic, $s = s(x)$, $d = \lfloor s/2 \rfloor$, $Q = 1/4$, $\delta_n = 2^{-(n+1)}$, and $q_n = \mathrm{up}_n(x)$.

6.1 Proof of the main theorem

Proof of Theorem 1.1. Fix $n \geq s$.

Step 1: one polynomial on the whole tail window. By Theorem 3.4 (equivalently Theorem 4.12), x is the midpoint of a knot cell of up_n: the closed window $[x - \delta_n, x + \delta_n]$ contains no knot in its interior, and Theorem 2.5 makes up_n a single polynomial $P = P_{n,x}$ of degree at most n on it, with $P(x) = \mathrm{up}_n(x)$.

Step 2: exact local deconvolution. By Theorem 2.4, $\mathrm{up} = \mathrm{up}_n * T_{\delta_n}$ with $T_\delta = \delta^{-1}\,\mathrm{up}(\cdot/\delta)$ supported in $[-\delta, \delta]$. The convolution at x, and all its derivatives at x, therefore see only the polynomial P, and

Theorem 4.13 gives, at $r = 0$,

$$\mathrm{up}_n(x) = P(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m a_m\, \delta_n^{2m}\, \mathrm{up}^{(2m)}(x), \qquad \delta_n^{2m} = 4^{-m(n+1)}. \tag{68}$$

This is an identity, with no remainder: the reciprocal series acts on a polynomial of degree at most n and stops by itself.

Step 3: termination. By Theorem 5.4(i), $\mathrm{up}^{(r)}(x) = 0$ for every $r > s$, so the terms of Equation (68) with $2m > s$, that is with $m > d$, vanish; and $n \ge s$ gives $\lfloor n/2 \rfloor \ge d$, so all terms with $m \le d$ are present. Hence

$$\mathrm{up}_n(x) = \sum_{m=0}^{d} (-1)^m a_m\, 4^{-m(n+1)}\, \mathrm{up}^{(2m)}(x) \qquad (n \ge s), \tag{69}$$

which is Equation (5). Writing $4^{-m(n+1)} = 4^{-m} \cdot 4^{-mn}$ and separating the term $m = 0$ gives Equation (4) with $C_m(x) = (-1)^m a_m 4^{-m}\, \mathrm{up}^{(2m)}(x)$.

Step 4: rationality. The samples q_n are rational by Theorem 2.6, and the a_m are positive rationals by Theorem 4.6. Read Equation (69) at the $d+1$ levels $n = s, \ldots, s+d$ as a linear system for the $d+1$ unknowns $\mathrm{up}^{(2m)}(x)$, $0 \le m \le d$: its matrix $\big((-1)^m a_m 4^{-m(n+1)}\big)_{n,m}$ is a Vandermonde matrix in the distinct nodes $4^{-s}, \ldots, 4^{-(s+d)}$ times an invertible diagonal matrix, hence invertible over $\mathbb{Q}$ (Theorem 9.1 gives the determinant). So every $\mathrm{up}^{(2m)}(x)$, $0 \le m \le d$, and every $C_m(x)$ is a rational combination of rational samples, and all quantities in Equation (4) lie in $\mathbb{Q} = \mathbb{Q}$. □

Remark 6.1 (What the proof used, and what it did not). The proof uses no property of up beyond the head–tail factorization, the knot geometry of the finite splines, the evenness of the tail, and the termination of the dyadic jet. In particular it does not use the classical rationality of up at dyadic points [1, 2], which it re-derives (Theorem 6.4), nor any convergence statement: the only limit in the volume is the one that defines up itself. The same four ingredients, with the level $n = s-1$ in place of $n \ge s$, are what Section 7.2 analyses; the difference is that at $n = s-1$ the window contains two polynomial pieces.

6.2 The derivative version

The identity Equation (54) holds for every derivative order, and the jet terminates at the same order s for all of them, so the law has a derivative version with fewer modes. For $n \ge s$ and $0 \le r \le n$ write $\mathrm{up}_n^{(r)}(x)$ for the r-th derivative at x of the cell polynomial $P_{n,x}$; for $r \le n-1$ this is the ordinary r-th derivative of $\mathrm{up}_n \in C^{n-1}$, and for $r = n$ it is the common one-sided value of the n-th derivative on the open cell.

Corollary 6.2 (Eventual law for the derivatives). *Let $0 \le r \le s$ and $d_r = \lfloor (s-r)/2 \rfloor$. For every $n \ge s$,*

$$\boxed{\mathrm{up}_n^{(r)}(x) = \sum_{m=0}^{d_r} (-1)^m a_m\, 4^{-m(n+1)}\, \mathrm{up}^{(r+2m)}(x) = \mathrm{up}^{(r)}(x) + \sum_{m=1}^{d_r} C_{r,m}(x)\, 4^{-mn},} \tag{70}$$

with $C_{r,m}(x) = (-1)^m a_m 4^{-m}\, \mathrm{up}^{(r+2m)}(x) \neq 0$ for $1 \le m \le d_r$; for $r > s$ and $n \ge s$ both $\mathrm{up}_n^{(r)}(x)$ and $\mathrm{up}^{(r)}(x)$ vanish. Consequently the extraction row of order d_r (Theorem 8.7 with d_r in place of d) recovers $\mathrm{up}^{(r)}(x)$ exactly from the $d_r + 1$ consecutive derivative samples $\mathrm{up}_N^{(r)}(x), \ldots, \mathrm{up}_{N+d_r}^{(r)}(x)$, for any $N \ge s$.

Proof. Equation (54) at order r reads

$$P^{(r)}(x) = \sum_{m \le \lfloor (n-r)/2 \rfloor} (-1)^m a_m \delta_n^{2m}\, \mathrm{up}^{(r+2m)}(x),$$

and $\mathrm{up}^{(r+2m)}(x) = 0$ once $r + 2m > s$ (Theorem 5.4(i)), which cuts the sum at $m = d_r$; since $n \geq s$, $\lfloor (n-r)/2 \rfloor \geq \lfloor (s-r)/2 \rfloor = d_r$, so no term is lost. The nonvanishing of the modes is Theorem 5.4(iii): $\mathrm{up}^{(r+2m)}(x) \neq 0$ for $r + 2m \leq s$. For $r > s$ the cell polynomial has degree at most n, and the same identity with an empty right side gives $P^{(r)}(x) = 0$, since every $\mathrm{up}^{(r+2m)}(x)$ vanishes. The extraction statement is the moment identity Equation (90) of the row of order d_r applied to the finite sum Equation (70), exactly as in the proof of Theorem 8.7. ☐

Remark 6.3 (Parity of the derivative order). For odd r the derivatives on the right of Equation (70) are the odd-order derivatives $\mathrm{up}^{(r)}, \mathrm{up}^{(r+2)}, \ldots$, so the derivative law is not a consequence of the value law at other points; it is the value law for the odd part of the local jet. The mode ratios are the same powers of $1/4$ in every case, because they come from the even tail and not from the order of differentiation.

6.3 Rationality

Corollary 6.4 (Exact rationality)**.** *For every dyadic $x \in [-1, 1]$, the value $\mathrm{up}(x)$, every derivative $\mathrm{up}^{(r)}(x)$, and every mode coefficient $C_m(x)$ are rational numbers, and each is a fixed rational combination of $d + 1$ rational finite-spline values:*

$$\mathrm{up}(x) = \sum_{j=0}^{d} w_j^{(d)}\, q_{N+j}, \qquad \mathrm{up}^{(2m)}(x) = \frac{(-1)^m 4^{m(N+1)}}{a_m} \sum_{j=0}^{d} q_{N+j}\, \frac{(-1)^{d-m} e_{d-m}^{(j)}}{D_j} \qquad (N \geq s), \tag{71}$$

with the weights $w_j^{(d)}$ of Theorem 8.3 and the quantities D_j, $e_r^{(j)}$ of Theorem 9.2. The odd derivatives are rational by the Thue–Morse tiling Theorem 5.2, and every derivative of order above s is zero.

Proof. The first identity is Theorem 8.7, the second is Theorem 9.3; both rest only on Theorem 1.1 and on the rationality of the samples (Theorem 2.6). The statements about odd and high orders are Theorems 5.2 and 5.4. ☐

The classical proof of rationality [1, 2] runs through the functional-differential equation and explicit recurrences for the values at dyadic points; the present one runs through finite convolution data alone, and it returns the value rather than a rationality certificate. The size of the denominators is the subject of Theorem 10.2: the extraction adds to the denominators of the samples only the odd factor $P_d = \prod_{k=1}^{d}(4^k - 1)$.

6.4 The tail generating function and the annihilating recurrence

The finite geometric law is equivalent to the rationality of a generating function, and its coefficient form is the annihilating recurrence of Section 9.

Proposition 6.5 (Rational tail generating function)**.** *Let $N \geq s$. Then, for $|z| < 1$,*

$$\sum_{k=0}^{\infty} q_{N+k}\, z^k = \frac{\mathrm{up}(x)}{1 - z} + \sum_{m=1}^{d} \frac{C_m(x)\, Q^{mN}}{1 - Q^m z}, \tag{72}$$

a rational function of z whose denominator divides $(1 - z) \prod_{m=1}^{d}(1 - Q^m z)$ and whose numerator has degree at most d; for $s \geq 2$ the denominator is exactly this product, since every partial fraction has a nonzero numerator. Conversely, a sequence whose generating function has this form for some N satisfies the law Equation (4) for all $n \geq N$.

Proof. Insert Equation (4) at $n = N + k$ and sum the $d + 1$ geometric series; the ratios 1 and Q^m are all less than $1/|z|$ when $|z| < 1$. Distinct poles $1, Q^{-1}, \ldots, Q^{-d}$ with nonzero residues (Theorem 7.1) give the exact denominator. The converse is the uniqueness of partial-fraction expansions. ☐

Corollary 6.6 (Eventually annihilating recurrence). *The sequence $(q_n)_{n\geq s}$ is annihilated by the product of shifts $\prod_{m=0}^{d}(\mathsf{E}-Q^m)$, that is, by the recurrence Equation (113) of Theorem 9.9, with the integer form Equation (114); for $s\geq 2$ no recurrence of lower order with constant coefficients annihilates it. The recurrence is* eventual*: Section 7.2 shows that at even depth its residual at $n=s-1$ is a nonzero rational whose value is known exactly.*

Proof. Multiplying Equation (72) by the denominator gives a polynomial of degree at most d, which is the statement that the coefficients of z^k, $k\geq d+1$, of the product vanish; those coefficients are the residuals $\rho_{N+k-d-1}$ of Equation (115). Minimality is Theorem 9.102 together with $C_d(x)\neq 0$. ☐

Section summary

Theorem 1.1 is proved: for $n\geq s$ the sample q_n is $\mathrm{up}(x)$ plus exactly $d=\lfloor s/2\rfloor$ geometric modes $C_m(x)\,4^{-mn}$, with no remainder, because the window seen by the omitted tail is one polynomial cell (Section 3), the tail is deconvolved by a finite operator (Section 4), and the jet stops at order s (Section 5). The same argument gives the law for every derivative order $r\leq s$ with $\lfloor (s-r)/2\rfloor$ modes (Theorem 6.2), rationality of the whole jet from finite convolution data (Theorem 6.4), a rational tail generating function with the exact denominator $(1-z)\prod_{m\leq d}(1-Q^m z)$ (Theorem 6.5), and the eventually annihilating recurrence (Theorem 6.6).

7 Sharpness: the number of modes and the onset

Theorem 1.1 makes three quantitative claims: the tail has $d=\lfloor s/2\rfloor$ modes, the law holds from level s on, and (through Theorem 8.7) $d+1$ samples suffice. This section shows that the first is exact (every mode is present), settles the second completely at even depth by computing the defect at the last level before the onset in closed form, records what is proved and what is verified at odd depth, shows that the third is minimal within the mode model, and makes precise the relation to the asymptotic statements of the Exponents volume [9].

7.1 The number of modes is exactly d

Theorem 7.1 (All d modes are present). *Let $x\in(-1,1)$ be dyadic with $s=s(x)\geq 2$ and $d=\lfloor s/2\rfloor$. Then $C_m(x)\neq 0$ for every $1\leq m\leq d$; the polynomial $P_N(z)=\sum_{m=0}^{d}B_m z^m$ of Theorem 9.2, whose values at $z=Q^k$ are the samples q_{N+k}, has exact degree d; the least order of a constant-coefficient recurrence annihilating $(q_n)_{n\geq s}$ is d; and the top coefficient has the magnitude*

$$|C_d(x)|=\begin{cases} a_d\,2^{\binom{s+1}{2}-2d}, & s=2d,\\ a_d\,2^{\binom{s}{2}-2d-1}, & s=2d+1.\end{cases}\tag{73}$$

For $s\in\{0,1\}$ (the points $0,\pm1,\pm1/2$) there is no mode: $q_n=\mathrm{up}(x)$ for every $n\geq s$.

Proof. $C_m(x)=(-1)^m a_m 4^{-m}\,\mathrm{up}^{(2m)}(x)$ with $a_m>0$ (Theorem 4.6) and $\mathrm{up}^{(2m)}(x)\neq 0$ for $2m\leq s$ (Theorem 5.5); the magnitude is Equation (67). The degree statement is $B_d=C_d(x)\,Q^{dN}\neq 0$ (Equation (109)), and the recurrence order is Theorem 9.102. For $s\leq 1$ the sum in Equation (4) is empty. ☐

Remark 7.2 (Which source proved what). Sources S1 and S3 prove only that the top mode $C_d(x)$ is nonzero; S6 proves that all d modes are nonzero, from the vanishing pattern of the even jet, and Theorem 5.5 is its argument. The exhaustive exact verification of the volume (Section 12) found all 684 mode coefficients at the 252 interior points of depth 2 to 7 nonzero, as the theorem requires.

7.2 The onset: the exact defect at the knot level

At the level $n = s - 1$ the point x is a knot of up_n, not the centre of a cell (Theorem 3.4(ii)), and the proof of the law does not apply. Whether the law nevertheless holds there was, in the sources, a matter of computation: S3 proved that it does at odd depth (Theorem 4.15), S4's verification found that it fails at every point of even depth through depth 7, and no source proved the failure. The following theorem settles the question at every depth by computing the defect in closed form: it is a Thue–Morse sign times a Bernoulli number.

Write, for $s \geq 1$ and $n = s - 1$,

$$\Delta_s(x) := \mathrm{up}_{s-1}(x) - \sum_{m=0}^{d} (-1)^m a_m\, 4^{-ms}\, \mathrm{up}^{(2m)}(x) \tag{74}$$

for the difference between the sample at the knot level and the value the law Equation (5) would predict there (note $4^{-m(n+1)} = 4^{-ms}$). For $s \geq 2$ the spline up_{s-1} is continuous, so $\Delta_s(x)$ is unambiguous; for $s = 1$ it depends on the value assigned to the box up_0 at its jumps, as in Theorem 4.15.

We use the moment generating function $M(z) = \sum_n m_n^{\mathrm{Rv}} z^n/n!$ of Theorem 4.3, $1/M(z) = \sum_m (-1)^m a_m z^{2m}$ (Equation (39)), and its half-range companion

$$M_+(z) := \int_0^1 \mathrm{e}^{zu}\, \mathrm{up}(u)\, \mathrm{d}u = \sum_{j=0}^{\infty} \frac{M_j}{2} \frac{z^j}{j!}, \qquad M_j := \int_{-1}^{1} |u|^j\, \mathrm{up}(u)\, \mathrm{d}u, \tag{75}$$

so that $M(z) = M_+(z) + M_+(-z)$, $M_0 = 1$, and $M_j = m_j^{\mathrm{Rv}}$ for even j.

Lemma 7.3 (Half-range moment generating function). *M_+ is entire and*

$$\boxed{z\, M_+(z) = \mathrm{e}^{z/2}\, M(z/2) - 1, \qquad M_+(z) + \frac{1}{z} = \frac{\mathrm{e}^{z/2} M(z/2)}{z}.} \tag{76}$$

Consequently

$$\frac{M_+(z) + z^{-1}}{M(z)} = \frac{\mathrm{e}^{z/2}}{2\sinh(z/2)} = \frac{1}{2}\left(1 + \coth\frac{z}{2}\right) = \frac{1}{2} + \frac{1}{z} + \sum_{j=1}^{\infty} \frac{B_{2j}}{(2j)!} z^{2j-1}, \tag{77}$$

where B_{2j} are the Bernoulli numbers.

Proof. Integrating by parts, with $\mathrm{up}(0) = 1$ and $\mathrm{up}(1) = 0$ (Theorem 2.8),

$$z\, M_+(z) = \left[\mathrm{e}^{zu}\, \mathrm{up}(u)\right]_0^1 - \int_0^1 \mathrm{e}^{zu}\, \mathrm{up}'(u)\, \mathrm{d}u = -1 - \int_0^1 \mathrm{e}^{zu}\, \mathrm{up}'(u)\, \mathrm{d}u.$$

Rvachev's equation $\mathrm{up}'(u) = 2\,\mathrm{up}(2u+1) - 2\,\mathrm{up}(2u-1)$ (Theorem 2.8) reduces on $0 < u < 1$ to $\mathrm{up}'(u) = -2\,\mathrm{up}(2u-1)$, because $2u + 1 > 1$ lies outside the support. Substituting $v = 2u - 1$,

$$-\int_0^1 \mathrm{e}^{zu}\, \mathrm{up}'(u)\, \mathrm{d}u = 2\int_0^1 \mathrm{e}^{zu}\, \mathrm{up}(2u-1)\, \mathrm{d}u = \int_{-1}^{1} \mathrm{e}^{z(v+1)/2}\, \mathrm{up}(v)\, \mathrm{d}v = \mathrm{e}^{z/2}\, M(z/2),$$

which is the first identity; the second is the first divided by z. For Equation (77), the product formula Equation (38) gives $M(z) = \dfrac{\sinh(z/2)}{z/2} M(z/2)$, so

$$\frac{M_+(z) + z^{-1}}{M(z)} = \frac{\mathrm{e}^{z/2} M(z/2)}{z} \cdot \frac{z/2}{\sinh(z/2)\, M(z/2)} = \frac{\mathrm{e}^{z/2}}{2\sinh(z/2)} = \frac{\cosh(z/2) + \sinh(z/2)}{2\sinh(z/2)},$$

and $\frac{z}{2}\coth\frac{z}{2} = \sum_{j\geq 0} B_{2j} z^{2j}/(2j)!$ is the generating function of the even Bernoulli numbers. □

Theorem 7.4 (Exact defect at the knot level)**.** *Let $x \in (-1,1)$ be dyadic with depth $s \geq 2$, $d = \lfloor s/2 \rfloor$, and let $k = \big(2^s(x+1)-1\big)/2 \in \{0,\dots,2^s-1\}$ be the index of x in the knot lattice $t_{s-1,k} = -1+(2k+1)2^{-s}$ of up_{s-1} (Theorem 3.3). Then*

$$\boxed{\Delta_s(x) = -\varepsilon_k\, 2^{-\binom{s}{2}}\,\beta_s, \qquad \beta_s = \begin{cases} \dfrac{B_s}{s!}, & s \text{ even},\\[4pt] 0, & s \text{ odd}.\end{cases}} \tag{78}$$

In particular, at every point of even depth s the law fails at $n = s-1$ by the universal amount $|\Delta_s(x)| = 2^{-\binom{s}{2}}\,|B_s|/s!$, which is $\frac{1}{24}$, $\frac{1}{46080}$, $\frac{1}{990904320}$, $\frac{1}{324699527577600}$ for $s = 2,4,6,8$; and at every point of odd depth $s \geq 3$ the law already holds at $n = s-1$.

Proof. Put $n = s-1 \geq 1$ and $\delta = \delta_n = 2^{-s}$, so that $4^{-ms} = \delta^{2m}$ and $\Delta_s(x) = \mathrm{up}_n(x) - \sum_{m=0}^{d}(-1)^m a_m \delta^{2m}\,\mathrm{up}^{(2m)}(x)$.

Two pieces on the window. By Theorem 3.4(ii), $x = t_{n,k}$ is the only knot of up_n in the window $[x-\delta, x+\delta]$. By Theorem 2.5 the n-th derivative of up_n is a step function whose jump at $t_{n,k}$, right value minus left value, is $J := 2^{\binom{n+1}{2}}\varepsilon_k = 2^{\binom{s}{2}}\varepsilon_k$, and up_n has degree at most n on each of the two cells. Hence on the window

$$\mathrm{up}_n(t) = P(t) + \frac{J}{n!}\,(t-x)_+^n, \tag{79}$$

where P is the polynomial piece on the left cell, and, since $n \geq 1$ makes up_n continuous, $\mathrm{up}_n(x) = P(x)$.

Transfer of the extra piece through the tail. Write $\mathcal{A}_\delta P = P * T_\delta$ for the tail average of Section 4. By Theorem 2.4, $\mathrm{up} = \mathrm{up}_n * T_\delta$, so for every $r \geq 0$

$$\mathrm{up}^{(r)}(x) = (\mathcal{A}_\delta P)^{(r)}(x) + J\,X_r, \qquad X_r := \frac{1}{n!}\,\frac{\mathrm{d}^r}{\mathrm{d}t^r}\Big[(t-x)_+^n * T_\delta\Big](x), \tag{80}$$

and we compute X_r in the three ranges of r.

(a) $0 \leq r \leq n-1$. The truncated power is C^{n-1}, so the derivative falls on it:

$$X_r = \frac{1}{(n-r)!}\int_{-\delta}^{0}(-y)^{n-r}\,T_\delta(y)\;\mathrm{d}y = \frac{\delta^{n-r}}{(n-r)!}\int_{-1}^{0}|u|^{n-r}\,\mathrm{up}(u)\;\mathrm{d}u = \frac{\delta^{n-r}\,M_{n-r}}{2\,(n-r)!},$$

by the substitution $y = \delta u$ and the evenness of up.

(b) $r = n$. The $(n-1)$-st derivative of the truncated power is absolutely continuous, so the n-th derivative of its convolution with the smooth T_δ is the convolution of its a.e. derivative $\mathbf{1}_{t>x}$ with T_δ, and $X_n = \int_{-\delta}^{0} T_\delta(y)\;\mathrm{d}y = \frac{1}{2} = \delta^0 M_0/(2\cdot 0!)$, the same formula as in (a).

(c) $r \geq n+1$. Now the derivatives fall on T_δ: $X_r = \frac{1}{n!}\int_{-\delta}^{0}(-y)^n\,T_\delta^{(r)}(y)\;\mathrm{d}y$. Integrate by parts $n+1$ times. Every boundary term at $y = -\delta$ vanishes because all derivatives of T_δ vanish at $-\delta$ (Theorem 5.4 at depth 0), and at $y = 0$ the derivatives of $(-y)^n$ of order below n vanish while the n-th is the constant $(-1)^n n!$; the $(n+1)$-st derivative of $(-y)^n$ is zero, so no integral remains, and

$$X_r = \frac{1}{n!}\cdot(-1)^n\cdot(-1)^n n!\;T_\delta^{(r-n-1)}(0) = T_\delta^{(r-n-1)}(0) = \delta^{n-r}\,\mathrm{up}^{(r-n-1)}(0) = \delta^{-1}\,[\,r = n+1\,],$$

because $\mathrm{up}(0) = 1$ and every derivative of positive order vanishes at the origin (Theorem 5.4 at depth 0).

In all three ranges,

$$X_r = \delta^{n-r}\,c_{n-r}, \qquad c_j := \frac{M_j}{2\,j!}\;(j \geq 0), \qquad c_{-1} := 1, \qquad c_j := 0\;(j \leq -2). \tag{81}$$

The defect. Insert Equation (80) at $r = 2m$ into the definition of $\Delta_s(x)$:

$$\Delta_s(x) = \Big[P(x) - \sum_{m=0}^{d}(-1)^m a_m \delta^{2m}(\mathcal{A}_\delta P)^{(2m)}(x)\Big] - J\sum_{m=0}^{d}(-1)^m a_m\,\delta^{2m}X_{2m}.$$

The bracket vanishes: by Theorem 4.10 the finite deconvolution $\sum_m(-1)^m a_m\delta^{2m}(\mathcal{A}_\delta P)^{(2m)}$ returns P for every polynomial P, the sum extending over $2m \le \deg P \le n$; the terms with $2m \le n$ are all among $m \le d$, and the possible extra term $m = d$ with $2d = n+1$ (even s) is zero because $\deg P \le n < 2d$. By Equation (81),

$$\Delta_s(x) = -J\,\delta^n\sum_{m=0}^{d}(-1)^m a_m\,c_{n-2m},$$

and the sum runs over exactly those m with $n - 2m \ge -1$, that is $m \le \lfloor (n+1)/2\rfloor = \lfloor s/2\rfloor = d$, so it is the full coefficient of z^n in the product of the two series $\sum_m(-1)^m a_m z^{2m} = 1/M(z)$ and $\sum_{j\ge -1} c_j z^j = M_+(z) + z^{-1}$ (Equation (75)):

$$\Delta_s(x) = -J\,\delta^n\,[z^n]\,\frac{M_+(z)+z^{-1}}{M(z)} = -J\,\delta^n\,[z^n]\,\frac{1}{2}\Big(1+\coth\frac{z}{2}\Big)$$

by Theorem 7.3. For $n \ge 1$ the coefficient of z^n in Equation (77) is $B_{n+1}/(n+1)!$ when n is odd and 0 when n is even; with $n = s-1$ this is β_s. Finally $J\delta^n = 2^{\binom{s}{2}}\varepsilon_k\,2^{-s(s-1)} = \varepsilon_k\,2^{-\binom{s}{2}}$.

The numerical values: $B_2/2! = \frac{1}{12}$, $B_4/4! = -\frac{1}{720}$, $B_6/6! = \frac{1}{30240}$, $B_8/8! = -\frac{1}{1209600}$, multiplied by $2^{-1}, 2^{-6}, 2^{-15}, 2^{-28}$. ☐

Remark 7.5 (Depth one, and a check). The computation applies verbatim at $s = 1$, $n = 0$, $x = \pm 1/2$, if $\mathrm{up}_0(x)$ is read as the left-cell value $P(x)$: the coefficient of z^0 in Equation (77) is $\frac{1}{2}$, so $\Delta_1 = -J/2$ with $J = \varepsilon_k$ the jump of the box, that is $\Delta_1(1/2) = \frac{1}{2}$ and $\Delta_1(-1/2) = -\frac{1}{2}$; with the Fourier-inversion value $\mathrm{up}_0(\pm 1/2) = \frac{1}{2}$ the defect is 0, and with the right-cell value it is $+J/2$. This is Theorem 4.15's remark on the symmetric-rectangle convention. At $x = 1/4$ ($s = 2$, $k = 2$, $\varepsilon_2 = -1$) the theorem gives $\Delta_2 = +\frac{1}{24}$, and indeed $\mathrm{up}_1(1/4) = 1$ while the law predicts $\frac{23}{24}$ (Theorem 1.6); at $x = 3/4$ ($k = 3$, $\varepsilon_3 = +1$) it gives $-\frac{1}{24}$, and $\mathrm{up}_1(3/4) = 0$ against a predicted $\frac{1}{24}$. The formula Equation (78) was checked in exact rational arithmetic at all 508 interior dyadic points of depth 2 to 8 (Section 12).

Corollary 7.6 (The exact onset). *Let $n_0(x)$ be the least level from which the law Equation (4) holds for all $n \ge n_0(x)$.*

(i) *If s is even, $s \ge 2$, then $n_0(x) = s$: the onset stated in Theorem 1.1 is exact at every point of even depth.*

(ii) *If s is odd, $s \ge 3$, then $n_0(x) \le s-1$, and $n_0(x) = s-1$ exactly when the law fails at $n = s-2$. This failure holds at every interior point of depth 3, 5 and 7, with $|\Delta| = \frac{1}{96}, \frac{7}{737280}, \frac{31}{63417876480}$ respectively, independent of the point (verified in exact arithmetic); a proof for all odd depths is open.*

(iii) *For $s \in \{0, 1\}$, $n_0(x) = s$ with the Fourier-inversion convention at $s = 1$, and $n_0(\pm 1/2) = 1$ with either one-sided convention.*

Proof. (i) is Theorem 7.4 with $B_s \ne 0$ for even s. (ii) is Theorem 7.4 with $\beta_s = 0$ (equivalently Theorem 4.15); at $n = s-2$ the window contains one knot strictly inside, at distance 2^{-s} from x (Theorem 3.4(iii)), and the two-piece computation above does not apply as it stands, because the extra truncated power is centred off x. The listed values are from the exhaustive verification. (iii) is Theorem 7.5 and Theorem 3.5. ☐

Remark 7.7 (The two-knot levels). The defects at $n = s - 2$ recorded in Theorem 7.6(ii), and the analogous even-depth values $\frac{1}{15360}$ ($s = 4$) and $\frac{1}{198180864}$ ($s = 6$) at $n = s - 2$, are again independent of the point within a depth, which suggests that a closed form like Equation (78) exists at every level below the onset, with the half-range moments of Theorem 7.3 replaced by partial moments $\int_{-1}^{\pm 1/2} (\pm\frac{1}{2} - u)^j \operatorname{up}(u)\, \mathrm{d}u$ of the tail over the off-centre knot. The numerators $1, 7, 31 = 2^{s-2} - 1$ at odd depth and the denominators containing $\prod_{k \le d}(4^k - 1)$ point to such a formula. It is not needed for the results of this volume and is left open.

7.3 Minimal number of samples

Theorem 7.8 ($d+1$ samples are necessary and sufficient within the mode model)**.** *Let $s \ge 2$.*

(i) *The row of order d returns $\operatorname{up}(x)$ exactly from any $d+1$ consecutive samples at levels $\ge s$ (Theorem 8.7), and any row of order $D \ge d$ does the same from $D+1$ samples (Theorem 8.8).*

(ii) *The sequence $(q_n)_{n \ge s}$ has exact degree d as a polynomial in Q^n, and for any $k \le d$ levels $n_0 < \cdots < n_{k-1}$ (all $\ge s$) there are two sequences of the model class $\mathcal{M}_d = \left\{ L + \sum_{r=1}^{d} A_r Q^{rn} \right\}$ that agree at these levels and have different constants L. Hence no rule, linear or not, that reads only $k \le d$ samples returns the constant on all of $\mathcal{M}_d$, and d cannot be replaced by $d - 1$ for the up-function sequence at x.*

Proof. (i) is quoted. (ii): the exact degree is Theorem 7.1; the two sequences and the conclusion are Theorem 9.5, whose hypothesis $C_d(x) \ne 0$ is Theorem 7.1. ☐

Remark 7.9 (Outside the model class). The theorem is a statement about rules that see only the samples. A rule that also knows x may exploit relations among the coefficients, for instance the fact that the top mode is determined up to sign by s (Equation (73)), and S1's remark that fewer samples "can succeed only by using additional point-specific information" is exactly this caveat. Within the volume no such rule is used: the extraction is universal in x and in the window.

7.4 Relation to the Exponents volume: exact versus asymptotic

The Exponents volume [9] treats the same finite splines, under the bridge $\operatorname{up}_n = p_{n+1}$ of Equation (22), at a *general* point. Three of its statements are relevant here, and each becomes exact at a dyadic point.

Proposition 7.10 (The sharp C^r law is attained at dyadic points)**.** *Let $r \ge 0$ and let $x \in (-1, 1)$ be dyadic of depth $s = r + 2$. Then for every $n \ge s$,*

$$\operatorname{up}_n^{(r)}(x) - \operatorname{up}^{(r)}(x) = -\frac{a_1}{4^{n+1}} \operatorname{up}^{(r+2)}(x) = -\varepsilon_k \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^{n+1}}, \tag{82}$$

with k the Thue–Morse index of Equation (61). The right-hand side has absolute value $2^{\binom{r+3}{2}-1}/(9 \cdot 4^N)$ with $N = n+1$, which is the value of $\left\| p_N^{(r)} - \operatorname{up}^{(r)} \right\|_\infty$ in the sharp ordinary-prefix law of the Exponents volume (its `p3:thm:sharp-prefix-law`, valid for $N \ge r + 3$): the supremum in that law is attained at every dyadic point of depth $r + 2$, for every level from the onset on, and the one-mode law Equation (82) is the exact form of the extremal error there.

Proof. Theorem 6.2 with $s = r + 2$ has $d_r = 1$ and a single mode,

$$\operatorname{up}_n^{(r)}(x) - \operatorname{up}^{(r)}(x) = -a_1 4^{-(n+1)} \operatorname{up}^{(r+2)}(x);$$

here $a_1 = \frac{1}{18}$ and, by Theorem 5.4(ii), $\operatorname{up}^{(s)}(x) = 2^{\binom{s+1}{2}} \varepsilon_k$ with $s + 1 = r + 3$. Then $2^{\binom{r+3}{2}}/18 = 2^{\binom{r+3}{2}-1}/9$. The hypothesis $N \ge r + 3$ of the cited law is $n \ge r + 2 = s$ under the bridge, which is exactly the onset. ☐

Proposition 7.11 (The exact CDF law is the depth-two density law)**.** *For every $n \ge 0$ and every real x,*

$$\mathrm{up}_{n+1}(x) = \int_{2x-1}^{2x+1} \mathrm{up}_n(t)\ \mathrm{d}t, \qquad \mathrm{up}(x) = \int_{2x-1}^{2x+1} \mathrm{up}(t)\ \mathrm{d}t. \tag{83}$$

Consequently, with $P_n(x) = \int_{-\infty}^{x} \mathrm{up}_n$ and $P(x) = \int_{-\infty}^{x} \mathrm{up}$,

$$P_n(\tfrac{1}{2}) = \mathrm{up}_{n+1}(\tfrac{1}{4}), \qquad P(\tfrac{1}{2}) = \mathrm{up}(\tfrac{1}{4}) = \tfrac{67}{72}, \qquad P_n(\tfrac{1}{2}) - P(\tfrac{1}{2}) = \frac{1}{9\cdot 4^{n+1}} \quad (n \ge 1). \tag{84}$$

The last identity is the exact CDF law $1/(9\cdot 4^N)$ of the Exponents volume (its `p3:cor:kolmogorov`, at its extremizer $x = \frac{1}{2}$) under $N = n+1$: it is the case $s = 2$, $d = 1$ of Theorem 1.1 at $x = \frac{1}{4}$, transported to the distribution function by Equation (83).

Proof. On the Fourier side, $\Phi_{n+1}(\xi) = \prod_{k=0}^{n+1} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^k) = \mathrm{sinc}_{\mathrm{rad}}(\pi\xi) \prod_{j=0}^{n} \mathrm{sinc}_{\mathrm{rad}}\big(\pi(\xi/2)/2^j\big) = \mathrm{sinc}_{\mathrm{rad}}(\pi\xi)\,\Phi_n(\xi/2)$. The inverse transform of $\mathrm{sinc}_{\mathrm{rad}}(\pi\xi)$ is the box $\mathbf{1}_{[-1/2,1/2]}$ (Theorem 2.2) and that of $\Phi_n(\xi/2)$ is $2\,\mathrm{up}_n(2\,\cdot)$, so $\mathrm{up}_{n+1}(x) = \int_{x-1/2}^{x+1/2} 2\,\mathrm{up}_n(2t)\ \mathrm{d}t = \int_{2x-1}^{2x+1} \mathrm{up}_n(u)\ \mathrm{d}u$. Letting $n \to \infty$ (or integrating Rvachev's equation) gives the second identity. For Equation (84), evenness and total mass 1 give $P_n(\frac{1}{2}) = 1 - P_n(-\frac{1}{2}) = \int_{-1/2}^{1} \mathrm{up}_n = \int_{-1/2}^{3/2} \mathrm{up}_n = \mathrm{up}_{n+1}(\frac{1}{4})$, using the support $[-1,1]$; likewise for P. Then Theorem 1.1 at $x = \frac{1}{4}$, where $s = 2$, $d = 1$, $C_1(\frac{1}{4}) = \frac{1}{9}$ (Theorem 1.6), gives $\mathrm{up}_{n+1}(\frac{1}{4}) - \mathrm{up}(\frac{1}{4}) = \frac{1}{9}\cdot 4^{-(n+1)}$ for $n+1 \ge 2$. ☐

Remark 7.12 (Two laws at the point $\frac{1}{2}$). At $x = \frac{1}{2}$ itself the density law has no mode: $s(\frac{1}{2}) = 1$, and $\mathrm{up}_n(\frac{1}{2}) = \mathrm{up}(\frac{1}{2}) = \frac{1}{2}$ for every $n \ge 1$ (Theorem 3.5). The CDF discrepancy $1/(9\cdot 4^{n+1})$ at the same point is not a contradiction: the distribution function at $\frac{1}{2}$ is the density at $\frac{1}{4}$, one level up, and $\frac{1}{4}$ has depth 2. In the Lean corpus the CDF form of this one-mode law at the quarter point is kernel-verified (`Fabius.fabiusUniformSpline_quarter_eq_quadratic`; see Section 2.5 and the register in Section 14).

Corollary 7.13 (Richardson acceleration is exact at dyadic points)**.** *In the notation of the Exponents volume, let $R_{n,M} = \sum_{j=0}^{M-1} w_{M,j}\, p_{n+j}$ be its signed geometric Richardson combination with the quarter-base Lagrange weights $w_{M,j}$ (its `p3:eq:richardson-combination` and `p3:eq:richardson-weights`). Let x be dyadic of depth s and $d = \lfloor s/2 \rfloor$. Then for every $n \ge s+1$ and every $M \ge d+1$,*

$$R_{n,M}(x) = \mathrm{up}(x) \tag{85}$$

exactly: the leading term $-a_M q^{\binom{M}{2}} \delta_n^{2M}\, \mathrm{up}^{(2M)}(x)$ and the $\mathcal{O}_{n\to\infty}\big(\delta_n^{2M+2}\big)$ remainder of its asymptotic expansion (its `p3:eq:richardson-leading`) both vanish identically, because $\mathrm{up}^{(2M)}(x) = 0$ for $2M > s$. The same holds for the derivative combinations $R_{n,M}^{(r)}$ with $M \ge \lfloor (s-r)/2 \rfloor + 1$.

Proof. Under the bridge $p_{n+j} = \mathrm{up}_{n+j-1}$, and $w_{M,j} = w_j^{(M-1)}$ is the weight of Theorem 8.3 of order $M-1 \ge d$; the levels $n-1, \dots, n+M-2$ are all $\ge s$. Theorem 8.8 gives the value law, and Theorem 6.2 with the same row gives the derivative law. ☐

Remark 7.14 (Away from dyadic points). At a non-dyadic x two things fail at once: the window $[x - \delta_n, x + \delta_n]$ contains a knot of up_n for every n, so no single polynomial describes the convolution neighbourhood, and the jet of up at x does not terminate. The all-orders expansion of the Exponents volume and the asymptotic Richardson acceleration remain valid, and its sharp prefix law is the general statement; the finite exact law is an arithmetic property of the dyadic points. Between the two lie the rationals with odd denominator and the other non-dyadic points, at which the finite splines are still polynomial pieces but the tail sees a knot; nothing in this volume applies there.

Section summary

Every one of the d modes is present for $s \geq 2$ (Theorem 7.1). At the last level before the onset the defect is known in closed form, $\Delta_s(x) = -\varepsilon_k 2^{-\binom{s}{2}}\beta_s$ with $\beta_s = B_s/s!$ for even s and 0 for odd s (Theorem 7.4), which proves that the onset $n \geq s$ is exact at even depth and reproves the one-level improvement at odd depth (Theorem 7.6); the failure at $n = s-2$ for odd s is verified through depth 7 and left open. Within the mode model $d+1$ samples are necessary and sufficient (Theorem 7.8). The sharp C^r law of the Exponents volume is attained at every dyadic point of depth $r+2$ (Theorem 7.10), its exact CDF law at $\frac{1}{2}$ is the $s=2$ density law at $\frac{1}{4}$ (Theorem 7.11), and its signed Richardson combination is exact, not asymptotic, at dyadic points (Theorem 7.13).

8 The quarter-base Gaussian-binomial extraction row

The preceding part of this volume proves that the finite spline values at a dyadic point obey an *exact* finite geometric law once the level has reached the depth of the point. From here on that law is the only input: everything in the present three sections is finite-dimensional linear algebra over the coefficient field $\mathbb{Q} = \mathbb{Q}$, and every identity is an equality of rational numbers. We first fix the standing data.

Standing hypotheses and data. Throughout, the Fourier transform is $\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi\mathrm{i}x\xi}\,\mathrm{d}x$, $\Phi_n(\xi) = \prod_{k=0}^{n}\operatorname{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$ has $n+1$ factors, $\mathrm{up}_n = \mathcal{F}_{2\pi}^{-1}[\Phi_n]$, and $\mathrm{up} = \mathcal{F}_{2\pi}^{-1}[\Phi]$ with $\Phi = \prod_{k\geq 0}\operatorname{sinc}_{\mathrm{rad}}(\pi\xi/2^k)$. Fix a dyadic point $x \in [-1,1]$ of depth $s = s(x)$ (the least $s \geq 0$ with $2^s x \in \mathbb{Z}$), put

$$d := \lfloor s/2 \rfloor, \qquad Q := \tfrac{1}{4}, \qquad q_n := \mathrm{up}_n(x) \quad (n \geq 0), \tag{86}$$

and recall the exact eventual geometric law, Theorem 1.1, proved in Section 6: for every $n \geq s$,

$$q_n = \mathrm{up}(x) + \sum_{r=1}^{d} C_r(x)\, Q^{rn}, \qquad C_r(x) = (-1)^r a_r\, 4^{-r}\, \mathrm{up}^{(2r)}(x), \tag{87}$$

where $a_r \in \mathbb{Q}_{>0}$ are the reciprocal-product coefficients, $1/\Phi(z) = \sum_{m\geq 0} a_m (2\pi z)^{2m}$. For $d = 0$ (that is, $s \in \{0,1\}$, $x \in \{0, \pm\frac{1}{2}, \pm 1\}$) the sum is empty and Equation (87) says $q_n = \mathrm{up}(x)$ for all $n \geq s$. Nothing below uses the specific values of the $C_r(x)$; only the shape of Equation (87) matters. Consequently every statement is proved for an arbitrary sequence of the form $q_n = L + \sum_{r=1}^{d} A_r Q^{rn}$ with L, A_r in a field, and the up-function enters only through Equation (87). Where the base $Q = 1/4$ is inessential we write a general q and say what is needed of it.

8.1 Geometric Lagrange interpolation at the nodes $1, Q, \ldots, Q^d$

Fix a starting level $N \geq s$ and rewrite Equation (87) at the $d+1$ consecutive levels $N, N+1, \ldots, N+d$:

$$q_{N+j} = \sum_{r=0}^{d} B_r \left(Q^j\right)^r, \qquad B_0 = \mathrm{up}(x), \quad B_r = C_r(x)\, Q^{rN}\ (1 \leq r \leq d), \qquad 0 \leq j \leq d. \tag{88}$$

Thus the samples are the values of the polynomial $P_N(z) := \sum_{r=0}^{d} B_r z^r$, of degree at most d, at the geometric nodes $z = 1, Q, Q^2, \ldots, Q^d$, and the number wanted is $P_N(0) = B_0 = \mathrm{up}(x)$. Extrapolating a degree-d interpolant from $d+1$ nodes to the origin is Lagrange interpolation; the whole content of this section is that at geometric nodes the Lagrange weights are q-Pochhammer quotients.

Definition 8.1 (Geometric Lagrange weights)**.** Let K be a field and $q \in K$ with $q \neq 0$ and $(q;q)_d = \prod_{h=1}^{d}(1-q^h) \neq 0$; equivalently, the $d+1$ nodes $q^0, q^1, \ldots, q^d$ are pairwise distinct. For $0 \leq j \leq d$ let

$$\ell_j^{(d)}(z) := \prod_{\substack{0\leq k\leq d\\ k\neq j}} \frac{z-q^k}{q^j-q^k}$$

be the Lagrange basis polynomial of the node q^j, and define the *geometric Lagrange weight*

$$w_j^{(d)} := \ell_j^{(d)}(0) = \prod_{\substack{0\le k\le d\\ k\ne j}} \frac{-q^k}{q^j - q^k}. \tag{89}$$

When $K = \mathbb{Q}$ and $q = Q = 1/4$ we call $(w_0^{(d)}, \dots, w_d^{(d)})$ the *quarter-base row of order* d. For $d = 0$ the product is empty and $w_0^{(0)} = 1$.

The equivalence claimed in the definition is elementary: $q^i = q^j$ with $i < j \le d$ and $q \ne 0$ holds exactly when $q^{j-i} = 1$ with $1 \le j - i \le d$, that is, exactly when some factor of $(q;q)_d$ vanishes. For $q = 1/4$ the condition holds for every d.

Lemma 8.2 (Moment identities of a geometric Lagrange row)**.** *Under the hypotheses of Theorem 8.1:*

1. *For every polynomial $p \in K[z]$ of degree at most d, $\sum_{j=0}^d w_j^{(d)}\, p(q^j) = p(0)$. In particular*

$$\sum_{j=0}^{d} w_j^{(d)}\, q^{jm} = \delta_{m0} \qquad (0 \le m \le d). \tag{90}$$

2. *The row $(w_0^{(d)}, \dots, w_d^{(d)})$ is the* only *vector in K^{d+1} satisfying Equation (90).*
3. *For every $r \ge 0$, writing h_r for the complete homogeneous symmetric polynomial of degree r,*

$$\sum_{j=0}^{d} w_j^{(d)}\, q^{j(d+1+r)} = (-1)^d q^{\binom{d+1}{2}}\, h_r(1, q, \dots, q^d). \tag{91}$$

In particular the first uncancelled moment is $\sum_j w_j^{(d)} q^{j(d+1)} = (-1)^d q^{\binom{d+1}{2}}$.

Proof. 1: the polynomial $p - \sum_j p(q^j)\ell_j^{(d)}$ has degree at most d and vanishes at the $d+1$ distinct nodes $q^0, \dots, q^d$ (because $\ell_j^{(d)}(q^k) = \delta_{jk}$), hence is identically zero; evaluate at $z = 0$. Taking $p = z^m$ gives Equation (90).

2: the system Equation (90) is $V^{\mathsf T} w = e_0$ with $V = (q^{jm})_{0\le j,m\le d}$ the Vandermonde matrix of the distinct nodes, and $\det V = \prod_{0\le i<k\le d}(q^k - q^i) \ne 0$.

3: put $\omega(z) = \prod_{k=0}^d (z - q^k)$ and divide with remainder, $z^{d+1+r} = \omega(z)h(z) + \rho(z)$ with $\deg\rho \le d$. Since $\omega(q^j) = 0$, $\rho(q^j) = q^{j(d+1+r)}$, so by 1 $\sum_j w_j^{(d)} q^{j(d+1+r)} = \rho(0) = -\omega(0)h(0)$. Here $\omega(0) = \prod_{k=0}^d(-q^k) = (-1)^{d+1} q^{\binom{d+1}{2}}$. For $h(0)$, expand at $z = \infty$:

$$\frac{z^{d+1+r}}{\omega(z)} = z^r \prod_{k=0}^{d} \left(1 - q^k z^{-1}\right)^{-1} = z^r \sum_{i\ge 0} h_i(1, q, \dots, q^d) z^{-i},$$

because $\prod_k (1 - x_k t)^{-1} = \sum_i h_i(x) t^i$ is the generating function of the complete homogeneous symmetric polynomials. Since $\rho/\omega = O(z^{-1})$, the quotient h is the polynomial part of this Laurent series, $h(z) = \sum_{i=0}^r h_i z^{r-i}$, whose constant term is h_r. Therefore the moment equals $(-1)^d q^{\binom{d+1}{2}} h_r$. $\square$

Theorem 8.3 (Closed form of the geometric Lagrange weights)**.** *Under the hypotheses of Theorem 8.1, for $0 \le j \le d$,*

$$\boxed{w_j^{(d)} = \frac{(-1)^{d-j}\, q^{\binom{d-j+1}{2}}}{(q;q)_j\ (q;q)_{d-j}} = \frac{(-1)^{d-j}\, q^{\binom{d-j+1}{2}}}{(q;q)_d} \begin{bmatrix} d \\ j \end{bmatrix}_q.} \tag{92}$$

Proof. Split the product Equation (89) into its numerator and the factors with $k < j$ and $k > j$. The numerator is

$$\prod_{\substack{0\le k\le d\\ k\ne j}} (-q^k) = (-1)^d q^{\binom{d+1}{2}-j}.$$

For $k < j$ write $q^j - q^k = -q^k(1-q^{j-k})$; the j factors give $(-1)^j q^{\binom{j}{2}} \prod_{h=1}^{j}(1-q^h) = (-1)^j q^{\binom{j}{2}} (q;q)_j$. For $k > j$ write $q^j - q^k = q^j(1-q^{k-j})$; the $d-j$ factors give $q^{j(d-j)} (q;q)_{d-j}$. Hence

$$w_j^{(d)} = \frac{(-1)^d q^{\binom{d+1}{2}-j}}{(-1)^j q^{\binom{j}{2}+j(d-j)} (q;q)_j (q;q)_{d-j}},$$

and the exponent is $\binom{d+1}{2} - j - \binom{j}{2} - j(d-j) = \frac{1}{2}\left(d^2 + d - j - 2dj + j^2\right) = \frac{1}{2}(d-j)(d-j+1) = \binom{d-j+1}{2}$. This is the first form. The second follows from $\begin{bmatrix} d \\ j \end{bmatrix}_q = (q;q)_d / ((q;q)_j (q;q)_{d-j})$, which is the catalogue's definition and is legitimate because $(q;q)_d \ne 0$. □

The row has a compact generating polynomial, from which the moment identities can be read off a second time. We first record the finite q-binomial theorem with its short proof, since the same identity is used four times below (row polynomial, integer form, annihilating recurrence, Lebesgue factor).

Lemma 8.4 (Finite q-binomial theorem). *In any commutative ring, for $n \ge 0$,*

$$(z;q)_n = \prod_{h=0}^{n-1}(1 - zq^h) = \sum_{k=0}^{n}(-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \tag{93}$$

where $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the Gaussian polynomial (the catalogue's polynomial-in-q reading). In particular $\begin{bmatrix} n \\ k \end{bmatrix}_q$ has non-negative integer coefficients, so it takes integer values at every integer q.

Proof. The Gaussian polynomials satisfy the Pascal recurrence $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$ for $1 \le k \le n$: in the quotient form, both terms on the right have the common denominator $(q;q)_k (q;q)_{n-k}$ after multiplying the first by $(1-q^{n-k})$ and the second by $(1-q^k)$, and $(1-q^{n-k}) + q^{n-k}(1-q^k) = 1 - q^n$; so the sum is $(q;q)_{n-1}(1-q^n)/((q;q)_k (q;q)_{n-k}) = \begin{bmatrix} n \\ k \end{bmatrix}_q$. Together with $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = \begin{bmatrix} n \\ n \end{bmatrix}_q = 1$ the recurrence shows by induction on n that the Gaussian polynomials lie in $\mathbb{Z}_{\ge 0}[q]$.

Now induct on n; the case $n = 0$ reads $1 = 1$. Multiplying the identity for $n-1$ by $(1 - zq^{n-1})$ gives, as coefficient of z^k,

$$(-1)^k \left(q^{\binom{k}{2}} \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{\binom{k-1}{2}+n-1} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q \right) = (-1)^k q^{\binom{k}{2}} \left(\begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q \right),$$

because $\binom{k-1}{2} + n - 1 - \binom{k}{2} = n - k$. The bracket is $\begin{bmatrix} n \\ k \end{bmatrix}_q$ by the Pascal recurrence. □

Proposition 8.5 (Row polynomial and mode annihilation). *Under the hypotheses of Theorem 8.1, the generating polynomial of the row and its reverse are*

$$W_d(z) := \sum_{j=0}^{d} w_j^{(d)} z^j = \prod_{r=1}^{d} \frac{z - q^r}{1 - q^r} = \frac{z^d\, (q/z;q)_d}{(q;q)_d}, \qquad \sum_{j=0}^{d} w_{d-j}^{(d)} z^j = \frac{(qz;q)_d}{(q;q)_d} = \prod_{r=1}^{d} \frac{1 - q^r z}{1 - q^r}. \tag{94}$$

Consequently $W_d(1) = 1$ and $W_d(q^r) = 0$ for $1 \le r \le d$: the roots of the row polynomial are exactly the ratios $q, q^2, \ldots, q^d$ of the decaying modes, and the roots of the reversed polynomial are their inverses.

Proof. By Theorem 8.4 with z replaced by q/z,

$$\prod_{r=1}^{d}(z-q^r) = z^d\prod_{h=0}^{d-1}\Bigl(1-\frac{q}{z}q^h\Bigr) = \sum_{k=0}^{d}(-1)^k q^{\binom{k}{2}+k}\begin{bmatrix}d\\k\end{bmatrix}_q z^{d-k} = \sum_{j=0}^{d}(-1)^{d-j}q^{\binom{d-j+1}{2}}\begin{bmatrix}d\\j\end{bmatrix}_q z^j,$$

using $\binom{k}{2}+k=\binom{k+1}{2}$, $j=d-k$ and the symmetry $\begin{bmatrix}d\\d-j\end{bmatrix}_q=\begin{bmatrix}d\\j\end{bmatrix}_q$. Dividing by $\prod_{r=1}^{d}(1-q^r)=(q;q)_d$ and comparing with Equation (92) gives the first display. Reversing, with $z\mapsto qz$ in Equation (93), $\sum_j w_{d-j}^{(d)}z^j = (q;q)_d^{-1}\sum_k(-1)^k q^{\binom{k}{2}}\begin{bmatrix}d\\k\end{bmatrix}_q(qz)^k = (qz;q)_d\,/\,(q;q)_d$. The values $W_d(1)=1$ and $W_d(q^r)=0$ are read off the product form. □

Remark 8.6 (Two proofs of the moment identities). $W_d(q^m)=\sum_j w_j^{(d)}q^{jm}$, so the root statement of Theorem 8.5 *is* Equation (90): once the closed form Equation (92) is known, the finite q-binomial theorem proves the moment identities without any interpolation. This is the route of S6 and of S2's appendix; the interpolation route of Theorem 8.2 has the advantage of giving the uncancelled moments Equation (91) as well, which the row polynomial does not see. In operator language, with E the forward shift $(\mathsf{E}\,q)_n=q_{n+1}$, the identity $\sum_j w_j^{(d)}q_{N+j}=\bigl(W_d(\mathsf{E})q\bigr)_N$ and the factorization $W_d(\mathsf{E})=\prod_{r=1}^{d}(\mathsf{E}-q^r)/(1-q^r)$ say that the row is the product of the elementary filters $(\mathsf{E}-q^r)/(1-q^r)$, each of which kills one mode and preserves constants; this is the form in which the Romberg tableau of Section 9 evaluates the row.

8.2 The extraction theorem

For $Q=1/4$ it is convenient to have the finite products as integers. Put

$$P_k := \prod_{r=1}^{k}(4^r-1) \qquad (P_0=1,\ P_1=3,\ P_2=45,\ P_3=2835,\ P_4=722925,\ P_5=739552275), \tag{95}$$

so that $(Q;Q)_k=\prod_{r=1}^{k}(1-4^{-r})=4^{-\binom{k+1}{2}}P_k$. Every P_k is odd.

Theorem 8.7 (Single Gaussian-binomial extraction formula). *Let $x\in[-1,1]$ be dyadic of depth $s=s(x)$, let $d=\lfloor s/2\rfloor$, $Q=1/4$, and let $N\geq s$ be any integer. Then, with $q_n=\mathrm{up}_n(x)$ in the $(n+1)$-factor convention,*

$$\boxed{\mathrm{up}(x)=\sum_{j=0}^{d}\frac{(-1)^{d-j}\,Q^{\binom{d-j+1}{2}}}{(Q;Q)_j\,(Q;Q)_{d-j}}\,q_{N+j}=\frac{1}{(Q;Q)_d}\sum_{j=0}^{d}(-1)^{d-j}\,Q^{\binom{d-j+1}{2}}\begin{bmatrix}d\\j\end{bmatrix}_Q\mathrm{up}_{N+j}(x).} \tag{96}$$

The same row can be written in any of the following equivalent ways:

$$\mathrm{up}(x)=\frac{1}{(Q;Q)_d}\sum_{k=0}^{d}(-1)^k\,Q^{\binom{k+1}{2}}\begin{bmatrix}d\\k\end{bmatrix}_Q q_{N+d-k} \qquad \textit{(reversed order, S6)}, \tag{97}$$

$$\mathrm{up}(x)=\sum_{j=0}^{d}(-1)^{d-j}\frac{4^{\binom{j+1}{2}}}{P_j\,P_{d-j}}\,q_{N+j} \qquad \textit{(integer-product weights, S3)}, \tag{98}$$

$$\mathrm{up}(x)=\frac{1}{P_d}\sum_{j=0}^{d}(-1)^{d-j}\,4^{\binom{j+1}{2}}\begin{bmatrix}d\\j\end{bmatrix}_4 q_{N+j} \qquad \textit{(common odd denominator, S4)}. \tag{99}$$

In Equation (99) every numerator coefficient is an integer. All four displays are identities in $\mathbb{Q}$: with exact rational inputs $q_N,\ldots,q_{N+d}$ they return the exact rational number $\mathrm{up}(x)$, with no remainder and no limit.

Proof. By Equation (88) the samples are $q_{N+j} = P_N(Q^j)$ for a polynomial P_N of degree at most d with $P_N(0) = \operatorname{up}(x)$. The hypotheses of Theorem 8.1 hold for $q = Q = 1/4$ and every d (no positive power of $1/4$ equals 1), so Theorem 8.21 gives $\operatorname{up}(x) = P_N(0) = \sum_j w_j^{(d)} q_{N+j}$, and Theorem 8.3 gives both forms in Equation (96). Explicitly, with $\sum_j w_j^{(d)} Q^{jr} = \delta_{r0}$:

$$\sum_{j=0}^{d} w_j^{(d)} q_{N+j} = \sum_{r=0}^{d} B_r \sum_{j=0}^{d} w_j^{(d)} Q^{jr} = B_0 = \operatorname{up}(x).$$

Equation (97) is Equation (96) with the summation index $k = d - j$ and $\left[{d \atop d-k}\right]_Q = \left[{d \atop k}\right]_Q$; S6 writes the samples in the order $q_{N+d}, q_{N+d-1}, \dots, q_N$, which is the same formula.

Equation (98): substitute $(Q;Q)_k = 4^{-\binom{k+1}{2}} P_k$ into the first form of Equation (96); the power of 4 in the j-th weight becomes $4^{-\binom{d-j+1}{2}+\binom{j+1}{2}+\binom{d-j+1}{2}} = 4^{\binom{j+1}{2}}$.

Equation (99): the inversion identity $\left[{d \atop j}\right]_{q^{-1}} = q^{-j(d-j)} \left[{d \atop j}\right]_q$ holds because

$$\begin{bmatrix} d \\ j \end{bmatrix}_{q^{-1}} = \prod_{i=1}^{j} \frac{1 - q^{-(d-j+i)}}{1 - q^{-i}} = \prod_{i=1}^{j} \frac{q^{-(d-j+i)}\left(q^{d-j+i} - 1\right)}{q^{-i}\left(q^i - 1\right)} = q^{-j(d-j)} \begin{bmatrix} d \\ j \end{bmatrix}_q.$$

With $q = 4$ this gives $\left[{d \atop j}\right]_Q = 4^{-j(d-j)} \left[{d \atop j}\right]_4$, and $(Q;Q)_d = 4^{-\binom{d+1}{2}} P_d$, so the j-th weight in the second form of Equation (96) is

$$(-1)^{d-j}\, 4^{\binom{d+1}{2}-\binom{d-j+1}{2}-j(d-j)}\, \frac{\left[{d \atop j}\right]_4}{P_d},$$

where

$$\binom{d+1}{2} - \binom{d-j+1}{2} - j(d-j) = \tfrac{1}{2}\left(2dj - j^2 + j\right) - \left(dj - j^2\right) = \binom{j+1}{2}.$$

The numerators are integers by Theorem 8.4, and P_d is a product of odd numbers. ☐

Exact, not asymptotic: the same row as in the Exponents volume

The weights Equation (92) at $q = 1/4$ are exactly the signed geometric Richardson weights $w_{M,j}$ of the Exponents volume [9], Part III (labels `p3:eq:richardson-weights`, `p3:eq:richardson-moments`), with $M = d + 1$ weights on the nodes q^j, $0 \le j < M$; under the bridge $\operatorname{up}_n = p_{n+1}$ our row $\sum_{j=0}^{d} w_j^{(d)} \operatorname{up}_{N+j}$ is their combination $R_{N+1,d+1}$ (label `p3:eq:richardson-combination`). That volume proves, for *general* x, only the asymptotic statement $R_{n,M} = \operatorname{up} - a_M q^{\binom{M}{2}} \delta_n^{2M} \operatorname{up}^{(2M)} + O(\delta_n^{2M+2})$ (label `p3:eq:richardson-leading`); the sign and the factor $q^{\binom{M}{2}}$ of its leading term are the first uncancelled moment $(-1)^d q^{\binom{d+1}{2}}$ of Equation (91) applied to the mode $r = d + 1$. The contribution of the present volume is that at a dyadic point of depth s the mode expansion Equation (87) *terminates* at $r = d$ as soon as $n \ge s$, so the very same row is an exact identity (Theorem 8.7), not an acceleration with a remainder. Do not read the $O(\delta_n^{2M+2})$ of the Exponents volume as a bound on an error of Equation (96): at a dyadic point of depth s and $N \ge s$ that error is zero.

Corollary 8.8 (Over-ordering, other windows, other level sets). *In the situation of Theorem 8.7:*

1. *For every $M \ge d$ and every $N \ge s$, $\operatorname{up}(x) = \sum_{j=0}^{M} w_j^{(M)} q_{N+j}$: a row of higher order than necessary is still exact.*
2. *Any two windows $N, N' \ge s$ (of orders $M, M' \ge d$) return the same rational number.*

3. *For distinct offsets* $0 \le t_0 < t_1 < \cdots < t_d$ *and* $N \ge s$,

$$\operatorname{up}(x) = \sum_{j=0}^{d} q_{N+t_j} \prod_{\substack{0 \le k \le d \\ k \ne j}} \frac{-Q^{t_k}}{Q^{t_j} - Q^{t_k}}; \tag{100}$$

for the stride $t_j = \sigma j$ *this is* Equation (96) *with* Q^σ *in place of* Q.

Proof. (1) The samples q_{N+j}, $0 \le j \le M$, are still values of the polynomial P_N of degree at most $d \le M$ at the nodes Q^j, and Theorem 8.21 applies with M in place of d. (2) Both windows return $\operatorname{up}(x)$. (3) The values $q_{N+t_j} = P_N(Q^{t_j})$ sit at the distinct nodes Q^{t_j}, and the displayed product is the Lagrange weight of the node Q^{t_j} for evaluation at 0; for $t_j = \sigma j$ the nodes are the powers of Q^σ, which satisfy Theorem 8.1, and Theorem 8.3 applies with $q = Q^\sigma$. Only in the consecutive case does the common factor Q^N cancel to a q-Pochhammer quotient. □

Corollary 8.9 (One completely explicit finite formula). *Insert into* Equation (96) *the Thue–Morse truncated-power representation of the finite spline,* Theorem 2.5, *which in the present indexing reads, for* $n \ge 0$ *and* $x \in [-1, 1]$,

$$\operatorname{up}_n(x) = \frac{2^{\binom{n+1}{2}}}{n!} \sum_{k=0}^{2^{n+1}-1} \varepsilon_k \left(x + 1 - 2^{-(n+1)} - \frac{k}{2^n}\right)_+^n,$$

with $\varepsilon_k = (-1)^{(\text{binary digit sum of } k)}$ *and* $t_+^n = \max(t, 0)^n$. *Then, for every* $N \ge s$,

$$\operatorname{up}(x) = \sum_{j=0}^{d} \frac{(-1)^{d-j}\, 4^{\binom{j+1}{2}}}{P_j P_{d-j}} \frac{2^{\binom{N+j+1}{2}}}{(N+j)!} \sum_{k=0}^{2^{N+j+1}-1} \varepsilon_k \left(x + 1 - 2^{-(N+j+1)} - \frac{k}{2^{N+j}}\right)_+^{N+j}. \tag{101}$$

Every sum and product is finite, every term is rational, and $N = s$ *gives the shortest guaranteed instance, with exactly* $d + 1$ *finite splines.*

Proof. Substitute the representation of $\operatorname{up}_{N+j}(x)$ into Equation (98). No passage to a limit remains. □

8.3 The first rows, verified by hand

Example 8.10 (The rows of order 1, 2, 3). With $Q = 1/4$: $(Q;Q)_1 = \frac{3}{4}$, $(Q;Q)_2 = \frac{3}{4} \cdot \frac{15}{16} = \frac{45}{64}$, $(Q;Q)_3 = \frac{45}{64} \cdot \frac{63}{64} = \frac{2835}{4096}$. Evaluating the first form of Equation (92):

$$d = 1: \quad w_0^{(1)} = \frac{-Q^1}{(Q;Q)_1} = -\frac{1/4}{3/4} = -\frac{1}{3}, \qquad w_1^{(1)} = \frac{1}{(Q;Q)_1} = \frac{4}{3};$$

$$d = 2: \quad w_0^{(2)} = \frac{Q^3}{(Q;Q)_2} = \frac{1/64}{45/64} = \frac{1}{45}, \qquad w_1^{(2)} = \frac{-Q}{(Q;Q)_1^2} = -\frac{1/4}{9/16} = -\frac{4}{9}, \qquad w_2^{(2)} = \frac{1}{(Q;Q)_2} = \frac{64}{45};$$

$$d = 3: \quad w_0^{(3)} = \frac{-Q^6}{(Q;Q)_3} = -\frac{1/4096}{2835/4096} = -\frac{1}{2835},$$

$$w_3^{(3)} = \frac{1}{(Q;Q)_3} = \frac{4096}{2835},$$

$$w_1^{(3)} = \frac{Q^3}{(Q;Q)_1\,(Q;Q)_2} = \frac{1/64}{135/256} = \frac{4}{135},$$

$$w_2^{(3)} = \frac{-Q}{(Q;Q)_2\,(Q;Q)_1} = -\frac{1/4}{135/256} = -\frac{64}{135}.$$

The rows sum to 1: $-\frac{1}{3}+\frac{4}{3}=1$; $\frac{1}{45}-\frac{20}{45}+\frac{64}{45}=1$; $\frac{-1+84-1344+4096}{2835}=\frac{2835}{2835}=1$. The moments $1\le m\le d$ vanish: $-\frac{1}{3}+\frac{4}{3}\cdot\frac{1}{4}=0$; $\frac{1}{45}-\frac{4}{9}\cdot\frac{1}{4}+\frac{64}{45}\cdot\frac{1}{16}=\frac{1-5+4}{45}=0$; and for $d=3$

$$\begin{aligned} m=1: &\quad \frac{-1+84\cdot 4^{-1}-1344\cdot 4^{-2}+4096\cdot 4^{-3}}{2835}=\frac{-1+21-84+64}{2835}=0,\\ m=2: &\quad \frac{-1+84\cdot 4^{-2}-1344\cdot 4^{-4}+4096\cdot 4^{-6}}{2835}=\frac{-1+\frac{21}{4}-\frac{21}{4}+1}{2835}=0,\\ m=3: &\quad \frac{-1+84\cdot 4^{-3}-1344\cdot 4^{-6}+4096\cdot 4^{-9}}{2835}=\frac{-1+\frac{21}{16}-\frac{21}{64}+\frac{1}{64}}{2835}\\ &\quad =\frac{(-64+84-21+1)/64}{2835}=0. \end{aligned}$$

The common-denominator form Equation (99) for $d=3$ uses $\begin{bmatrix}3\\0\end{bmatrix}_4=\begin{bmatrix}3\\3\end{bmatrix}_4=1$ and $\begin{bmatrix}3\\1\end{bmatrix}_4=\begin{bmatrix}3\\2\end{bmatrix}_4=\frac{1-4^3}{1-4}=21$, giving the numerators $(-1,\ 4\cdot 21,\ -4^3\cdot 21,\ 4^6)=(-1,84,-1344,4096)$ over $P_3=2835$, in agreement with $\frac{4}{135}=\frac{84}{2835}$ and $\frac{64}{135}=\frac{1344}{2835}$. These vectors coincide with the weight vectors printed in the Exponents volume [9] after `p3:eq:richardson-leading` for $M=2,3,4$, namely $(-\frac{1}{3},\frac{4}{3})$, $(\frac{1}{45},-\frac{4}{9},\frac{64}{45})$ and $(-\frac{1}{2835},\frac{4}{135},-\frac{64}{135},\frac{4096}{2835})$; their M is our $d+1$.

Table 2: The quarter-base rows through order 5, in the common-denominator form Equation (99). The integer vector multiplies $(q_N,q_{N+1},\dots,q_{N+d})$ and the result is divided by P_d. Rows $d\le 5$ agree with the verification outputs of S1 (`m=0..5`) and S4 (`d=0..4`); the rightmost numerator is always $4^{\binom{d+1}{2}}$ and the leftmost is $(-1)^d$.

d	integer numerator vector	P_d
0	(1)	1
1	$(-1,\ 4)$	3
2	$(1,\ -20,\ 64)$	45
3	$(-1,\ 84,\ -1344,\ 4096)$	2835
4	$(1,\ -340,\ 22848,\ -348160,\ 1048576)$	722925
5	$(-1,\ 1364,\ -371008,\ 23744512,\ -357564416,\ 1073741824)$	739552275

The rows depend on x only through $d=\lfloor s(x)/2\rfloor$: one row serves every dyadic point of depths $2d$ and $2d+1$, and every admissible starting level $N\ge s$. The sample count therefore grows by one only every second time the depth grows: depths $0,1$ need one sample, depths $2,3$ need two, depths $4,5$ need three, and so on.

Remark 8.11 (Lean crosswalk). The row and its algebra are formalized in the repository [10]; in Lean the order d is the parameter p and the nodes are `q j`, `j <= p`.

- The weight of Theorem 8.1 is `Fabius.geometricLagrangeWeight` (the evaluation-at-zero Lagrange weight on the nodes $q^0,\dots,q^d$), with its literal product form `Fabius.geometricLagrangeWeight_eq_product` (`GeometricLagrangeWeights.lean`).
- Theorem 8.3 is `Fabius.geometricLagrangeWeight_eq_geometricQPochhammer` over any field with $q\ne 0$; at the quarter base, unconditionally, `Fabius.quarterLagrangeWeight_eq_qPochhammer` and `Fabius.quarterLagrangeWeight_eq_qBinomial` (`GeometricLagrangeQBinomial.lean`). The Gaussian form carries the explicit hypothesis $(q;q)_d\ne 0$ in its field-generic version, discharged at $q=1/4$ by `Fabius.quarter_qPochhammer_self_ne_zero`.
- The moment identities Equation (90) are `Fabius.sum_quarterLagrangeWeight_eq_one` and `Fabius.sum_quarterLagrangeWeight_mul_pow_eq_zero`; the uncancelled moments Equation (91) are `Fabius.sum_geometricLagrangeWeight_mul_pow_card_add` (with the complete homogeneous polynomial) and `Fabius.sum_quarterLagrangeWeight_firstUncancelled`.

- The row polynomial identity Equation (94) is `Fabius.quarterLagrangeWeightPolynomial_eq_forwardRichardson`, whose right side is the normalized root polynomial with roots $q, \ldots, q^d$.
- The filter's action on the *infinite* product is formalized as `Fabius.quarterLagrange_rvachevFourierProduct_eq_gaussian_tsum` (`RvachevQBinomialFilter.lean`): for the infinite sinc product Φ in the repository's normalization,

$$\sum_{j=0}^{d} w_j^{(d)}\, \Phi(2^{-j}z) = 1 + \sum_{r\geq 0}(-1)^d Q^{\binom{d+1}{2}} \begin{bmatrix} d+r \\ d \end{bmatrix}_Q \mathrm{m}_{d+1+r}(z),$$

where $\mathrm{m}_k(z)$ is the k-th even Taylor mode of Φ: the quarter-base row cancels the first d nonconstant even moments of the sinc product and multiplies every surviving mode by an explicit Gaussian coefficient. This is the Fourier-side counterpart of Equation (91); on the function side it is the asymptotic statement of the Exponents volume.

What is *not* formalized is Theorem 8.7 itself, because its input Equation (87) (the exact eventual geometric law at a dyadic point) has no Lean proof yet; once it has one, the theorem is the moment identities applied to a finite sum and requires no new analysis.

Section summary

Lagrange extrapolation from the geometric nodes $1, Q, \ldots, Q^d$ to the origin has the q-Pochhammer weights $w_j^{(d)} = (-1)^{d-j} Q^{\binom{d-j+1}{2}} / ((Q;Q)_j\,(Q;Q)_{d-j})$ (Theorem 8.3); they annihilate the moments Q^{jm}, $1 \leq m \leq d$, preserve constants, and multiply the mode $Q^{j(d+1)}$ by $(-1)^d Q^{\binom{d+1}{2}}$ (Theorem 8.2). Applied to $d+1$ consecutive finite-spline values at a dyadic point of depth $s \leq N$, the row returns $\mathrm{up}(x)$ exactly (Theorem 8.7), in four equivalent notations, with an odd common denominator $P_d = \prod_{r\leq d}(4^r - 1)$ and integer numerators $(-1)^{d-j} 4^{\binom{j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_4$.

9 Recovering every mode and the Romberg tableau

The extraction row is the first row of the inverse of a geometric Vandermonde matrix. This section writes down the whole inverse in closed form, so that the same $d+1$ samples return every mode coefficient $C_r(x)$ and hence every even derivative $\mathrm{up}^{(2r)}(x)$ that occurs in Equation (87); it then organizes the row as a triangular Romberg-type tableau, and turns the annihilating recurrence of the sequence into an exact test for the onset level.

9.1 The geometric Vandermonde system and its exact inverse

With $B_r = C_r(x)\, Q^{rN}$ as in Equation (88), the $d+1$ samples $q_N, \ldots, q_{N+d}$ satisfy

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & Q & Q^2 & \cdots & Q^d \\ 1 & Q^2 & Q^4 & \cdots & Q^{2d} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & Q^d & Q^{2d} & \cdots & Q^{d^2} \end{pmatrix}}_{V=(Q^{ij})_{0\leq i,j\leq d}} \begin{pmatrix} B_0 \\ B_1 \\ B_2 \\ \vdots \\ B_d \end{pmatrix} = \begin{pmatrix} q_N \\ q_{N+1} \\ q_{N+2} \\ \vdots \\ q_{N+d} \end{pmatrix}. \tag{102}$$

The matrix V is symmetric.

Lemma 9.1 (Determinant). *For any field element q with distinct powers $q^0, \ldots, q^d$,*

$$\det\left(q^{ij}\right)_{0\leq i,j\leq d} = \prod_{0\leq i<k\leq d} \left(q^k - q^i\right) = (-1)^{\binom{d+1}{2}}\, q^{\binom{d+1}{3}} \prod_{k=1}^{d} (q;q)_k \neq 0. \tag{103}$$

Proof. The first equality is the Vandermonde determinant at the nodes q^k. For fixed k, $\prod_{i<k}(q^k - q^i) = \prod_{i<k}(-q^i)(1-q^{k-i}) = (-1)^k q^{\binom{k}{2}} (q;q)_k$; multiply over $k = 0, \ldots, d$ and use $\sum_{k=0}^{d} k = \binom{d+1}{2}$ and $\sum_{k=0}^{d} \binom{k}{2} = \binom{d+1}{3}$. Each factor $(q;q)_k$ is nonzero by the distinctness of the powers. □

Theorem 9.2 (Closed inverse of the geometric Vandermonde system; mode projectors)**.** *Let $N \geq s$ and let $\ell_j^{(d)}$ be the Lagrange basis polynomials of Theorem 8.1 at $q = Q$. Put*

$$\begin{aligned} D_j &:= \prod_{\substack{0 \leq k \leq d \\ k \neq j}} (Q^j - Q^k) = (-1)^j \, Q^{\binom{j}{2} + j(d-j)} \, (Q;Q)_j \, (Q;Q)_{d-j}\,, \\ e_r^{(j)} &:= e_r\big(1, Q, \ldots, \widehat{Q^j}, \ldots, Q^d\big), \end{aligned} \tag{104}$$

where e_r is the elementary symmetric polynomial of degree r and the hat omits the node Q^j. Then:

1. *$P_N(z) = \sum_{j=0}^{d} q_{N+j} \, \ell_j^{(d)}(z)$, and for $0 \leq m \leq d$*

$$B_m = \sum_{j=0}^{d} q_{N+j} \, [z^m] \ell_j^{(d)}(z), \qquad \left(V^{-1}\right)_{mj} = [z^m]\ell_j^{(d)}(z) = [z^j]\ell_m^{(d)}(z) = \frac{(-1)^{d-m} \, e_{d-m}^{(j)}}{D_j}. \tag{105}$$

2. *The elementary symmetric functions of the punctured node set have the Gaussian expansion*

$$e_r^{(j)} = [t^r] \, \frac{(-t;Q)_{d+1}}{1 + tQ^j} = \sum_{h=0}^{r} (-1)^h \, Q^{jh + \binom{r-h}{2}} \begin{bmatrix} d+1 \\ r-h \end{bmatrix}_Q. \tag{106}$$

3. *(Mode projector.) With E the forward shift on the sample sequence, for $0 \leq m \leq d$,*

$$B_m = \frac{1}{D_m} \Big[\prod_{\substack{0 \leq k \leq d \\ k \neq m}} (\mathsf{E} - Q^k) \, q \Big]_N, \tag{107}$$

and the operator expands as

$$\Big[\prod_{\substack{0 \leq k \leq d \\ k \neq m}} (\mathsf{E} - Q^k) \, q \Big]_N = \sum_{a=0}^{m} \sum_{b=0}^{d-m} (-1)^{a+b} \, Q^{\binom{a}{2} + \binom{b}{2} + (m+1)b} \begin{bmatrix} m \\ a \end{bmatrix}_Q \begin{bmatrix} d-m \\ b \end{bmatrix}_Q q_{N+d-a-b}. \tag{108}$$

4. *Consequently every mode coefficient is a rational combination of the same $d+1$ samples,*

$$C_m(x) = Q^{-mN} B_m = 4^{mN} \sum_{j=0}^{d} q_{N+j} \, \frac{(-1)^{d-m} e_{d-m}^{(j)}}{D_j} \qquad (0 \leq m \leq d), \tag{109}$$

independent of the admissible window $N \geq s$; for $m = 0$ this is Equation (96).

Proof. 1: P_N has degree at most d and $P_N(Q^j) = q_{N+j}$, so $P_N = \sum_j q_{N+j} \ell_j^{(d)}$ as in the proof of Theorem 8.2; comparing the coefficient of z^m gives $B_m = \sum_j q_{N+j} [z^m] \ell_j^{(d)}$, that is $(V^{-1})_{mj} = [z^m]\ell_j^{(d)}$, because this expresses the solution of Equation (102) linearly in the right-hand side for every right-hand side. Since V is symmetric, so is V^{-1}, whence $[z^m]\ell_j^{(d)} = [z^j]\ell_m^{(d)}$. Finally $\ell_j^{(d)}(z) = D_j^{-1} \prod_{k \neq j}(z - Q^k)$ and, by Vieta's formulas, $\prod_{k \neq j}(z - Q^k) = \sum_{m=0}^{d} (-1)^{d-m} e_{d-m}^{(j)} z^m$. The closed form of D_j is the denominator computation in the proof of Theorem 8.3: the factors with $k < j$ give $(-1)^j Q^{\binom{j}{2}} (Q;Q)_j$, those with $k > j$ give $Q^{j(d-j)} (Q;Q)_{d-j}$.

2: $\prod_{k=0}^{d}(1+tQ^k) = (-t;Q)_{d+1} = \sum_{i=0}^{d+1} Q^{\binom{i}{2}} \left[{d+1 \atop i}\right]_Q t^i$ by Theorem 8.4 with $z=-t$, and $\prod_{k\neq j}(1+tQ^k) = (-t;Q)_{d+1}\cdot(1+tQ^j)^{-1}$ with $(1+tQ^j)^{-1} = \sum_{h\geq 0}(-1)^h Q^{jh} t^h$ as a formal power series. The coefficient of t^r in the left side is $e_r^{(j)}$; in the right side it is the displayed convolution.

3: put $u_i = q_{N+i} = \sum_r B_r (Q^r)^i$. The shift acts on the sequence $i \mapsto (Q^r)^i$ by the scalar Q^r, so $\left[\prod_{k\neq m}(\mathsf{E}-Q^k)u\right]_0 = \sum_{r=0}^{d} B_r \prod_{k\neq m}(Q^r - Q^k)$; for $r\neq m$ the product contains the factor $k=r$ and vanishes, and for $r=m$ it is D_m. This is Equation (107). For the expansion, split the operator into the factors below and above m and apply Theorem 8.4 to each:

$$\prod_{k=0}^{m-1}(\mathsf{E}-Q^k) = \mathsf{E}^m\left(\mathsf{E}^{-1};Q\right)_m = \sum_{a=0}^{m}(-1)^a Q^{\binom{a}{2}}\begin{bmatrix} m \\ a\end{bmatrix}_Q \mathsf{E}^{m-a},$$

$$\begin{aligned}\prod_{k=m+1}^{d}(\mathsf{E}-Q^k) &= \prod_{i=1}^{d-m}\left(\mathsf{E}-Q^{m+i}\right) = \mathsf{E}^{d-m}\left(Q^{m+1}\mathsf{E}^{-1};Q\right)_{d-m} \\ &= \sum_{b=0}^{d-m}(-1)^b Q^{\binom{b}{2}+(m+1)b}\begin{bmatrix} d-m \\ b\end{bmatrix}_Q \mathsf{E}^{d-m-b}.\end{aligned}$$

(The operators are polynomials in the single commuting symbol E, so the formal manipulation with E^{-1} is legitimate after multiplying back by the displayed power of E.) Multiplying and applying E^{d-a-b} to u at index 0 gives $u_{d-a-b} = q_{N+d-a-b}$. Comparing Equation (107) with Equation (105) is the symmetry $[z^i]\ell_m^{(d)} = [z^m]\ell_i^{(d)}$ once more, since $D_m^{-1}\prod_{k\neq m}(\mathsf{E}-Q^k) = \ell_m^{(d)}(\mathsf{E})$.

4: $B_m = C_m(x)\,Q^{mN}$ by definition, and $C_m(x)$ does not depend on N; the right side of Equation (109) is therefore the same rational number for every window $N\geq s$. ☐

Corollary 9.3 (Even-jet recovery). *For $1\leq m\leq d$ and any $N\geq s$,*

$$\mathrm{up}^{(2m)}(x) = \frac{(-1)^m 4^m}{a_m}C_m(x) = \frac{(-1)^m 4^{m(N+1)}}{a_m}\sum_{j=0}^{d} q_{N+j}\,\frac{(-1)^{d-m}e_{d-m}^{(j)}}{D_j} \in \mathbb{Q}. \tag{110}$$

Thus the $d+1$ consecutive finite-spline values at a dyadic point of depth s determine, besides $\mathrm{up}(x)$, every even derivative $\mathrm{up}^{(2)}(x),\ldots,\mathrm{up}^{(2d)}(x)$ that occurs in the terminating expansion, and all of them are rational.

Proof. Solve $C_m(x) = (-1)^m a_m 4^{-m}\,\mathrm{up}^{(2m)}(x)$ for the derivative; $a_m\neq 0$ because the a_m are positive. Insert Equation (109). Rationality: the samples, the nodes, and a_m are rational. ☐

Example 9.4 (The two modes at $x=1/16$). For $x=1/16$ one has $s=4$, $d=2$, and the window $N=4$ gives

$$q_4 = \frac{24575}{24576},\qquad q_5 = \frac{1965959}{1966080},\qquad q_6 = \frac{94365509}{94371840}.$$

Solving Equation (102) exactly, $(B_0,B_1,B_2) = \left(\frac{2073457}{2073600}, \frac{5}{165888}, -\frac{31}{16588800}\right)$, hence

$$\mathrm{up}\left(\tfrac{1}{16}\right) = \frac{2073457}{2073600},\qquad C_1\left(\tfrac{1}{16}\right) = 4^4 B_1 = \frac{5}{648},\qquad C_2\left(\tfrac{1}{16}\right) = 4^8 B_2 = -\frac{248}{2025},$$

and the window $N=5$ returns the same three numbers. By Equation (110) with $a_1=\frac{1}{18}$, $a_2=\frac{31}{16200}$: $\mathrm{up}''(1/16) = -4\cdot 18\cdot\frac{5}{648} = -\frac{5}{9}$ and $\mathrm{up}^{(4)}(1/16) = 16\cdot\frac{16200}{31}\cdot\left(-\frac{248}{2025}\right) = -1024$. The first value has an independent check from the functional-differential equation $\mathrm{up}'(x) = 2\,\mathrm{up}(2x+1) - 2\,\mathrm{up}(2x-1)$: $\mathrm{up}''(1/16) = 4\left(\mathrm{up}'(9/8)-\mathrm{up}'(-7/8)\right) = -4\,\mathrm{up}'(-7/8) = -4\cdot 2\ \mathrm{up}(-3/4) = -8\cdot\frac{5}{72} = -\frac{5}{9}$, using $\mathrm{up}(3/4) = 5/72$ (itself the $d=1$ row at $x=3/4$).

Proposition 9.5 (Why $d+1$ samples are needed within the mode model)**.** *Consider the class $\mathcal{M}_d$ of all sequences $n \mapsto L + \sum_{r=1}^{d} A_r Q^{rn}$ with $L, A_1, \ldots, A_d \in \mathbb{Q}$.*

1. *Any $d+1$ samples at distinct levels determine $(L, A_1, \ldots, A_d)$ uniquely (Theorem 9.1 for consecutive levels; for arbitrary distinct levels $n_0 < \cdots < n_d$ the matrix (Q^{rn_i}) is the Vandermonde matrix of the distinct nodes Q^{n_i}).*
2. *For any $k \le d$ levels $n_0 < \cdots < n_{k-1}$ there exist two sequences in $\mathcal{M}_d$ with identical values at these k levels and different constants L. Hence no rule whatsoever (linear or not) that reads only $k \le d$ samples returns L on all of $\mathcal{M}_d$.*
3. *If $C_d(x) \neq 0$, the up-function sequence q_n ($n \ge s$) has exact degree d in Q^n; so $\mathcal{M}_d$ cannot be replaced by $\mathcal{M}_{d-1}$ for this x, and the order-d row is the minimal universal rule.*

Proof. (1) as stated. (2) Take a sequence $q_n = \sum_r B_r Q^{r(n-N)}$ of the class, $N := n_0$, and let $v(z) = \prod_{i=0}^{k-1}(z - Q^{n_i - N}) = \sum_{r=0}^{k} v_r z^r$, a polynomial of degree $k \le d$. The sequence $q'_n = q_n + \sum_{r=0}^{k} v_r Q^{r(n-N)}$ belongs to $\mathcal{M}_d$, agrees with q_n at every n_i because $v(Q^{n_i-N}) = 0$, and its constant differs from L by $v(0) = (-1)^k Q^{\sum_i (n_i - N)} \neq 0$. (3) If $C_d(x) \neq 0$, the polynomial P_N of Equation (88) has degree exactly d, so the sequence is not in $\mathcal{M}_{d-1}$: a nontrivial combination of the modes $1, Q^n, \ldots, Q^{dn}$ with distinct ratios is not the zero sequence (Vandermonde again). ☐

The nonvanishing $C_d(x) \neq 0$ for every dyadic x of depth $s \ge 2$ is Theorem 5.5, restated as Theorem 7.1; here it enters only as a hypothesis. Statement (2) concerns the abstract model class: it does not exclude that a rule exploiting point-specific information about the particular numbers $C_r(x)$ could use fewer samples, but no such information is available without first knowing the answer.

9.2 The Romberg tableau

The row can be evaluated without forming its large numerators, one mode at a time, by the classical Romberg pattern (Richardson's deferred approach to the limit [11], in the geometric-node form treated in [3]).

Algorithm 9.6 (Exact quarter-base q-Richardson tableau)**.** Given exact samples q_n for $n \ge N$, set $T_{0,n} = q_n$ and, for $m = 1, 2, \ldots$,

$$\boxed{T_{m,n} = \frac{T_{m-1,n+1} - Q^m\, T_{m-1,n}}{1 - Q^m} = \frac{4^m\, T_{m-1,n+1} - T_{m-1,n}}{4^m - 1}.} \tag{111}$$

Entry $T_{m,n}$ uses the samples $q_n, \ldots, q_{n+m}$. Output $T_{d,N}$.

Theorem 9.7 (What the tableau computes)**.** *For every $m \ge 0$ and $n \ge N$:*

1. $T_{m,n} = \sum_{j=0}^{m} w_j^{(m)} q_{n+j} = \Big[\prod_{i=1}^{m} \frac{\mathsf{E} - Q^i}{1 - Q^i}\, q\Big]_n$*: the m-th column of the tableau is the extraction row of order m applied at level n.*
2. *$T_{m,n}$ is the value at $z = 0$ of the polynomial of degree at most m interpolating the data (Q^j, q_{n+j}), $0 \le j \le m$; the recursion Equation (111) is Neville's recurrence for these nodes.*
3. *If $n \ge s$, then*
$$T_{m,n} = \mathrm{up}(x) + \sum_{r=m+1}^{d} C_r(x)\, Q^{rn} \prod_{i=1}^{m} \frac{Q^r - Q^i}{1 - Q^i}. \tag{112}$$
In particular $T_{m,n} = \mathrm{up}(x)$ for every $m \ge d$ and every $n \ge s$, while for $m < d$ the column $T_{m,\cdot}$ is a nonconstant function of n whenever $C_{m+1}(x) \neq 0$, its deviation from $\mathrm{up}(x)$ being an exact finite sum whose leading term is $C_{m+1}(x) \prod_{i=1}^{m} \frac{Q^{m+1} - Q^i}{1 - Q^i}\, Q^{(m+1)n}$.

Proof. 1: induction on m. For $m=0$ both sides are q_n. If the operator identity holds for $m-1$, then Equation (111) says $T_{m,\cdot} = \frac{\mathsf{E}-Q^m}{1-Q^m} T_{m-1,\cdot}$, and composing with the operator for $m-1$ gives the operator for m. Its expansion in powers of E is the row of order m by Theorem 8.5: $W_m(\mathsf{E}) = \sum_j w_j^{(m)} \mathsf{E}^j$.

2: by 1 and Theorem 8.21 (with m in place of d), $T_{m,n} = \sum_j w_j^{(m)} p(Q^j) = p(0)$ for the interpolant p of $(Q^j, q_{n+j})_{0\le j\le m}$, which is the first claim. For the recurrence, let $p_{[0,m]}$ be that interpolant, $p_{[0,m-1]}$ the interpolant of the first m data and $p_{[1,m]}$ that of the last m. Neville's identity $p_{[0,m]}(z) = \frac{(z-Q^0)p_{[1,m]}(z)-(z-Q^m)p_{[0,m-1]}(z)}{Q^m-Q^0}$ (both sides have degree at most m and agree at all $m+1$ nodes) gives at $z=0$: $p_{[0,m]}(0) = \frac{p_{[1,m]}(0)-Q^m p_{[0,m-1]}(0)}{1-Q^m}$. Here $p_{[0,m-1]}(0) = T_{m-1,n}$ by the first claim, and $p_{[1,m]}(0) = T_{m-1,n+1}$ because $p_{[1,m]}(Qz)$ interpolates $(Q^j, q_{n+1+j})_{0\le j\le m-1}$ and has the same value at 0.

3: apply the operator of 1 to Equation (87), valid for all indices $\ge n \ge s$: the constant is multiplied by $\prod_i (1-Q^i)/(1-Q^i) = 1$, and the mode Q^{rn} by $\prod_{i=1}^m (Q^r - Q^i)/(1-Q^i)$, which vanishes for $1 \le r \le m$ and is nonzero for $r > m$. If $C_{m+1}(x) \ne 0$ the right side of Equation (112) is a nonzero combination of the distinct geometric sequences Q^{rn}, $m < r \le d$, hence nonconstant in n. ☐

The tableau costs $\binom{d+1}{2}$ entries, each one multiplication, one subtraction and one division in $\mathbb{Q}$; the intermediate denominators are products of the odd numbers $4^m - 1$ times the sample denominators, never larger than in the closed row. The columns are also the natural diagnostic display (Theorem 9.10 below): column d is constant along $n \ge s$, and column $d-1$ is not.

Example 9.8 (Tableau at $x = 1/16$)**.** With the samples of Theorem 9.4 and q_7, q_8:

m	$T_{m,4}$	$T_{m,5}$	$T_{m,6}$
0	24575/24576	1965959/1966080	94365509/94371840
1	1474459/1474560	70774001/70778880	226476797/226492416
2	2073457/2073600	2073457/2073600	2073457/2073600

Column 2 is constant, equal to up(1/16); column 1 is not, because $C_2(1/16) = -248/2025 \ne 0$; for instance $T_{1,4} - \mathrm{up}(1/16) = C_2(1/16)\, Q^8 \frac{Q^2-Q}{1-Q} = -\frac{248}{2025} \cdot \frac{1}{65536} \cdot \left(-\frac{1}{4}\right) = \frac{31}{66355200}$, which is $\frac{1474459}{1474560} - \frac{2073457}{2073600} = \frac{66350655-66350624}{66355200}$ exactly.

9.3 The annihilating recurrence and onset diagnostics

Theorem 9.9 (Annihilating recurrence)**.** *For every $n \ge s$,*

$$\left[\prod_{m=0}^{d} (\mathsf{E} - Q^m)\, q\right]_n = \sum_{j=0}^{d+1} (-1)^{d+1-j} Q^{\binom{d+1-j}{2}} \begin{bmatrix} d+1 \\ j \end{bmatrix}_Q q_{n+j} = 0, \tag{113}$$

or, after multiplication by $4^{\binom{d+1}{2}}$, with integer coefficients

$$\sum_{j=0}^{d+1} (-1)^{d+1-j} 4^{\binom{j}{2}} \begin{bmatrix} d+1 \\ j \end{bmatrix}_4 q_{n+j} = 0 \qquad (n \ge s). \tag{114}$$

For $d=1$ this reads $q_n - 5q_{n+1} + 4q_{n+2} = 0$, for $d=2$ it reads $-q_n + 21q_{n+1} - 84q_{n+2} + 64q_{n+3} = 0$.

Proof. Each factor $\mathsf{E} - Q^m$ annihilates the mode Q^{mn} of Equation (87) and the factors commute, so the product annihilates the whole sequence from index s on. For the expansion, $\prod_{m=0}^d (\mathsf{E} - Q^m) = \mathsf{E}^{d+1} \left(\mathsf{E}^{-1}; Q\right)_{d+1} = \sum_{k=0}^{d+1} (-1)^k Q^{\binom{k}{2}} \begin{bmatrix} d+1 \\ k \end{bmatrix}_Q \mathsf{E}^{d+1-k}$ by Theorem 8.4; put $j = d+1-k$ and use the symmetry of the Gaussian coefficient. The integer form follows from $\begin{bmatrix} d+1 \\ j \end{bmatrix}_Q = 4^{-j(d+1-j)} \begin{bmatrix} d+1 \\ j \end{bmatrix}_4$ (inversion identity in the proof of Theorem 8.7) and $\binom{d+1}{2} - \binom{d+1-j}{2} - j(d+1-j) = \binom{j}{2}$. The displayed rows are $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_4 = 5$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}_4 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}_4 = 21$ with $4^{\binom{2}{2}} = 4$, $4^{\binom{3}{2}} = 64$. ☐

Note the difference from the extraction row: the recurrence contains the factor $\mathsf{E}-1$ and annihilates *everything*, including the constant; the row omits that factor and therefore preserves the constant while killing the rest. Write

$$\rho_n := \sum_{j=0}^{d+1} (-1)^{d+1-j}\, 4^{\binom{j}{2}} \begin{bmatrix} d+1 \\ j \end{bmatrix}_4 q_{n+j} \tag{115}$$

for the residual of Equation (114) at level n.

Proposition 9.10 (Onset and model-order diagnostics). *Let x be dyadic of depth s and $d = \lfloor s/2 \rfloor$.*

1. *(Onset detection is a finite test.) Let $0 \le n_0 \le s$. If $\rho_n = 0$ for every n with $n_0 \le n \le s-1$, then the geometric law Equation (87) already holds for all $n \ge n_0$, and every extraction row and every mode formula of Theorems 8.7 and 9.2 is exact from any window $N \ge n_0$. Conversely, if the law holds for all $n \ge n_0$ then $\rho_n = 0$ for all $n \ge n_0$.*
2. *(Model order.) For the sequence restricted to $n \ge s$, the least k such that $\prod_{m=0}^{k}(\mathsf{E} - Q^m)$ annihilates it equals $\max\{r : C_r(x) \ne 0\}$ (and 0 if all modes vanish). In particular, when $C_d(x) \ne 0$ the least annihilating order is exactly $d = \lfloor s/2 \rfloor$, so exact data at an* unreduced *dyadic fraction reveal d, and s up to parity, without reducing the fraction.*
3. *(Certificate.) In exact arithmetic, $\rho_n \ne 0$ for some $n \ge s$ is impossible; a nonzero residual signals a wrong depth, a wrong indexing convention (for instance p_n in place of up_n), or an arithmetic error. Two windows $N \ne N'$ both $\ge s$ must return identical rationals.*
4. *(Floating point.) With rounded samples $\tilde{q}_n = q_n + e_n$ and $n \ge s$, the computed residual is $\tilde{\rho}_n = \sum_j c_j e_{n+j}$ with the integer coefficients c_j of Equation (115), so $|\tilde{\rho}_n| \le \left(\sum_j |c_j|\right) \max_j |e_{n+j}|$ with $\sum_j |c_j| = 4^{\binom{d+1}{2}} (-1;Q)_{d+1}$ ($= 10$ for $d = 1$, 170 for $d = 2$), and a residual larger than this multiple of the sample accuracy proves that level n is still in the transient range.*

Proof. 1: by Theorem 9.9, $\rho_n = 0$ for $n \ge s$; with the hypothesis, $\rho_n = 0$ for all $n \ge n_0$. The sequence $(q_n)_{n \ge n_0}$ therefore satisfies a linear recurrence of order $d+1$ with characteristic polynomial $\prod_{m=0}^{d}(z - Q^m)$, whose roots $1, Q, \dots, Q^d$ are distinct. The solution space of such a recurrence on $n \ge n_0$ is $(d+1)$-dimensional (a solution is determined by its first $d+1$ values) and contains the $d+1$ linearly independent geometric sequences Q^{mn}; so $q_n = L' + \sum_{r=1}^{d} A'_r Q^{rn}$ for all $n \ge n_0$ with some constants. On $n \ge s$ this representation and Equation (87) both hold; their difference is a combination of the $d+1$ distinct geometric sequences vanishing on $d+1$ consecutive indices, hence (Theorem 9.1) all coefficients agree: $L' = \mathrm{up}(x)$, $A'_r = C_r(x)$. So Equation (87) holds for $n \ge n_0$, and the proofs of Theorems 8.7 and 9.2 go through verbatim with n_0 in place of s. The converse is the proof of Theorem 9.9.

2: $\prod_{m=0}^{k}(\mathsf{E} - Q^m)$ kills the modes $r \le k$ and multiplies the mode Q^{rn}, $r > k$, by the nonzero scalar $\prod_{m \le k}(Q^r - Q^m)$; the image is a combination of the distinct geometric sequences Q^{rn}, $k < r \le d$, with coefficients $C_r(x) \prod_{m \le k}(Q^r - Q^m)$, which is the zero sequence exactly when all $C_r(x)$ with $r > k$ vanish.

3 restates Theorem 9.9 and Theorem 8.8.

4: linearity, the triangle inequality, and

$$\sum_j |c_j| = 4^{\binom{d+1}{2}} \sum_{j=0}^{d+1} Q^{\binom{d+1-j}{2}} \begin{bmatrix} d+1 \\ j \end{bmatrix}_Q = 4^{\binom{d+1}{2}} \sum_{k=0}^{d+1} Q^{\binom{k}{2}} \begin{bmatrix} d+1 \\ k \end{bmatrix}_Q = 4^{\binom{d+1}{2}} (-1;Q)_{d+1}$$

by Theorem 8.4 at $z = -1$. For $d = 1$: $1 + 5 + 4 = 10$; for $d = 2$: $1 + 21 + 84 + 64 = 170$. □

Remark 9.11 (What the residual sees below the depth). At $n = s - 1$ the point x is a knot of the spline up_n, not the centre of a polynomial cell, and the mechanism that proves Equation (87) does not apply; the residual ρ_{s-1} need not vanish, and in general it does not: for $x = 1/16$ ($s = 4$, $d = 2$)

exact computation gives $\rho_2 = -\frac{1}{1920}$, $\rho_3 = \frac{1}{46080}$, $\rho_4 = 0$. For odd depth the picture is different: at $x = 7/8, 3/8, 1/8$ ($s = 3$) and at $x = 3/32, 1/32$ ($s = 5$) one finds $\rho_{s-1} = 0$ but $\rho_{s-2} \neq 0$, in agreement with Theorem 4.15, according to which the law Equation (87) holds from $n = s-1$ on when $s \geq 3$ is odd. The residual at the knot level is in fact known in closed form at every depth: since the law holds at the levels $s, \ldots, s+d$, only the first term of Equation (115) survives, and $\rho_{s-1} = (-1)^{d+1}\Delta_s(x) = (-1)^d \varepsilon_k\, 2^{-\binom{s}{2}}\beta_s$ with $\beta_s = B_s/s!$ for even s and 0 for odd s (Theorem 7.4); at $x = 1/16$ this is $(-1)^2 \cdot (-1) \cdot 2^{-6} \cdot (-\frac{1}{720}) = \frac{1}{46080}$, the value computed above. By Theorem 9.101 the residual test detects this earlier onset automatically, and then the row may start at $N = s-1$; the guaranteed onset $N \geq s$ of Theorem 8.7 remains the safe choice when no test is run.

Section summary

The $d+1$ samples $q_N, \ldots, q_{N+d}$, $N \geq s$, determine every mode: $(V^{-1})_{mj} = [z^m]\ell_j^{(d)}(z) = (-1)^{d-m} e_{d-m}^{(j)}/D_j$ with $D_j = (-1)^j Q^{\binom{j}{2}+j(d-j)}\,(Q;Q)_j\,(Q;Q)_{d-j}$ (Theorem 9.2), hence $C_m(x)$ and $\mathrm{up}^{(2m)}(x) = (-1)^m 4^m C_m(x)\,/a_m$ are rational combinations of the samples (Theorem 9.3); fewer than $d+1$ samples cannot do this within the mode model (Theorem 9.5). The Romberg recursion $T_{m,n} = (4^m T_{m-1,n+1} - T_{m-1,n})/(4^m - 1)$ evaluates the rows of all orders at once, and its column d is constant from $n = s$ on (Theorem 9.7). The order-$(d+1)$ recurrence with integer coefficients $(-1)^{d+1-j}4^{\binom{j}{2}}\left[{d+1 \atop j}\right]_4$ annihilates the sequence for $n \geq s$; a finite residual test certifies an earlier onset and reveals the true number of modes (Theorems 9.9 and 9.10).

10 Stability of the extraction

Exact rational input makes the extraction exact, and that is how it should be used. This section records what happens when the input is inexact: the row is uniformly well conditioned in the depth, the full mode recovery is not, and the only genuine floating-point hazard lies upstream, in the evaluation of the samples.

10.1 The Lebesgue factor of the row

Proposition 10.1 (Exact ℓ^∞ noise amplification). *For $0 < q < 1$ and $d \geq 0$ let $\Lambda_d(q) := \sum_{j=0}^{d}\left|w_j^{(d)}\right|$. Then*

$$\boxed{\Lambda_d(q) = \frac{(-q;q)_d}{(q;q)_d} = \prod_{h=1}^{d}\frac{1+q^h}{1-q^h},} \qquad \Lambda_0 < \Lambda_1 < \Lambda_2 < \cdots < \Lambda_\infty(q) := \frac{(-q;q)_\infty}{(q;q)_\infty} < \infty, \tag{116}$$

and $\Lambda_d(q) \to \Lambda_\infty(q)$. At the quarter base,

$$\Lambda_\infty(\tfrac14) = \frac{(-1/4;1/4)_\infty}{(1/4;1/4)_\infty} = 1.96926035366826943194719369696726567006618996 23\ldots \tag{117}$$

Consequently, if the samples are perturbed, $\tilde q_{N+j} = q_{N+j} + e_j$, the row Equation (96) returns $\widetilde{\mathrm{up}}(x)$ with

$$|\widetilde{\mathrm{up}}(x) - \mathrm{up}(x)| \leq \Lambda_d(\tfrac14)\max_j |e_j| < 1.96927\,\max_j |e_j|\,, \tag{118}$$

uniformly in the depth, and the first inequality is attained by $e_j = \varepsilon\,\mathrm{sign}\,w_j^{(d)}$.

Proof. For $0 < q < 1$ every $(q;q)_k$ is positive, so by Equation (92) the sign of $w_j^{(d)}$ is $(-1)^{d-j}$ and $\left|w_j^{(d)}\right| = q^{\binom{d-j+1}{2}}\left[{d \atop j}\right]_q/\,(q;q)_d$. With $k = d-j$ and the symmetry of the Gaussian coefficient,

$$\Lambda_d(q) = \frac{1}{(q;q)_d}\sum_{k=0}^{d} q^{\binom{k+1}{2}}\begin{bmatrix} d \\ k \end{bmatrix}_q = \frac{1}{(q;q)_d}\sum_{k=0}^{d}(-1)^k q^{\binom{k}{2}}\begin{bmatrix} d \\ k \end{bmatrix}_q (-q)^k = \frac{(-q;q)_d}{(q;q)_d},$$

the last step being Theorem 8.4 at $z = -q$; and $(-q;q)_d \,/\, (q;q)_d = \prod_{h=1}^{d}(1+q^h)/(1-q^h)$. Each new factor $(1+q^d)/(1-q^d)$ exceeds 1, so the sequence is strictly increasing; both infinite products converge to positive limits because $\sum_h q^h < \infty$, so the limit is finite. The error bound is the triangle inequality applied to $\widetilde{\mathrm{up}}(x) - \mathrm{up}(x) = \sum_j w_j^{(d)} e_j$, with equality for the displayed choice of signs. The numerical value is the partial product with 400 factors; the omitted factors change it by a relative amount below $2\sum_{h>400} 4^{-h} < 4^{-399}$. □

Table 3: Lebesgue factors $\Lambda_d(1/4) = \sum_j \left|w_j^{(d)}\right|$ of the quarter-base rows. The same numbers appear in the Exponents volume [9] as `p3:eq:richardson-stability` with $M-1=d$.

d	$\Lambda_d(1/4)$ exact	decimal
0	1	1
1	$5/3$	1.66667
2	$17/9$	1.88889
3	$1105/567$	1.94885
4	$3341/1701$	1.96414
5	$3424525/1740123$	1.96798
∞	$(-1/4;1/4)_\infty \,/\, (1/4;1/4)_\infty$	1.96926

The bounded Lebesgue factor is a property of the geometric nodes, not of extrapolation in general: the nodes $1, \frac{1}{4}, \frac{1}{16}, \ldots$ are equally spaced on a logarithmic scale and the target 0 lies at their accumulation point, whereas polynomial extrapolation from nearly coincident nodes has weights that grow without bound. The base matters: at the half base $q = 1/2$, the natural base for samples taken one dyadic level apart when the modes decay like 2^{-n}, the same formula gives $\Lambda_\infty(\frac{1}{2}) = 8.25598793577825\ldots$, four times larger; the quarter base is available here precisely because only even derivatives occur in Equation (87), so consecutive levels are two powers of $1/2$ apart in every mode. For general $q \in (0,1)$ everything in this proposition holds verbatim; as $q \to 1^-$ the factor $\Lambda_d(q)$ tends to $\binom{2d}{d}$-type growth, since each factor $(1+q^h)/(1-q^h)$ diverges.

10.2 Exact rational arithmetic versus floating point

Proposition 10.2 (Denominator bookkeeping). *Let x be dyadic of depth s, $N \geq s$, and let Δ be the least common denominator of $q_N, \ldots, q_{N+d}$. Then*

$$\mathrm{up}(x) = \frac{\textit{integer}}{P_d\,\Delta}, \qquad P_d = \prod_{r=1}^{d}(4^r - 1) \ \textit{odd}, \tag{119}$$

so the denominator of $\mathrm{up}(x)$ divides $P_d\Delta$; in particular the extraction row never introduces a new power of 2, and the 2-adic valuation of $\mathrm{up}(x)$ is at least $-\nu_2(\Delta)$.

Proof. Write Equation (99) over the common denominator $P_d\Delta$; the numerators $(-1)^{d-j}4^{\binom{j+1}{2}}\begin{bmatrix} d \\ j \end{bmatrix}_4$ are integers and $q_{N+j}\Delta$ is an integer. Every 4^r-1 is odd. □

Remark 10.3 (Where floating point actually fails). Suppose the samples are available only as floating-point numbers with absolute errors $|e_j| \leq \varepsilon$, and the dot product Equation (96) is evaluated in IEEE arithmetic with unit roundoff u ($u = 2^{-53}$ in double precision) from correctly rounded weights. The standard forward error bound for a $(d+1)$-term dot product, together with Theorem 10.1, gives a total error at most

$$\Lambda_d(\tfrac{1}{4})\Big(\varepsilon + \gamma_{d+2}\max_j |\tilde{q}_{N+j}|\Big), \qquad \gamma_k = \frac{ku}{1-ku},$$

that is, essentially 2ε plus a few units of roundoff. The row is therefore not the problem. The problem is ε itself when the samples are computed from the Thue–Morse truncated-power representation (Theorem 8.9) in floating point: in the scaled form $\mathrm{up}_n(x) = \left(2^{\binom{n+1}{2}} n!\right)^{-1} \sum_{0\le k\le y} \varepsilon_k (2y-2k)^n$ with $y = 2^n(x+1-2^{-(n+1)})$, the summands are integers of size up to $(2y)^n \le 2^{n(n+2)}$ while their alternating sum is at most $2^{\binom{n+1}{2}} n! \max|\mathrm{up}_n|$, so up to $n(n+2) - \binom{n+1}{2} - \log_2 n! = \frac{1}{2}n(n+3) - \log_2 n!$ bits can cancel: about 23 bits at $n = 7$ and about 43 bits at $n = 10$, close to the entire double-precision mantissa. Whether this worst case is attained at a given point is beside the point; the remedy is free. With $2y$ an odd integer (midpoint alignment at $n \ge s$) the sum is an exact integer computation followed by one rational normalization, and then the extraction is exact by Theorem 8.7. If, conversely, only a high-precision floating approximation $\tilde v$ of $\mathrm{up}(x)$ is at hand, the exact value can still be recovered: Theorem 10.2 bounds the denominator by $D = P_d\Delta$, and a rational a/b with $b \le D$ is determined uniquely by any approximation with $|\tilde v - a/b| < 1/(2D^2)$, the continued-fraction convergents of $\tilde v$ producing it.

10.3 Conditioning of the full Vandermonde recovery

The row is the first line of V^{-1}; the other lines recover the decaying modes and are necessarily much larger, because they must undo the factors Q^{mN}.

Proposition 10.4 (Norms of the geometric Vandermonde inverse). *Let $V = (Q^{ij})_{0\le i,j\le d}$ and $(V^{-1})_{mj} = [z^m]\ell_j^{(d)}(z)$ as in Theorem 9.2. Write $\Lambda_d^{(m)} := \sum_{j=0}^{d} \left|(V^{-1})_{mj}\right|$ for the absolute row sums. Then:*

1. *$\|V\|_\infty = d+1$ (the first row), and $\Lambda_d^{(0)} = \Lambda_d(\frac{1}{4})$.*
2. *The last row sums exactly:*

$$\Lambda_d^{(d)} = \sum_{j=0}^{d} \frac{1}{|D_j|} = 4^{\binom{d}{2}} \frac{(-1;Q)_d}{(Q;Q)_d} = 2\cdot 4^{\binom{d}{2}} \frac{(-Q;Q)_{d-1}}{(Q;Q)_d} \qquad (d \ge 1). \tag{120}$$

3. *The absolute column sums are*

$$\sum_{m=0}^{d} \left|(V^{-1})_{mj}\right| = \left|\ell_j^{(d)}(-1)\right| = \frac{(-1;Q)_{d+1}}{(1+Q^j)\,|D_j|}, \qquad \left\|V^{-1}\right\|_1 = \max_j \frac{(-1;Q)_{d+1}}{(1+Q^j)\,|D_j|}. \tag{121}$$

4. *Hence $\kappa_\infty(V) = \|V\|_\infty \left\|V^{-1}\right\|_\infty \ge (d+1)\Lambda_d^{(d)} \ge 2(d+1)4^{\binom{d}{2}}$: the recovery of all modes is ill conditioned in absolute terms, at least like $4^{d^2/2}$.*

If the samples carry absolute errors at most ε, the recovered B_m carries an absolute error at most $\Lambda_d^{(m)}\varepsilon$ (sharp), and the mode coefficient $C_m(x) = 4^{mN}B_m$ an absolute error at most $4^{mN}\Lambda_d^{(m)}\varepsilon$.

Proof. (1) The i-th row sum of V is $\sum_j Q^{ij} \le d+1$ with equality only for $i = 0$; the first row of V^{-1} is the extraction row.

(2) $(V^{-1})_{dj} = [z^d]\ell_j^{(d)} = 1/D_j$. By Equation (104), $|D_j| = Q^{\binom{j}{2}+j(d-j)}(Q;Q)_j(Q;Q)_{d-j}$, and $\binom{j}{2} + j(d-j) = \frac{1}{2}j(2d-j-1) = \binom{d}{2} - \binom{d-j}{2}$. Hence

$$\sum_j \frac{1}{|D_j|} = \frac{Q^{-\binom{d}{2}}}{(Q;Q)_d} \sum_{j=0}^{d} Q^{\binom{d-j}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_Q = \frac{Q^{-\binom{d}{2}}}{(Q;Q)_d} (-1;Q)_d\,,$$

by Theorem 8.4 at $z = -1$ (with $k = d-j$). Finally $(-1;Q)_d = (1+1)\prod_{h=1}^{d-1}(1+Q^h) = 2\,(-Q;Q)_{d-1}$ and $Q^{-\binom{d}{2}} = 4^{\binom{d}{2}}$.

(3) $\ell_j^{(d)}(z) = D_j^{-1}\prod_{k\ne j}(z-Q^k)$, and the coefficients of $\prod_{k\ne j}(z-Q^k)$ are $(-1)^{d-m}e_{d-m}^{(j)}$ with $e_r^{(j)} > 0$ (elementary symmetric functions of positive numbers). Therefore $\sum_m \left|[z^m]\ell_j^{(d)}\right| =$

$|D_j|^{-1} \sum_r e_r^{(j)} = |D_j|^{-1} \prod_{k \neq j}(1+Q^k) = \left|\ell_j^{(d)}(-1)\right|$, and $\prod_{k \neq j}(1+Q^k) = (-1;Q)_{d+1} \,/\, (1+Q^j)$. The 1-norm is the largest column sum.

(4) $\|V^{-1}\|_\infty = \max_m \Lambda_d^{(m)} \geq \Lambda_d^{(d)}$, and $(-1;Q)_d \,/\, (Q;Q)_d \geq 2$. The error statements are the triangle inequality on $B = V^{-1}q$, row by row, with equality for $e_j = \varepsilon \operatorname{sign}(V^{-1})_{mj}$. □

Table 4: Absolute row sums $\Lambda_d^{(m)}$ of V^{-1} for the quarter base, exact, and the condition number $\kappa_\infty(V) = (d+1) \max_m \Lambda_d^{(m)}$. Row $m = 0$ is the Lebesgue factor of Table 3; row $m = d$ is Equation (120). Note that for $d \geq 4$ the largest row is $m = d-1$, not $m = d$.

d	$\Lambda_d^{(0)}, \Lambda_d^{(1)}, \ldots, \Lambda_d^{(d)}$	$\kappa_\infty(V)$
1	$\frac{5}{3}, \frac{8}{3}$	$\frac{16}{3} \approx 5.33$
2	$\frac{17}{9}, \frac{136}{9}, \frac{128}{9}$	$\frac{136}{3} \approx 45.3$
3	$\frac{1105}{567}, \frac{1768}{27}, \frac{8320}{27}, \frac{139264}{567}$	$\frac{33280}{27} \approx 1.23 \cdot 10^3$
4	$\frac{3341}{1701}, \frac{454376}{1701}, \frac{427648}{81}, \frac{35790848}{1701}, \frac{27262976}{1701}$	$\frac{178954240}{1701} \approx 1.05 \cdot 10^5$
5	$\frac{3424525}{1740123}, \frac{5479240}{5103}, \frac{438339200}{5103}, \frac{7337123840}{5103}, \frac{27944550400}{5103}, \frac{7174742867968}{1740123}$	$\frac{55889100800}{1701} \approx 3.29 \cdot 10^7$

The growth is inherent, not an artefact of the method: the mode $B_m = C_m(x)\, 4^{-mN}$ is exponentially small in N, so any procedure that reads it off samples of absolute accuracy ε needs $\varepsilon \ll |C_m(x)|\, 4^{-mN}$, and the higher modes are recoverable in floating point only from samples at the smallest admissible level $N = s$ and with high working precision. With exact rational samples the question does not arise: V^{-1} is a fixed rational matrix, and Equation (109) returns each $C_m(x)$ as an exact rational number, as in Theorem 9.4. In terms of cost, the closed row needs $O(d)$ rational operations once the P_k are tabulated, the tableau and the closed inverse $O(d^2)$, and a dense solve of Equation (102) $O(d^3)$; all are negligible against the evaluation of the samples themselves, whose truncated-power sums have 2^{n+1} terms.

Section summary

The quarter-base row has Lebesgue factor $\Lambda_d(1/4) = (-1/4;1/4)_d \,/\, (1/4;1/4)_d < 1.969260353668\ldots$, so inexact samples are amplified by less than a factor 2 at every depth (Theorem 10.1); the extraction adds only the odd factor P_d to the sample denominators (Theorem 10.2); the floating-point hazard is the alternating Thue–Morse sum that produces the samples, not the row (Theorem 10.3); and recovering the decaying modes themselves is exponentially ill conditioned in absolute terms, with the exact last-row sum $2 \cdot 4^{\binom{d}{2}}\, (-1/4;1/4)_{d-1} \,/\, (1/4;1/4)_d$ (Theorem 10.4), which is why the mode recovery of Section 9 is meant for exact rational input.

11 Worked exact examples

The six sources between them work out about twenty dyadic points. This section consolidates them into one graded set, in the volume's indexing (up_n has $n+1$ factors, $q_n = \mathrm{up}_n(x)$), with every number an exact rational. Every displayed value was recomputed for this volume in exact arithmetic by an independent evaluator (Thue–Morse truncated powers for the samples, the closed row for the extraction, an exact Vandermonde solve for the modes, and the derivative tiling of Theorem 5.2 for the jet, the last two agreeing term by term), and cross-checked against the captured verification outputs of the six packages. Where a source's text disagreed with its own program output, the output was followed and the repair recorded in Section B.

11.1 The extraction rows

The rows of Theorem 8.3, $Q = 1/4$, for the orders that occur through depth 11:

d	$(w_0^{(d)}, \ldots, w_d^{(d)})$
0	(1)
1	$\left(-\frac{1}{3}, \frac{4}{3}\right)$
2	$\left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right)$
3	$\left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right)$
4	$\left(\frac{1}{722925}, -\frac{4}{8505}, \frac{64}{2025}, -\frac{4096}{8505}, \frac{1048576}{722925}\right)$
5	$\left(-\frac{1}{739552275}, \frac{4}{2168775}, -\frac{64}{127575}, \frac{4096}{127575}, -\frac{1048576}{2168775}, \frac{1073741824}{739552275}\right)$

Each row sums to 1 and annihilates $Q^n, \ldots, Q^{dn}$ (Theorem 8.2); the last entry is $4^{\binom{d+1}{2}}/P_d$ with $P_d = \prod_{k \le d}(4^k - 1) = 3, 45, 2835, 722925, 739552275$, and the alternating pattern $4^{\binom{j+1}{2}}$ in the numerators is Equation (98). The rows for $d \le 3$ are the Exponents volume's printed weight vectors for $M = d + 1$ (Section 8).

11.2 Symmetry: half the points suffice

Proposition 11.1 (Reflection). *For $n \ge 1$ and $0 \le x \le 1$,*

$$\mathrm{up}_n(x) + \mathrm{up}_n(1 - x) = 1, \qquad \mathrm{up}(x) + \mathrm{up}(1 - x) = 1. \tag{122}$$

Consequently, for dyadic $x \in (0, 1)$ of depth $s \ge 1$: $s(1 - x) = s$, $C_m(1 - x) = -C_m(x)$ for $1 \le m \le d$, $\Delta_s(1 - x) = -\Delta_s(x)$, and $\mathrm{up}^{(2m)}(1 - x) = -\mathrm{up}^{(2m)}(x)$ for $m \ge 1$. Together with evenness, every quantity at $\pm x$ and $\pm(1 - x)$ is determined by its value at x.

Proof. By Equation (83), $\mathrm{up}_n(x) = \int_{2x-1}^{2x+1} \mathrm{up}_{n-1} = P_{n-1}(2x+1) - P_{n-1}(2x-1) = 1 - P_{n-1}(2x-1)$ for $x \ge 0$, since $2x + 1 \ge 1$ is beyond the support; and $\mathrm{up}_n(1 - x) = 1 - P_{n-1}(1 - 2x) = 1 - \big(1 - P_{n-1}(2x - 1)\big) = P_{n-1}(2x - 1)$ by the evenness of up_{n-1}. The sum is 1. The same computation with Rvachev's equation, or the limit $n \to \infty$, gives the second identity. The depth is invariant under $x \mapsto 1 - x$ by Theorem 3.2. Differentiating Equation (122) $2m$ times gives the derivative statement, and Equation (4) at x and $1 - x$ then gives the modes and the defects. □

Thus the pairs $(\frac{1}{4}, \frac{3}{4})$, $(\frac{1}{8}, \frac{7}{8})$, $(\frac{1}{16}, \frac{15}{16})$, $(\frac{5}{16}, \frac{11}{16})$, $(\frac{1}{64}, \frac{63}{64})$, $(\frac{1}{128}, \frac{127}{128})$ that the sources treat separately are one example each.

11.3 Depth two and three: one mode

Example 11.2 ($x = \frac{1}{4}$ and $x = \frac{3}{4}$; $s = 2$, $d = 1$)**.** The point $\frac{1}{4}$ is worked out in Theorem 1.6: $q_2 = \frac{15}{16}$, $q_3 = \frac{179}{192}$, $\mathrm{up}(\frac{1}{4}) = -\frac{1}{3} \cdot \frac{15}{16} + \frac{4}{3} \cdot \frac{179}{192} = \frac{67}{72}$, $C_1\left(\frac{1}{4}\right) = \frac{1}{9}$, $\mathrm{up}''(\frac{1}{4}) = -8$. By reflection, at $x = \frac{3}{4}$: $q_2 = \frac{1}{16}$, $q_3 = \frac{13}{192}$, $\mathrm{up}(\frac{3}{4}) = \frac{5}{72}$, $C_1\left(\frac{3}{4}\right) = -\frac{1}{9}$, $\mathrm{up}''(\frac{3}{4}) = 8$, so that $q_n = \frac{5}{72} - \frac{1}{9}\, 4^{-n}$ for all $n \ge 2$ (S4's and S6's example). At the knot level $n = 1$: $q_1 = \mathrm{up}_1(\frac{3}{4}) = 0$ (the trapezoid up_1 vanishes for $|x| \ge \frac{3}{4}$) while the law would give $\frac{5}{72} - \frac{1}{36} = \frac{1}{24}$; the defect $-\frac{1}{24}$ is Theorem 7.4 with $k = 3$, $\varepsilon_3 = +1$.

Example 11.3 ($x = \frac{1}{8}, \frac{3}{8}, \frac{7}{8}$; $s = 3$, $d = 1$)**.** Three points of depth 3, all with the single mode ratio $\frac{1}{4}$ and the row $(-\frac{1}{3}, \frac{4}{3})$ applied at $N = 3$:

x	q_3	q_4	$\mathrm{up}(x)$	$C_1(x)$	$\mathrm{up}''(x)$
$\frac{1}{8}$	$\frac{383}{384}$	$\frac{1531}{1536}$	$\frac{287}{288}$	$\frac{1}{18}$	-4
$\frac{3}{8}$	$\frac{287}{384}$	$\frac{1147}{1536}$	$\frac{215}{288}$	$\frac{1}{18}$	-4
$\frac{7}{8}$	$\frac{1}{384}$	$\frac{5}{1536}$	$\frac{1}{288}$	$-\frac{1}{18}$	4

The full jets (up, up′, up″, up‴) are: at $\frac{1}{8}$, $(\frac{287}{288}, -\frac{5}{36}, -4, -64)$; at $\frac{3}{8}$, $(\frac{215}{288}, -\frac{67}{36}, -4, 64)$; at $\frac{7}{8}$, $(\frac{1}{288}, -\frac{5}{36}, 4, -64)$; and $\mathrm{up}^{(4)} = 0$ at all three, as Theorem 5.4 requires at depth 3. The top derivative $\pm 64 = \pm 2^{\binom{4}{2}}$ is Equation (61). Odd depth: the law already holds at $n = s - 1 = 2$ (Theorem 4.15): $q_2(\frac{1}{8}) = 1$, which is $\frac{287}{288} + \frac{1}{18} \cdot \frac{1}{16}$; $q_2(\frac{3}{8}) = \frac{3}{4}$, $q_2(\frac{7}{8}) = 0$, each equal to the predicted value. At $n = s - 2 = 1$ it fails by $\mp\frac{1}{96}$ at all three points ($q_1 = 1, \frac{3}{4}, 0$ against $\frac{97}{96}, \frac{73}{96}, -\frac{1}{96}$).

11.4 Depth four and five: two modes

Example 11.4 ($x = \frac{1}{16}$; $s = 4, d = 2$)**.** Three samples at $N = 4$ and the row $(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45})$:

$$q_4 = \frac{24575}{24576}, \qquad q_5 = \frac{1965959}{1966080}, \qquad q_6 = \frac{94365509}{94371840},$$

$$\mathrm{up}(\tfrac{1}{16}) = \frac{1}{45}\, q_4 - \frac{4}{9}\, q_5 + \frac{64}{45}\, q_6 = \frac{2073457}{2073600},$$

and the modes, from the exact Vandermonde solve and, independently, from the formula $C_m(x) = (-1)^m a_m 4^{-m}\, \mathrm{up}^{(2m)}(x)$, with the tiled derivatives $\mathrm{up}''(\frac{1}{16}) = -\frac{5}{9}$ and $\mathrm{up}^{(4)}(\frac{1}{16}) = -1024$:

$$C_1\big(\tfrac{1}{16}\big) = \frac{5}{648}, \qquad C_2\big(\tfrac{1}{16}\big) = -\frac{248}{2025}, \qquad q_n = \frac{2073457}{2073600} + \frac{5}{648}\, 4^{-n} - \frac{248}{2025}\, 4^{-2n} \quad (n \geq 4).$$

In S1's $(n+1)$-convention the same modes read $A_1 = \frac{5}{162}$, $A_2 = -\frac{3968}{2025}$, and its output line for this point, "extraction row $[1, -20, 64]/45$, modes $5/162, -3968/2025$", is the display above. The odd jet is $\mathrm{up}'(\frac{1}{16}) = -\frac{1}{144}$, $\mathrm{up}'''(\frac{1}{16}) = -32$; $\mathrm{up}^{(5)} = 0$. Even depth: at the knot level $n = 3$, $q_3 = \mathrm{up}_3(\frac{1}{16}) = 1$ while the law predicts $\frac{46081}{46080}$; the defect $-\frac{1}{46080}$ is Theorem 7.4 with $k = 8$, $\varepsilon_8 = -1$, $\beta_4 = -\frac{1}{720}$, 2^{-6}. The annihilating recurrence $-q_n + 21q_{n+1} - 84q_{n+2} + 64q_{n+3} = 0$ holds for $n \geq 4$; its residual at $n = 3$ is $-\frac{1}{46080}$ times the leading coefficient, and at $n = 2$ it is nonzero as well ($q_2 = 1$ against a predicted $\frac{15359}{15360}$).

Example 11.5 ($x = \frac{5}{16}$ and $x = \frac{15}{16}$; $s = 4, d = 2$)**.** At $x = \frac{5}{16}$: $q_4 = \frac{6987}{8192}$, $q_5 = \frac{1676281}{1966080}$, $q_6 = \frac{80454331}{94371840}$,

$$\mathrm{up}(\tfrac{5}{16}) = \frac{1767743}{2073600}, \qquad C_1\big(\tfrac{5}{16}\big) = \frac{67}{648}, \qquad C_2\big(\tfrac{5}{16}\big) = \frac{248}{2025},$$

with $\mathrm{up}'' = -\frac{67}{9}$, $\mathrm{up}^{(4)} = 1024$, $\mathrm{up}' = -\frac{215}{144}$, $\mathrm{up}''' = 32$. At the knot level $n = 3$, $q_3 = \frac{41}{48} = \frac{39360}{46080}$ against a predicted $\frac{39359}{46080}$: the defect is $+\frac{1}{46080}$, with $k = 10$ and $\varepsilon_{10} = +1$. By reflection, $x = \frac{15}{16}$ (S4's example) has $\mathrm{up}(\frac{15}{16}) = \frac{143}{2073600}$, $C_1\big(\frac{15}{16}\big) = -\frac{5}{648}$, $C_2\big(\frac{15}{16}\big) = \frac{248}{2025}$, and $x = \frac{11}{16}$ has $\frac{305857}{2073600}$, $-\frac{67}{648}$, $-\frac{248}{2025}$. Note that $\frac{1}{16}$ and $\frac{5}{16}$, which are not reflections of each other, share $|C_2(x)| = \frac{248}{2025} = a_2\, 2^{\binom{5}{2}-4}$, as Equation (73) says every point of depth 4 must.

Example 11.6 ($x = \frac{1}{32}$ and $x = \frac{3}{32}$; $s = 5, d = 2$)**.** Odd depth 5 with two modes. At $x = \frac{1}{32}$:

$$q_5 = \frac{3932159}{3932160}, \qquad q_6 = \frac{188743589}{188743680}, \qquad q_7 = \frac{1006632407}{1006632960},$$

$$\mathrm{up}(\tfrac{1}{32}) = \frac{33177581}{33177600}, \qquad C_1\big(\tfrac{1}{32}\big) = \frac{1}{2592}, \qquad C_2\big(\tfrac{1}{32}\big) = -\frac{124}{2025},$$

with $\mathrm{up}'' = -\frac{1}{36}$, $\mathrm{up}^{(4)} = -512$, the odd jet $\mathrm{up}' = -\frac{143}{1036800}$, $\mathrm{up}''' = -\frac{40}{9}$, $\mathrm{up}^{(5)} = -32768 = -2^{\binom{6}{2}}$, and $\mathrm{up}^{(6)} = 0$. The law holds already at $n = 4$: $q_4 = 1$ equals the predicted value exactly (odd onset), and fails at $n = 3$ by $\frac{7}{737280}$. At $x = \frac{3}{32}$: $q_5 = \frac{3929279}{3932160}$, $q_6 = \frac{188601509}{188743680}$, $q_7 = \frac{1005869527}{1006632960}$, $\mathrm{up}(\frac{3}{32}) = \frac{33152381}{33177600}$, $C_1\big(\frac{3}{32}\big) = \frac{73}{2592}$, $C_2\big(\frac{3}{32}\big) = -\frac{124}{2025}$, and again $q_4 = \frac{1535}{1536}$ is exactly the value the law predicts. The top mode $|C_2(x)| = \frac{124}{2025} = a_2 2^{\binom{5}{2}-5}$ is the odd-depth line of Equation (73).

11.5 Depth six and seven: three modes

Example 11.7 ($x = \frac{1}{64}$; $s = 6, d = 3$)**.** Four samples at $N = 6$ and the row $(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835})$:

$$q_6 = \frac{1509949439}{1509949440}, \quad q_7 = \frac{676457348027}{676457349120}, \quad q_8 = \frac{43293270259811}{43293270343680}, \quad q_9 = \frac{1662461577834037}{1662461581197312},$$

$$\mathrm{up}(\tfrac{1}{64}) = \frac{561842748287}{561842749440}, \qquad C_1(\tfrac{1}{64}) = \frac{143}{18662400}, \qquad C_2(\tfrac{1}{64}) = -\frac{31}{3645}, \qquad C_3(\tfrac{1}{64}) = \frac{4806656}{2679075},$$

with $\mathrm{up}'' = -\frac{143}{259200}$, $\mathrm{up}^{(4)} = -\frac{640}{9}$, $\mathrm{up}^{(6)} = -2097152 = -2^{\binom{7}{2}}$, and the odd jet $\mathrm{up}' = -\frac{19}{16588800}$, $\mathrm{up}''' = -\frac{2}{9}$, $\mathrm{up}^{(5)} = -16384$. Even depth: $q_5 = 1$ against a predicted $\frac{990904319}{990904320}$, a defect of $+\frac{1}{990904320} = 2^{-15} \cdot \frac{1}{30240}$ with $\varepsilon_k = -1$ ($k = 32$). By reflection, $x = \frac{63}{64}$ has $\mathrm{up} = \frac{1153}{561842749440}$ and the negated modes, and its knot-level sample $q_5 = 0$ falls short of the prediction by the same amount.

Example 11.8 ($x = \frac{1}{128}$ and $x = \frac{3}{128}$; $s = 7, d = 3$)**.** At $x = \frac{1}{128}$, four samples at $N = 7$:

$$q_7 = \frac{1352914698239}{1352914698240}, \qquad q_8 = \frac{17317308137431}{17317308137472},$$

$$q_9 = \frac{3324923162384629}{3324923162394624}, \qquad q_{10} = \frac{212795082392578741}{212795082393255936},$$

$$\mathrm{up}(\tfrac{1}{128}) = \frac{179789679820217}{179789679820800}, \qquad C_1(\tfrac{1}{128}) = \frac{19}{298598400},$$

$$C_2(\tfrac{1}{128}) = -\frac{31}{72900}, \qquad C_3(\tfrac{1}{128}) = \frac{2403328}{2679075},$$

with $\mathrm{up}'' = -\frac{19}{4147200}$, $\mathrm{up}^{(4)} = -\frac{32}{9}$, $\mathrm{up}^{(6)} = -1048576 = -2^{\binom{7}{2}-1}$ (the odd-depth top even derivative of Equation (62)), and $\mathrm{up}^{(7)} = -268435456 = -2^{\binom{8}{2}}$. Odd onset: $q_6 = 1$ equals the predicted value; at $n = 5$ the law fails by $\frac{31}{634178764800}$. At $x = \frac{3}{128}$: $q_7 = \frac{1352914622303}{1352914698240}$, $q_8 = \frac{86586535502003}{86586540687360}$, $q_9 = \frac{3324922960085749}{3324923162394624}$, $q_{10} = \frac{106397534697079540 1}{1063975411966279680}$, $\mathrm{up}(\frac{3}{128}) = \frac{179789668823441}{179789679820800}$, $C_1(\frac{3}{128}) = \frac{25219}{298598400}$, $C_2(\frac{3}{128}) = -\frac{2263}{72900}$, $C_3(\frac{3}{128}) = \frac{2403328}{2679075}$, and $q_6 = \frac{23592959}{23592960}$ is exactly the predicted value.

11.6 A table of exact laws

Table 5: Exact laws $q_n = \mathrm{up}(x) + \sum_{m=1}^{d} C_m(x)\, 4^{-mn}$, valid for all $n \geq s$ (and for $n \geq s - 1$ at odd s). Every entry is an exact rational, recomputed for this volume.

x	s	d	$\mathrm{up}(x)$	$C_1(x), \ldots, C_d(x)$
0	0	0	1	—
± 1	0	0	0	—
$\frac{1}{2}$	1	0	$\frac{1}{2}$	—
$\frac{1}{4}$	2	1	$\frac{67}{72}$	$\frac{1}{9}$
$\frac{3}{4}$	2	1	$\frac{5}{72}$	$-\frac{1}{9}$
$\frac{1}{8}$	3	1	$\frac{287}{288}$	$\frac{1}{18}$
$\frac{3}{8}$	3	1	$\frac{215}{288}$	$\frac{1}{18}$
$\frac{7}{8}$	3	1	$\frac{1}{288}$	$-\frac{1}{18}$
$\frac{1}{16}$	4	2	$\frac{2073457}{2073600}$	$\frac{5}{648}, -\frac{248}{2025}$
$\frac{3}{16}$	4	2	$\frac{2026943}{2073600}$	$\frac{67}{648}, \frac{248}{2025}$
$\frac{5}{16}$	4	2	$\frac{1767743}{2073600}$	$\frac{67}{648}, \frac{248}{2025}$
$\frac{11}{16}$	4	2	$\frac{305857}{2073600}$	$-\frac{67}{648}, -\frac{248}{2025}$
$\frac{15}{16}$	4	2	$\frac{143}{2073600}$	$-\frac{5}{648}, \frac{248}{2025}$

x	s	d	$\mathrm{up}(x)$	$C_1(x), \dots, C_d(x)$
$\frac{1}{32}$	5	2	$\frac{33177581}{33177600}$	$\frac{1}{2592}, -\frac{124}{2025}$
$\frac{3}{32}$	5	2	$\frac{33152381}{33177600}$	$\frac{73}{2592}, -\frac{124}{2025}$
$\frac{5}{32}$	5	2	$\frac{32842819}{33177600}$	$\frac{215}{2592}, \frac{124}{2025}$
$\frac{1}{64}$	6	3	$\frac{561842748287}{561842749440}$	$\frac{143}{18662400}, -\frac{31}{3645}, \frac{4806656}{2679075}$
$\frac{37}{64}$	6	3	$\frac{966420578371}{2809213747200}$	$-\frac{305857}{18662400}, \frac{2077}{18225}, \frac{4806656}{2679075}$
$\frac{1}{128}$	7	3	$\frac{179789679820217}{179789679820800}$	$\frac{19}{298598400}, -\frac{31}{72900}, \frac{2403328}{2679075}$
$\frac{3}{128}$	7	3	$\frac{179789668823441}{179789679820800}$	$\frac{25219}{298598400}, -\frac{2263}{72900}, \frac{2403328}{2679075}$

Two regularities are visible and both are theorems: within one depth the top coefficient $C_d(x)$ has a constant absolute value (Equation (73)), and reflection $x \mapsto 1-x$ negates every mode (Theorem 11.1). The denominators of $\mathrm{up}(x)$ are $72 = 2^3 \cdot 9$, $288 = 2^5 \cdot 9$, $2073600 = 2^{10} \cdot 2025$, $33177600 = 2^{14} \cdot 2025$, $561842749440 = 2^{21} \cdot 267915 \cdot \ldots$; their growth is the subject of Theorem 10.2.

Section summary

Twenty dyadic points through depth 7, every number exact and independently recomputed: the samples, the extraction, the value, every mode, and the jet. The examples exhibit the law holding from the onset, holding one level early at odd depth (Theorems 11.3, 11.6 and 11.8), failing at the knot level at even depth by exactly the Bernoulli amount of Theorem 7.4 (Theorems 11.2, 11.4, 11.5 and 11.7), and the reflection symmetry that halves the work (Theorem 11.1).

12 An exact algorithm and its verification

Each source ends with a procedure and a program. The procedures are the same procedure and the programs check the same identities; this section states the procedure once, with its preconditions and postconditions, and specifies what the single verifier retained with this volume checks.

12.1 The algorithm

Algorithm 12.1 (Exact dyadic extraction)**.** *Input.* A dyadic rational $x \in [-1, 1]$, given either in lowest terms or as a pair (m, n) with $x = m/2^n$; the arithmetic is that of $\mathbb{Q}$ with unbounded integers.

Output. The exact rational $\mathrm{up}(x)$; optionally every mode $C_1(x), \dots, C_d(x)$, every even derivative $\mathrm{up}^{(2)}(x), \dots, \mathrm{up}^{(2d)}(x)$, and a certificate.

1. (*Depth.*) Compute $s = s(x) = \max(0, n - \nu_2(m))$ (Theorem 3.2; for $x = 0$ take $s = 0$) and $d = \lfloor s/2 \rfloor$. If $|x| = 1$ return 0; if $x = 0$ return 1 (Theorem 3.5).
2. (*Window.*) Choose $N = s$; if $s \geq 3$ is odd, $N = s - 1$ is also admissible (Theorem 4.15), and any N above these is admissible but wasteful.
3. (*Samples.*) For $j = 0, \dots, d$ compute $q_{N+j} = \mathrm{up}_{N+j}(x)$ exactly from the scaled cell formula Equation (20): with $y = 2^n(x+1) - \frac{1}{2}$ and $n = N + j$,
$$q_n = \frac{2^{-\binom{n}{2}}}{n!} \sum_{k=0}^{\lfloor y \rfloor} \varepsilon_k \, (y-k)^n,$$
where, since $n \geq s$, $2y$ is an odd integer and the sum involves only integer powers of half-integers, so that $2^{\binom{n+1}{2}} n! \, q_n$ is an integer (Theorem 2.6).
4. (*Row.*) Compute $w_j^{(d)}$, $0 \leq j \leq d$, from Equation (98), $w_j^{(d)} = (-1)^{d-j} 4^{\binom{j+1}{2}} / (P_j P_{d-j})$ with $P_i = \prod_{k \leq i} (4^k - 1)$ accumulated once, or from the q-Pochhammer form Equation (96); the two agree (Theorem 8.3).

5. (*Value.*) Return $\mathrm{up}(x) = \sum_{j=0}^{d} w_j^{(d)}\, q_{N+j}$ (Theorem 8.7); equivalently run the triangular tableau of Theorem 9.7, whose last entry is the same rational.
6. (*Modes and jet, optional.*) Compute $C_m(x) = 4^{mN} \sum_j q_{N+j}(-1)^{d-m} e_{d-m}^{(j)} / D_j$ from Equation (109), or solve the $(d+1) \times (d+1)$ geometric Vandermonde system Equation (102) exactly; then $\mathrm{up}^{(2m)}(x) = (-1)^m 4^m C_m(x)\,/\,a_m$ (Theorem 9.3), with a_m from the recurrence Equation (44).
7. (*Certificate, optional.*) Any of: (a) repeat step 5. from the window $N+1$ and compare; (b) compute one further sample q_{N+d+1} and compare with the law; (c) evaluate the residual of the integer recurrence Equation (114) at $n = N$. Each must give exact equality (Theorem 9.103); a nonzero residual proves an indexing or arithmetic error, or that N is below the onset.

Postcondition. The returned rational is $\mathrm{up}(x)$, with no limit, tolerance or rational reconstruction; the modes and the jet are exact; and, for $s \geq 2$, every returned mode is nonzero (Theorem 7.1).

Correctness. Steps 1.–2. are Theorems 1.1, 3.2 and 4.15; step 3. is Theorem 2.6; steps 4.–5. are Theorems 8.3, 8.7 and 9.7; step 6. is Theorems 9.2 and 9.3; step 7. is Theorem 9.10. ☐

Listing 1: The core of Theorem 12.1 in exact rational arithmetic; `up_n(x, n)` is the sample $\mathrm{up}_n(x)$ of step 3 and `P` the products P_i.

```
s = depth(x); d = s // 2; N = s
samples = [up_n(x, N + j) for j in range(d + 1)]
P = [1]
for k in range(1, d + 1): P.append(P[-1] * (4**k - 1))
weights = [(-1)**(d-j) * Fraction(4**((j*(j+1))//2), P[j]*P[d-j])
           for j in range(d + 1)]
value = sum(w * q for w, q in zip(weights, samples))   # = up(x), exactly
```

12.2 Cost

The extraction itself is negligible: $O(d)$ rational operations for the value once the P_i are cached, $O(d^2)$ for the tableau or the closed inverse, $O(d^3)$ for a dense solve, with $d = \lfloor s/2 \rfloor$. The cost is in the samples: the truncated-power sum for q_n has up to 2^{n+1} terms, each a power of a half-integer of about $n \log_2(2^n) = n^2$ bits, so the reference evaluator is exponential in the depth and is intended for transparent verification at moderate depth (the exhaustive runs below reach depth 7 or 8 in seconds). Faster exact evaluators for the samples—binary block recurrences, repeated integration of the piecewise polynomial, or the dyadic arithmetic of [2] and of the Lean corpus [10]—can be substituted without changing any theorem, since the extraction sees only the $d+1$ rationals.

In floating point the picture reverses: the samples are the hazard (an alternating Thue–Morse sum, Theorem 10.3) and the row is benign, amplifying uniform absolute error by less than 1.97 (Theorem 10.1); if a denominator bound is known (Theorem 10.2), rational reconstruction from a high-precision result is possible but never necessary, since exact input is available.

12.3 The verification harness

Each of the six packages shipped a Python program using only the standard library and exact fractions (`fractions.Fraction`), together with its captured output. Consolidated, they checked the following, all as equalities of reduced fractions and never to a tolerance:

- S1: the q-Pochhammer and Gaussian-binomial forms of the row agree; interpolation of the mode polynomial and extraction of all modes; Bernoulli numbers, cumulants, the coefficients $\gamma_{2m}^{\mathrm{Rv}}$ and the dyadic derivatives by the Thue–Morse stencil; interpolated and differential-operator coefficients coincide; the expansion and the recurrence have zero residual;

- S2: the truncated-power formula, the weights, an exact Gauss–Jordan solve, prediction of unused levels, the recurrence; exhaustively at all 129 signed dyadic points of depth at most 6;
- S3: the evaluator, the weights, the Bell–Bernoulli recurrence for a_m, exact values and derivatives; 2376 identities at the 255 interior points of depth at most 7, including sliding-window invariance and the derivative formula at several consecutive levels;
- S4: extraction from four windows agrees; fitted coefficients reproduce six further levels; fitted coefficients equal the analytic $(-1)^m a_m 4^{-m}\,\mathrm{up}^{(2m)}(x)$ with tiled derivatives; the recurrence has zero residual at six consecutive levels; 2570 sequence and window identities, 941 coefficient entries and 1542 recurrence identities at all 257 points of depth at most 7;
- S5: the evaluator, the weights, the triangular table, an exact solve, overlapping-block and out-of-sample checks at 129 points of depth at most 6;
- S6: closed-row and recursive extraction, an exact solve, the reciprocal moments from Bernoulli numbers, assertions for two worked examples, and an optional exhaustive test through a chosen depth.

The volume retains one verifier, `verify_dyadic_up_extraction.py`, adapted from S4's (the most exhaustive) and extended by the claims that are specific to this volume. Its specification:

(V1) it evaluates the finite splines by the Thue–Morse truncated-power formula in the $(n+1)$-factor indexing, computes the quarter-base row in both closed forms, solves the geometric Vandermonde system exactly, and evaluates the tiled derivatives;

(V2) at every reduced dyadic point of depth at most 7 (257 points, including $0, \pm 1$) it checks the law from four windows, six further levels, the analytic-versus-fitted coefficients, and the recurrence residuals: 2570, 941 and 1542 identities;

(V3) it checks $a_0, \ldots, a_5$ against the printed table (Theorem 4.6);

(V4) at the 252 interior points of depth 2 to 7 it checks that all 684 mode coefficients are nonzero (Theorem 7.1), that the reversed row Equation (97) equals the forward row, and that the law holds at $n = s - 1$ at all 168 points of odd depth (Theorem 4.15);

(V5) at the same 252 points it checks the closed form of the knot-level defect, Theorem 7.4: the difference between $\mathrm{up}_{s-1}(x)$ and the law's prediction equals $-\varepsilon_k 2^{-\binom{s}{2}}\beta_s$, which is nonzero at all 84 points of even depth and zero at the 168 of odd depth;

(V6) it checks $\sum_j \left|w_j^{(d)}\right| = (-Q;Q)_d \,/\, (Q;Q)_d$ for $d \le 6$ (Theorem 10.1);

(V7) it writes nothing unless asked (`-out-dir`), runs in about twelve seconds, and accepts `-max-depth` for deeper runs.

The knot-defect formula was additionally checked through depth 8 (508 interior points) with a separate script during the consolidation. All these computations are evidence that the formulas are transcribed and indexed correctly; the proofs do not depend on them.

Section summary

One algorithm (Theorem 12.1): depth, window, $d+1$ exact samples, one row, one dot product; optionally all modes and the jet, and a certificate that is a finite exact test. The samples are the only expensive part. One retained verifier checks, at every reduced dyadic point of depth at most 7, the law, the coefficients, the recurrence, the nonvanishing of all modes, both forms of the row, the odd-depth onset, the Bernoulli formula for the knot-level defect, and the Lebesgue factor.

13 Consequences and extensions

13.1 What is universal and what is dyadic

Two logically distinct structures meet in Theorem 1.1. The algebra—the finite deconvolution by a reciprocal power series, the geometric Lagrange row, its moment identities, the annihilating recurrence—is universal: it holds for any sequence of the form $L + \sum_{r \le d} A_r Q^{rn}$ and for any base. The exactness is arithmetic: it needs the point to become the centre of a polynomial cell and the jet to terminate, and both are properties of dyadic points. The following proposition isolates the universal part.

Proposition 13.1 (Local deconvolution principle). *Let K be an even probability density on $\mathbb{R}$ with compact support, all of whose moments $\mu_{2j} = \int u^{2j} K$ exist (automatic), and let $B_K(z) = \sum_j \mu_{2j} z^{2j}/(2j)!$ be its moment generating function, $1/B_K(z) = \sum_j b_j z^{2j}$ its formal reciprocal. Let f be a function, x a point, $\delta > 0$, and p a polynomial such that*

(H1) *$f = g * K_\delta$ with $K_\delta = \delta^{-1} K(\cdot/\delta)$, for some locally integrable g;*

(H2) *$g = p$ on $[x - \delta', x + \delta']$ for some $\delta' > \delta$ with $\operatorname{supp} K_\delta \subseteq [-\delta', \delta']$;*

(H3) *the jet of f at x vanishes above order R: $f^{(r)}(x) = 0$ for $r > R$.*

Then

$$p(x) = \sum_{j=0}^{\lfloor R/2 \rfloor} b_j \, \delta^{2j} \, f^{(2j)}(x), \tag{123}$$

a finite identity. If moreover $\delta = \delta_n = c\alpha^n$ along a sequence of pairs (g_n, p_n) satisfying (H1)–(H2), then $n \mapsto p_n(x)$ is $f(x)$ plus a finite sum of geometric modes α^{2jn}, $1 \le j \le \lfloor R/2 \rfloor$, and the geometric Lagrange row at base $q = \alpha^2$ of order $\lfloor R/2 \rfloor$ returns $f(x)$ exactly from $\lfloor R/2 \rfloor + 1$ consecutive values of $p_n(x)$.

Proof. On polynomials, convolution with K_δ is the finite operator $B_K(\delta D)$, $D = \mathrm{d}/\mathrm{d}t$: for a monomial, $(t^k * K_\delta)(x) = \int (x-y)^k K_\delta(y)\,\mathrm{d}y = \sum_j \binom{k}{2j} \mu_{2j} \delta^{2j} x^{k-2j}$ by evenness, which is $B_K(\delta D) t^k$ evaluated at x. By (H1)–(H2), f and all its derivatives at x are those of $p * K_\delta = B_K(\delta D)p$: $f^{(r)}(x) = (B_K(\delta D)p)^{(r)}(x)$ for every r. The operator $B_K(\delta D)$ is invertible on polynomials with inverse $(1/B_K)(\delta D) = \sum_j b_j \delta^{2j} D^{2j}$, a finite sum on any polynomial, so $p(x) = \sum_j b_j \delta^{2j} (B_K(\delta D)p)^{(2j)}(x) = \sum_j b_j \delta^{2j} f^{(2j)}(x)$, the sum over $2j \le \deg p$; by (H3) the terms with $2j > R$ vanish, and if $\deg p < R$ the missing terms vanish on both sides. The geometric statement is Equation (123) with $\delta_n^{2j} = c^{2j}\alpha^{2jn}$, and the extraction is the moment identity of the row (Theorems 8.2 and 8.3 hold for every base $q \in (0,1)$, indeed for every q with $(q;q)_d \ne 0$). □

The Rvachev case is the cleanest instance: $K = \mathrm{up}$ itself (the tail is a scaled copy of the whole density), $\delta_n = 2^{-(n+1)}$ so $\alpha = \frac{1}{2}$ and $q = \frac{1}{4}$, $g = \mathrm{up}_n$ with the knot lattice aligned to the dyadic denominators, and the jet terminates at the order $s(x)$. For a general base $b > 1$ of sinc products the same principle gives $q = b^{-2}$; what would be missing is a terminating jet at the natural lattice, which is a property of the base-2 Rvachev function (Section 5) and is not known for other bases.

13.2 Other windows, strides and functionals

Proposition 13.2 (Arbitrary levels). *Let $n_0 < n_1 < \cdots < n_d$ be any $d+1$ levels with $n_0 \ge s$. Then*

$$\mathrm{up}(x) = \sum_{j=0}^{d} q_{n_j} \prod_{\substack{0 \le l \le d \\ l \ne j}} \frac{-Q^{n_l}}{Q^{n_j} - Q^{n_l}}. \tag{124}$$

In particular, for samples at a fixed stride $t \ge 1$, $n_j = N + jt$, the row is Equation (96) with Q^t in place of Q.

Proof. By Theorem 1.1 the samples are the values at the distinct nodes Q^{n_j} of the polynomial $L + \sum_r A_r z^r$ of degree at most d, whose constant term is $\mathrm{up}(x)$; Equation (124) is Lagrange interpolation at those nodes evaluated at $z = 0$. For $n_j = N + jt$ the common factor Q^N cancels and the nodes are $1, Q^t, \ldots, Q^{dt}$. □

The consecutive window is preferable only because the products collapse into the q-Pochhammer form and the Lebesgue factor is then smallest for the given d. More generally, any linear functional of the local jet of up_n at x—a derivative, a finite combination of values and derivatives at points of the same cell—inherits a finite geometric law by applying the functional to Equation (54) (Theorem 6.2 is the case of one derivative), and the same rows extract its limit.

13.3 The distribution-function version

The Lean corpus works with distribution functions (Fabius's function) rather than densities; the law transports.

Proposition 13.3 (CDF version). *Let $P_n(x) = \int_{-\infty}^x \mathrm{up}_n$ and $P(x) = \int_{-\infty}^x \mathrm{up}$. For $x \in [-1, 1]$ and $n \geq 0$,*

$$P_n(x) = 1 - \mathrm{up}_{n+1}\Big(\frac{x+1}{2}\Big), \qquad P(x) = 1 - \mathrm{up}\Big(\frac{x+1}{2}\Big). \tag{125}$$

Hence, if x is dyadic of depth s, the point $y = (x+1)/2$ is dyadic of depth $s + 1$ (depth 1 if $x = 0$), and for every $n + 1 \geq s(y)$,

$$P_n(x) = P(x) - \sum_{m=1}^{\lfloor s(y)/2 \rfloor} C_m(y)\, 4^{-m(n+1)} : \tag{126}$$

the distribution-function sequence at a dyadic point of depth s has exactly $\lfloor (s+1)/2 \rfloor$ modes, all nonzero for $s \geq 1$, holds from $n \geq s$ on (from $n \geq s - 1$ on when s is even, $s \geq 2$), and is extracted by the row of order $\lfloor (s+1)/2 \rfloor$.

Proof. Equation (83) at the point $y = (x+1)/2 \in [0, 1]$ gives $\mathrm{up}_{n+1}(y) = P_n(2y+1) - P_n(2y-1) = P_n(x+2) - P_n(x) = 1 - P_n(x)$, since $x + 2 \geq 1$; the same for P. The depth of $y = (x+1)/2$ is $s + 1$ for $s \geq 1$ by Theorem 3.2, and $s\big(\frac{1}{2}\big) = 1$. Then Theorem 1.1 at y, with its onset $n + 1 \geq s(y)$ and its odd-depth improvement (Theorem 7.6), transports through Equation (125); the number and nonvanishing of the modes are Theorem 7.1 at y. □

At $x = \frac{1}{2}$ this is Theorem 7.11: $y = \frac{3}{4}$, one mode, $P_n(\frac{1}{2}) - P(\frac{1}{2}) = -C_1\big(\frac{3}{4}\big)\, 4^{-(n+1)} = \frac{1}{9} 4^{-(n+1)}$. The Lean statement `Fabius.fabiusUniformSpline_quarter_eq_quadratic` (Section 14) is this one-mode CDF law at the quarter point of the Fabius interval, which is $y = \frac{3}{4}$ in the present coordinates (Section 2.5).

13.4 Non-dyadic points

At a non-dyadic point nothing of the exact structure survives (Theorem 7.14): the tail window always contains a knot, and the jet does not terminate. The Exponents volume's all-orders expansion and its sharp C^r law are the statements that hold there, and the signed Richardson combination accelerates with the asymptotic remainder of its `p3:eq:richardson-leading`. The contrast is not a matter of degree: at a dyadic point the remainder is identically zero from a computable level on; at every other point of $(-1, 1)$ it is nonzero for infinitely many n, because a sequence q_n that eventually satisfied a finite geometric law would make the tail generating function rational and the jet at x finite by Theorem 6.5 read backwards through Equation (54)—which requires x to be a cell centre for all large n, hence dyadic.

13.5 Open problems

(O1) *The levels below the knot level.* Theorem 7.4 gives the defect at $n = s - 1$ in closed form. At $n = s - 2$ the defect is observed to be independent of the point within a depth (Theorem 7.7); a closed form, presumably in terms of partial moments of up over $[-1, \pm\frac{1}{2}]$, would prove that the onset at odd depth is exactly $s - 1$ for all s, which is now known through depth 7.

(O2) *Formalization.* The general theorem is absent from the Lean corpus; the register in Section 14 lists what exists and what a proof would have to build.

(O3) *Fast exact samples.* The extraction is linear in d; the reference evaluator for the samples is exponential in n. An exact evaluator polynomial in n for up_n at dyadic points, in the spirit of the dyadic arithmetic of [2], would make the method practical at large depth.

(O4) *Other kernels.* Theorem 13.1 applies to any self-similar even kernel with a terminating jet on a lattice. Which kernels beyond up (other Rvachev-type solutions of functional-differential equations, other bases) have this property is not known to the authors of the six sources or to this volume.

(O5) *Extremizers of the sharp law.* Theorem 7.10 shows that the sup norm in the Exponents volume's sharp C^r law is attained at every dyadic point of depth $r+2$. Whether these are all the extremizers is not settled here.

Section summary

The algebra of the method is universal (Theorem 13.1); its exactness is arithmetic and belongs to dyadic points. Arbitrary levels and strides, derivatives and other jet functionals inherit the finite law (Theorem 13.2); the distribution-function version has $\lfloor (s+1)/2 \rfloor$ modes and is the form the Lean corpus uses (Theorem 13.3); away from dyadic points only the asymptotic statements of the Exponents volume remain. Five open problems are recorded.

14 Relation to the corpus and formalization register

14.1 The Exponents volume

Part III of the Exponents volume [9] studies the same finite splines at a general point; its objects and this volume's are related by the bridge $\mathrm{up}_n = p_{n+1}$, $\Phi_n = \Phi_{n+1}$, $\delta_n^{\text{here}} = \delta_{n+1}^{\text{there}}$ of Equation (22), recorded in its Part 0 dictionary (its table `p0:tab:dictionary`, row "finite products"). What it proves, and what becomes of it at a dyadic point:

Exponents volume (label cited by text)	At a dyadic point of depth s (this volume)
`p3:eq:prefix-product`: the prefix product $\Phi_n = \prod_{j<n} \mathrm{sinc}_{\mathrm{rad}}(\pi\xi/2^j)$ and p_n its inverse transform, of degree at most $n-1$	Theorem 2.1: the same object as up_{n-1}; the Thue–Morse formula Theorem 2.5 with its genuine knots
`p3:thm:sharp-prefix-law`: $\left\lVert p_n^{(r)} - \mathrm{up}^{(r)} \right\rVert_\infty = 2^{\binom{r+3}{2}-1}/(9 \cdot 4^n)$ for $n \geq r+3$	attained at every dyadic point of depth $r+2$, for every $n \geq r+3$, with the one-mode exact law (Theorem 7.10)
`p3:cor:kolmogorov`: the CDF discrepancy is exactly $1/(9 \cdot 4^n)$, at $x = \frac{1}{2}$	the $s = 2$ density law at $x = \frac{1}{4}$ transported by the finite functional equation (Theorem 7.11)
`p3:eq:richardson-combination`, `-weights`, `-moments`: $R_{n,M} = \sum_{j<M} w_{M,j} p_{n+j}$ with the quarter-base Lagrange weights	the same weights, $w_{M,j} = w_j^{(M-1)}$ (Theorem 8.3); the same moment identities (Theorem 8.2)
`p3:eq:richardson-leading`: $R_{n,M}^{(r)} = \mathrm{up}^{(r)} - a_M q^{\binom{M}{2}} \delta_n^{2M} \mathrm{up}^{(r+2M)} + O(\delta_n^{2M+2})$, asymptotic at a general point	exact, with zero remainder, for $n \geq s+1$ and $M \geq \lfloor s/2 \rfloor + 1$ (Theorem 7.13); the leading coefficient vanishes because the jet terminates
`p3:eq:richardson-stability`: $\sum_j \lvert w_{M,j} \rvert < 1.9693$	the same Lebesgue factor, in closed form $(-Q;Q)_{M-1} / (Q;Q)_{M-1}$ (Theorem 10.1)

The contribution of this volume relative to Part III is therefore a single sentence: *at a dyadic point the asymptotic expansion is finite and exact from the level $s(x)$ on, so the same Richardson row is an exact reconstruction operator*, together with the exact knowledge of when the finite law starts (Theorem 7.6).

14.2 The half-base row

The corpus also contains a general half-base Gaussian-binomial row for shifted finite splines, with row polynomial $(z/2;1/2)_n\,/\,(1/2;1/2)_n$, which annihilates modes 2^{-rn}. The quarter base is the natural and minimal one for the centred sequence of this volume because the tail is even: only even derivative orders $2k$ survive the deconvolution, with the scale factor $\delta_n^{2k} = 4^{-k(n+1)}$, so the modes are 4^{-kn} and the quarter-base row of order $\lfloor s/2 \rfloor$ uses $\lfloor s/2 \rfloor + 1$ samples, where a half-base row prepared for the absent odd modes would need about twice as many. A notational caution: in parts of the corpus the letter Q denotes a complex translation parameter of a shifted spline; the $Q = \frac{1}{4}$ of this volume is the geometric decay ratio of the centred sequence, and Section A keeps the two apart.

14.3 Lean register

Every name below was checked to resolve in the repository [10] on the day of consolidation, namespace-aware (the module is given in parentheses). Three statuses are used: CROSSWALKED, a declaration states the same fact (possibly in the corpus's CDF or Appell normalization, with the translation stated); NEIGHBOURING, a declaration a reader could mistake for the result, with the difference stated; ABSENT, no declaration found. The crosswalks stated in the body of the volume (Sections 2.5, 4 and 5, and Theorem 8.11) are collected here.

Table 6: Formalization register.

Result in this volume	Lean declaration(s)	Status
Theorem 2.1, Theorem 2.5: the finite spline and its Thue–Morse truncated-power form	`Fabius.fabiusUniformSpline` (`FabiusUniformSpline`), defined *as* the truncated-power sum in the centred-CDF variable; `Fabius.ProbabilityRepresentation.fabiusUniformSpline_eq_centeredPartialCDF` identifies it with the partial-sum distribution function	CROSSWALKED (CDF form; the density up_n is its derivative, Section 2.5)
Theorem 2.2, Theorem 2.3: the transform of the finite product and the random-sum model	`Fabius.centeredSincPartialProduct_dyadic_eq_thueMorse` (`CenteredRvachevThueMorseFourier`): the denominator-cleared centred sinc prefix equals the Thue–Morse sine polynomial; `Fabius.ProbabilityRepresentation.rvachevUp_eq_weightedSum_probability` and . . . `geometricUniformDensity_one_half_eq_rvachevUp`: up is the density of $\sum_k 2^{-k} V_k$	CROSSWALKED (the infinite sum); the finite head–tail split $Z = Z_n + \delta_n Z'$ of Theorem 2.4: ABSENT
Theorem 3.3, Theorem 3.4: knot lattice, genuine knots, cell centres	—	ABSENT
Theorem 4.2, Theorem 4.6(a),(b): the coefficients a_m, their recurrence and Bell form	`Fabius.rvachevReciprocalMomentRat`, . . . `_eq_completeBellPolynomial`, `Fabius.binomialConv_rvachevRawMomentRat_reciprocal` (`RvachevMomentAppell`): $a_m = (-1)^m \gamma_{2m}^{\mathrm{Rv}}/(2m)!$ is Lean-computable	CROSSWALKED; positivity (c) and growth (d): ABSENT

Result in this volume	Lean declaration(s)	Status
Theorem 4.10, Theorem 4.13: finite polynomial deconvolution	`Fabius.rvachevDeconvolvedPolynomial`, `Fabius.rvachevAppellPolynomialRat`, `Fabius.integral_eval_rvachevAppellPolynomial_sub_mul_rvachev` (`RvachevMomentAppell`), in the Appell normalization at scale 1	CROSSWALKED at scale 1; the dilation to δ_n and the cell identity: ABSENT
Theorem 5.1: Thue–Morse derivative stencil	`Fabius.iteratedDeriv_rvachevUp_eq_thueMorse_sum` (`Differential`); Rvachev's equation itself is `Fabius.deriv_rvachevUp` (same module)	CROSSWALKED
Theorem 5.4(i),(ii): the jet stops at the depth; the top derivative	`Fabius.iteratedDeriv_rvachevUp_dyadic_eq_zero`, `. . . _dyadic_critical`, `Fabius.dyadic_depth_eq_max_nonzero_iteratedDeriv` (`DyadicDerivativeFiltration`); `Fabius.iteratedDeriv_rvachev_centeredDyadic_eq_zero`, `. . . _ne_zero` (`OriginalPaperSupplement`); `Fabius.rvachevDyadicTaylorPolynomial_natDegree_eq`	CROSSWALKED; part (iii) for $0 < r < s$: ABSENT as a separate statement
Theorem 5.5, Equation (64): the even jet and the sharp amplitude on the matched grid	`Fabius.iteratedDeriv_fabiusReal_dyadicGrid_eq_ite`, `. . . _eq_zero_iff`, `Fabius.abs_iteratedDeriv_fabiusReal_dyadicGrid_of_odd` (`BoundedDerivatives`): on the mesh $m/2^k$ the k-th derivative vanishes at even m and has absolute value $2^{\binom{k+1}{2}}$ at odd m	CROSSWALKED (CDF form)
Theorem 6.2 at $r = s$: $\mathrm{up}_n^{(s)}(x) = \mathrm{up}^{(s)}(x)$ for all $n \geq s$	`Fabius.iteratedDeriv_fabiusUniformSpline_dyadic_center` (`PlateauLimit`): the r-th derivative of the level-p spline at the anchor 2^{-r} equals $2^{\binom{r+1}{2}}$ for every $p \geq r$	CROSSWALKED (one anchor per depth, CDF form)
Theorem 1.1 at $x = \frac{3}{4}$ ($s = 2$, one mode), CDF form	`Fabius.fabiusUniformSpline_quarter_eq_quadratic`, `Fabius.reportFiniteFabiusApproximant_quarter_eq_quadratic` (`QuarterSplineLocalPolynomial`): the level-n centred CDF near its quarter point is $\frac{5}{72} + z + 4z^2 - \frac{4}{9}4^{-n}$	CROSSWALKED INSTANCE: kernel-verified for one point and one mode; the statement for general depth: ABSENT
Theorem 1.1, Theorem 5.7: the exact eventual law at general depth	—	ABSENT
Theorem 7.4, Theorem 7.3, Theorem 7.6	—	ABSENT
Theorem 8.1, Theorem 8.3: the geometric Lagrange weights and their closed forms	`Fabius.geometricLagrangeWeight`, `. . . _eq_product` (`GeometricLagrangeWeights`); `. . . _eq_geometricQPochhammer`, `Fabius.quarterLagrangeWeight_eq_qPochhammer`, `. . . _eq_qBinomial`, `Fabius.quarter_qPochhammer_self_ne_zero` (`GeometricLagrangeQBinomial`)	CROSSWALKED

Result in this volume	Lean declaration(s)	Status
Theorem 8.2, Equation (91), Theorem 8.5: moment identities, uncancelled moments, row polynomial	`Fabius.sum_quarterLagrangeWeight_eq_one`, . . . `_mul_pow_eq_zero`, `Fabius.sum_geometricLagrangeWeight_mul_pow_card_add`, `Fabius.sum_quarterLagrangeWeight_firstUncancelled`, `Fabius.quarterLagrangeWeightPolynomial_eq_forwardRichardson` (`GeometricLagrangeWeights`)	CROSSWALKED
Theorem 8.7: the row applied to the up-function sequence	`Fabius.quarterLagrange_rvachevFourierProduct_eq_gaussian_tsum` (`RvachevQBinomialFilter`): the row applied to the *infinite* product on the Fourier side, cancelling the first d even moments	NEIGHBOURING (Fourier side, asymptotic content); the theorem itself: ABSENT, pending the law
Theorem 7.13: exactness of the Richardson combination	`Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_qBinomial` (`QuarterCatalanRichardson`): a formal power-series filter with the same coefficients; its header disclaims any asymptotic meaning	NEIGHBOURING
Theorem 9.2, Theorem 9.9, Theorem 10.1: closed inverse, annihilating recurrence, Lebesgue factor	— (the general-q Toeplitz form of the row exists in `GeometricLagrangeQBinomial`; no statement of the inverse, the recurrence, or the absolute row sum was found)	ABSENT

14.4 What a formalization would build, in order

The general theorem is absent, and the register shows why: the corpus has the algebraic layer (the row and its identities), the arithmetic layer (the jet termination and the derivative amplitudes, in several forms), the coefficients, and the deconvolution at scale 1; what it lacks is the analytic bridge between them. A proof of Theorem 1.1 would add, in this order:

(F1) the density up_n on $\mathbb{R}$ as the derivative of `fabiusUniformSpline`, or directly as the $(n+1)$-fold box convolution, with its knot lattice and the cell-centre lemma Theorem 3.4 (the genuine-knot statement is the nonvanishing of the Thue–Morse sign, which is elementary);

(F2) the head–tail factorization $\mathrm{up} = \mathrm{up}_n * T_{\delta_n}$ as an identity of measures, from `ProbabilityRepresentation.weightedSumDistribution` by splitting the weighted sum at index n;

(F3) the finite deconvolution at scale δ on polynomials, by dilating `rvachevDeconvolvedPolynomial`, and the cell identity Theorem 4.13 (a finite computation once (F1)–(F2) are available);

(F4) the assembly with `dyadic_depth_eq_max_nonzero_iteratedDeriv` (Section 6.1), which gives the law, and with the existing row identities, which gives Theorem 8.7 with no new analysis;

(F5) for Theorem 7.4: the half-range identity Theorem 7.3 from `deriv_rvachevUp` by one integration by parts, and the two-piece transfer, which is (F3) with one truncated power added.

Keeping (F1)–(F2) (measure theory) separate from (F3)–(F4) (finite algebra) is what makes the statements reusable for the other kernels of Item (O4).

Section summary

Relative to the Exponents volume the contribution is exactness at dyadic points, with the onset known; the quarter base is forced by the even tail. In Lean the row and its identities, the jet termination and its amplitudes, the coefficients and the deconvolution at scale 1 are kernel-verified, and the one-mode law at the quarter point is a verified instance; the finite splines' knot geometry, the head–tail split at finite level, the general law, and the knot-level defect are absent, and the five-step plan of Section 14.4 is the shortest route to them.

Appendices

A Notation, and a concordance with the six sources

A.1 The normative source

Every symbol here is defined normatively in the corpus notation catalogue: the entries `rvachev-up`, `up-fourier-product`, `sinc`, `fabius-dyadic-cells`, `q-pochhammer`, `gaussian-binomial`, `full-moments`, `splines`, `thue-morse-block-sum` and `rvachev-appell-deconvolution` cover the material of this volume. The preamble defines a few document-local shorthands, listed below; each expands to catalogue notation or names an object this volume defines in Section 2, and none carries a meaning the text does not state.

A.2 Symbols used throughout

Table 7: Principal notation.

Symbol	Alias	Meaning
up	—	Rvachev's up-function: even, $\int \mathrm{up} = 1$, support $[-1, 1]$, $\mathrm{up}(x) = F_{\mathrm{bd}}(1 - \lvert x \rvert)$
$\widehat{f}_{2\pi}$	—	the 2π-convention transform $\int f(x)\mathrm{e}^{-2\pi\mathrm{i}tx}\,\mathrm{d}x$
$\mathrm{sinc}_{\mathrm{rad}}(z)$	—	$\sin z/z$, value 1 at 0
$\Phi(t)$	`\PhiUp`	$\widehat{\mathrm{up}}_{2\pi}(t) = \prod_{k\geq 0} \mathrm{sinc}_{\mathrm{rad}}(\pi t/2^k)$
$\Phi_n(t)$	`\Phin`	the prefix product with factors $k = 0, \ldots, n$ ($n + 1$ factors)
up_n	`\upn`	the finite sinc-product spline $\mathcal{F}_{2\pi}^{-1}[\Phi_n]$; equals p_{n+1} of the Exponents volume and corresponds to `fabiusUniformSpline` n in Lean
$s(x)$	`\dep`	dyadic depth: least $s \geq 0$ with $2^s x \in \mathbb{Z}$
d	—	$\lfloor s(x)/2 \rfloor$, the number of geometric modes
Q	`\Qq`	the ratio base $1/4$
a_m	`\am`	coefficients of $1/\Phi(z) = \sum_m a_m(2\pi z)^{2m}$; positive rationals
$C_r(x)$	`\Cr`	the r-th mode coefficient, $(-1)^r a_r 4^{-r}\,\mathrm{up}^{(2r)}(x)$
q_n	`\qn`	the sample $\mathrm{up}_n(x)$ at the fixed point x
$w_j^{(d)}$	`\Wrow`	the extraction weight at node Q^j
$(a; q)_n$	—	q-Pochhammer symbol $\prod_{k=0}^{n-1}(1 - aq^k)$
$\begin{bmatrix} n \\ k \end{bmatrix}_q$	—	Gaussian binomial coefficient
ε_k	—	the signed Thue–Morse sequence $(-1)^{\text{binary digit sum of } k}$
$\mathbb{Q}$	`\Kk`	the coefficient field, $\mathbb{Q}$ throughout
δ_n	—	$2^{-(n+1)}$, the half-width of the knot cells of up_n and the scale of the omitted tail T_{δ_n}
P_d	—	$\prod_{k=1}^{d}(4^k - 1)$, the odd common denominator of the row (Section 8)
ρ_n	—	residual of the annihilating recurrence at level n (Section 9)
$M(z)$, $M_+(z)$	—	the moment generating function $\int \mathrm{e}^{zu}\,\mathrm{up}(u)\,\mathrm{d}u$ and its half-range companion $\int_0^1 \mathrm{e}^{zu}\,\mathrm{up}(u)\,\mathrm{d}u$ (Sections 4 and 7)
M_j	—	the absolute moment $\int \lvert u \rvert^j\,\mathrm{up}(u)\,\mathrm{d}u$
$\Delta_s(x)$	—	the knot-level defect: $\mathrm{up}_{s-1}(x)$ minus the value the law predicts at $n = s - 1$ (Section 7)

Symbol	Alias	Meaning
β_s	—	$B_s/s!$ for even s, 0 for odd s; B_s the Bernoulli numbers

A.3 Concordance with the absorbed sources

The six reports used four letters for the depth, four for the mode count, three for the base, three normalizations of the reciprocal coefficients, and three names for the up-function itself.

Table 8: Each source's choice, and this volume's. S1–S6 are as identified in Table 10.

	up-function	depth	modes	base	reciprocal coefficients
S1	`\up`	r	m	ρ	$\gamma_{2\ell}/(2\ell)!$ from $1/M(z)$, M the MGF
S2	`\upf`	d	M	Q	a_m
S3	`\up`	m	d	4^{-1}	a_j, Bell–Bernoulli
S4	`\upf`	s	d	ρ	a_m
S5	`\rv`	s	R	Q	α_r
S6	`\upf`	s	d	ρ	reciprocal moments
here	up	$s(x)$	d	Q	a_m

Two of these differences change formulas rather than glyphs.

1. **Where the factor 4^{-r} lives.** Writing the law as $\mathrm{up}_n(x) = \mathrm{up}(x) + \sum_r C_r(x)\, 4^{-rn}$ (S4, S5, S6) or as $\sum_r (-1)^r a_r\, \mathrm{up}^{(2r)}(x) 4^{-r(n+1)}$ (S1, S2, S3) is the same statement; the coefficient $C_r(x)$ absorbs one factor 4^{-r}. This volume prints both forms once, in Section 6, and uses the first elsewhere.
2. **The sign of the coefficients.** S1's $\gamma_{2\ell}/(2\ell)!$ is $(-1)^\ell a_\ell$, so S1's mode coefficients carry no explicit sign while everyone else's carry $(-1)^r$; Section 4 proves the identity. S6's "reciprocal moments" are the catalogue's binomial-convolution reciprocal $\gamma^{\mathrm{Rv}}_{2m}$, which the same section identifies with a_m up to the factor $(2\pi)^{2m}(2m)!$ and a sign.

Index conventions that do not agree across the corpus

All six sources count the finite spline by its largest factor index: up_n has the $n+1$ factors $k = 0, \dots, n$. The Exponents volume counts factors: its p_n has n factors, so $\mathrm{up}_n = p_{n+1}$. The Lean corpus's `fabiusUniformSpline` n is the centred CDF counterpart of up_n (its argument is the number of summands minus one). Every statement transported from the Exponents volume into this one shifts its index by one; Section 2 records the bridge once, and the Exponents volume's Part 0 dictionary table (`p0:tab:dictionary`) records it from the other side.

B Ledger of editorial repairs

Every correction this consolidation made to its sources is marked at the point of repair in this document's own LaTeX source with an `% ed.:` comment. The table below is generated from those marks, so it cannot drift out of step with the text. Repairs of pure notation — a symbol renamed to its catalogue command — are not listed; those are recorded once, in the concordance of Table 8. What is listed is every place where a source's mathematical statement, proof, hypothesis, index range, or cost claim was wrong, incomplete, or overstated.

Three kinds of defect recur often enough to name. The first is an *omitted hypothesis*: a statement true under conditions the source did not print, most often a missing empty-product convention, a missing guard-precision requirement, or a missing restriction to $n \geq 1$. The second is *overclaiming*: an analytic conclusion drawn from formal algebra alone, typically a convergence or uniformity assertion that the argument does not support. The third is a *cost or sharpness claim* asserted without the count that would establish it.

Table 9: Editorial repairs, in document order.

Location	Repair
Theorem 1.2	the proof of the main theorem is distributed over the volume; the section references below were attached at assembly.
Theorem 1.3	this section proved the row in full by geometric Lagrange interpolation, and the extraction section proves the identical theorem by the identical argument. One proof: the statement stays here, the proof lives there.
The result in one page	section titles below are descriptive; the orchestrator should replace them by \cref's to the assembled parts.
Fourier normalization and the finite spl	the sources say "integrate n+1 times from -infinity"; the closure (a polynomial vanishing on a half-line is zero) is supplied here, as is the genuine-knot statement (3), which the sources do not state and which \cref{due:lem:set-midpoint} needs for "x is a knot".
Theorem 2.7	the denominator bound is new; it is what makes the phrase "exact rational arithmetic" quantitative for the verifiers.
Fourier normalization and the finite spl	a breakable tcolorbox placed directly after a heading inherits the heading's \nobreak and is moved whole to the next page, leaving the heading orphaned; the lead-in paragraph above prevents that.
Fourier normalization and the finite spl	consolidates S2 "Index conversion to the repository convention", S4 "Factor-count and user indexing" and S6 "Exact indexing bridge"; the Lean correspondence was checked against the definition of fabiusUniformSpline and the module docstring of QuarterSplineLocalPolynomial.lean.
Theorem 3.3	the catalogue's audit rule demands invariance under (m,n) -> (2m,n+1) for any exact evaluator; none of the sources states it.
Dyadic depth and cell geometry	(i) merges S1 "Cell-center stabilization", S2 "A dyadic point becomes a cell midpoint", S4 "Exact midpoint alignment" and S5 "Dyadic midpoint lemma" (the latter two in factor-count indexing, N >= s+1, which is n >= s here); (ii) is S1's item 3; (iii) and the "if and only if" are added so that the onset is explained for every level below s, not only for s-1.
Theorem 3.7	S1's "Exact self-similar tail" and S4's "the tail sees only one polynomial piece" state the r = 0 case; S6 states the jet form "in the distributional sense". The differentiation under the integral for r <= n-1 and the absolute-continuity argument at r = n are supplied here, since the deconvolution section uses r = n.
Theorem 4.5	S1 derives the cumulants probabilistically (cumulants add under convolution and scale homogeneously); that argument is recorded here as a remark and the analytic derivation above is the proof, because it also supplies the domain of validity of every series used later.
Theorem 4.7	(d) is not in any source; it quantifies the size of the modes and is the analytic reason the extraction weights of the later sections need no growth hypothesis on the coefficients.
Theorem 4.8	the sources disagree on which argument proves positivity; the orchestrating question was "which proof establishes positivity, not just nonvanishing". This remark records the answer exactly.

Location	Repair
Theorem 4.14	S1, S2 and S6 differentiate the spline distributionally and restrict to r <= n; S4 and S5 differentiate the smooth kernel instead. The kernel route is adopted because it needs no distribution theory and holds for all r, which is what makes the second proof of jet termination in \cref{due:rmk:mech-second-proof} available.
Theorem 4.15	S3's odd-level onset is the one genuinely stronger statement among the six; the endpoint case s = 1 is separated from it because it depends on a convention for the value of a jump.
Theorem 5.4	the two values are derived from the first mechanism at d = 0, as S3 does, rather than from a separate argument; positivity follows S4.
Theorem 5.5	S6 states the cutoff for even derivatives in terms of half the depth; this corollary is the exact reconciliation, and it also carries S1's "highest even derivative is nonzero" with its magnitude.
Theorem 5.7	this equivalence is editorial. It is what makes S3's Prouhet cancellation and the other five sources' jet-termination arguments two proofs of one fact rather than two different theorems.
Termination of the dyadic derivative jet	rationality by the Vandermonde argument of S6, which needs no separate arithmetic theorem about up at dyadic points.
Theorem 6.1	none of the six sources isolates the rationality step from the classical arithmetic of Arias de Reyna; the volume derives it from the law itself, so the volume is self-contained on this point.
Theorem 6.3	S2 states the derivative law with the depth letter in the summation bound (its d is this volume's s) and the onset "n >= d"; S4 states it for p_{n+1}; S3 states it for the centred splines. All three are the same statement; the mode count floor((s-r)/2) is S4's.
Theorem 7.2	S6 is the source that proves all modes nonzero; S1's "sharp maximal mode" proves only the top one, and S3's "sharp mode" corollary is the same top-mode statement. The volume's proof (cor:mech-even-jet) is S6's argument, made exact by the tiling identity of cor:mech-tiling.
Theorem 7.3	this lemma is new to the volume; it is the one-line consequence of Rvachev's equation up'(x) = 2up(2x+1) - 2up(2x-1) that the sources' numerics were tracking without naming.
Theorem 7.5	the s = 1 case is the convention question of prop:mech-odd-onset seen once more: the same computation at n = 0 gives -J/2.
Theorem 7.7	the sources' onset statements (S1 "cell-center level", S2/S4/S5 n >= s or N >= s+1, S3 the odd-level sharpening) are all consistent with the corollary; none of them claimed exactness at even depth, which is new here.
Theorem 7.12	the cluster brief asked whether the CDF law 1/(9 4^n) and the density law at x = 1/2 are the same object; they are not (at x = 1/2 the density law has no mode at all), but the CDF law at 1/2 IS the density law at 1/4 through the finite functional equation, which is the precise statement.
Theorem 8.1	cross-reference to the main theorem attached at assembly.
Theorem 8.4	the six sources prove this identically (S1 App. B, S2 App. A, S3 App. A, S4 App. A, S5 App. A, S5 Thm "Universal q-Lagrange weights"); one proof.
The quarter-base Gaussian-binomial extra	S1 Thm 8.1, S2 Thm 7.1, S3 Thm 1.3, S4 Thm 8.1, S5 Thm 8.1, S6 Thm 1.2 are all this statement; the forward/reversed/integer/base-4 displays are reconciled here once.

Location	Repair
Theorem 8.10	the Thue–Morse truncated-power representation is proved in the setting part of this volume; it is quoted here in the (n+1)-factor indexing, and was re-validated in exact arithmetic (up(1/2)=1/2, up(3/4)=5/72, up(7/8)=1/288) before being printed.
Recovering every mode and the Romberg ta	S1 Thm "Full coefficient reconstruction", S2 Prop "Closed inverse for all modes", S3 Thm "Mode projector", S4 sec. "The Vandermonde system", S5 sec. 9, S6 sec. 5.1 – merged into one statement. The symmetry [z^m] l_j = [z^j] l_m (which reconciles S1/S2's row formula with S3's operator formula) is not remarked in any source; it is proved here.
Theorem 9.5	all rationals below recomputed exactly from the truncated-power representation and Gauss–Jordan on the 3x3 system; they agree with S1 verification_output.txt lines 16-17, S2 experiment_output.txt lines 25-27, and S5's table (5/648, -248/2025).
Theorem 9.7	S4 Prop "Why d+1 samples are generically necessary", S6 sec. 5.5, S1 sec. 11.3. The sources prove (2) only for consecutive levels and only for linear rules; the polynomial construction below covers both.
Theorem 9.7	the nonvanishing of C_d(x) for s >= 2 is S6's theorem (S3 proves the top mode only); it is proved in the jet section and used here only as a hypothesis.
Theorem 9.8	S1 Alg "Exact quarter-base tableau", S5 Alg "Exact q-Richardson table", S6 sec. 5.4 – one statement; the Neville identification (2) and the explicit mode formula (3) are supplied.
Theorem 9.9	recomputed exactly; row m=2 was checked constant for n = 4,5,6, which uses samples through q_8.
Theorem 9.11	S4 sec. "Annihilating recurrences and onset diagnostics", S2 Thm "Eventual annihilating recurrence" + diagnostics, S1 Cor "Annihilating recurrence" + Remark "Why initial values can fail", S3 Thm 10.1. The sources state onset detection informally ("holds on several successive blocks"); (1) makes it a finite, provably sufficient test.
Recovering every mode and the Romberg ta	exact residuals computed for this remark: x = 7/8, 3/8, 1/8 (s = 3), 3/32, 1/32 (s = 5): rho_{s-1} = 0, rho_{s-2} != 0; x = 15/16, 1/16 (s = 4): rho_{s-1} != 0. Cross-references attached at assembly; the closed form of rho_{s-1} is the sharpness section's knot-defect theorem.
Stability of the extraction	S1 Prop "Exact l-infinity noise amplification", S2 sec. 9, S3 Prop "Lebesgue factor", S4 Prop "Uniform weight bound", S5 sec. 9.2, S6 sec. 5.5 – one statement. Monotonicity and the limit are proved (the sources assert them); the constant is recomputed to 46 digits from the partial products (400 factors; the tail is below 4^{-400}).
Theorem 10.4	the cancellation estimate is an upper bound on the possible loss; the sources (S3 sec. 11.3, S4 sec. 11.4) only say "exact input remains preferable". The Thue–Morse representation is quoted through cor:ext-explicit, which cites thm:set-finite-spline.
Stability of the extraction	the sources only state the l-infinity factor of the first row; the closed forms (2)-(3) and the table are new, verified in exact arithmetic for d <= 5.
Theorem 11.1	S1's tables print the mode coefficients in the (n+1)-convention (its A_ell = C_ell 4^ell); S4's table prints C_m as here; S6 prints A_k = C_k 4^k with the argument folded to a = 1 - |x|. All numbers below are C_m in the volume's convention, with the source form noted once at x = 1/16.

Location	Repair
Theorem 11.2	S4's "Symmetry" subsection states the reflection for its examples; the proof through the finite functional equation is supplied here.
Worked exact examples	561842749440 = 2^21 * 3^5 * 5 * 7^2 * ... is not worth printing in full; the prose stops at the power of two.
An exact algorithm and its verification	the six pseudocode listings (S1, S3, S4 in Python; S2, S5, S6 in prose) are the same seven lines; one listing is kept.
Theorem 13.1	S1's "reusable local principle", S2's "general local-deconvolution principle" and S4's "general geometric scales" are the same statement; one proposition with a proof.
Consequences and extensions	the last clause is the converse direction; the sources assert it without argument. The argument: a cell centre of up_n for all large n is an odd multiple of 2^{-(n+1)} for all large n, hence dyadic. It is given here in one sentence and not as a theorem, because the step "finite law implies cell centre" is only sketched.
Relation to the corpus and formalization	S6's closing section, kept as the one place where the letter clash (Q as a translation parameter elsewhere in the corpus, Q = 1/4 here) is recorded.

Remark B.1 (What is not in this ledger). Three classes of change are deliberately excluded. Deduplication is not a repair: where several sources proved the same result, the strongest statement and the best proof were kept and the rest collapsed, which the provenance section records in aggregate rather than line by line. Restructuring is not a repair. And a source claim that this volume could neither prove nor refute has not been silently dropped: it is stated here as an explicitly labelled open problem or conjecture, with the source's assertion quoted as an assertion.

C Provenance of the consolidation

C.1 What was merged

This volume is an editorial consolidation, carried out on 3 September 2026, of six independently written reports that arrived as bare directories on 2 September 2026 in the intake commit `8f822212d` and were filed the same day under `inverse-and-sampling/dyadic-up-extraction/` as standalone quick-gate receipts. No archive or checksum ledger accompanied them. Every one of the six proves the same theorem — that at a dyadic point the finite sinc-product spline is, after finitely many levels, *exactly* the up-function value plus finitely many geometric modes of ratio $4^{-1}, 4^{-2}, \ldots$ — and every one derives the same quarter-base Gaussian-binomial extraction row from it. They differ in normalization, indexing, the letters used for the two key integers, and in the layers each adds around the shared core.

Editorial merge means: one statement of each result, in the strongest form the six jointly support and that we can prove; the best proof kept and gaps filled; the six normalizations and four index letters reconciled once and recorded; errors corrected and marked in the source with `% ed.:` comments; and every layer that is not redundant preserved. Repetition was collapsed, content was not.

C.2 Absorbed sources and their receipts

Each package also shipped an exact-arithmetic verifier in Python and its captured output; S4 shipped two figures and two CSV tables as well. The six verifiers were consolidated into the single script named in Section 12; the figures were not retained, since every quantity they plot is printed exactly in Section 11. Git history is the archive for all of it.

Table 10: The six absorbed reports as filed on 2 September 2026 and as merged (no byte changed between filing and merging). Page counts are of the retained arrival PDFs, which were not rebuilt.

Source	Lines	Bytes	Source SHA-256
S1 = `Dyadic-Up-Extraction/dyadic_up_extraction.tex` (22 A4 pp.)			
	1,725	58,695	`9ab6bd6d04f2725faf1b96a94c8b64d71e78c053665196ad7a4d4cdd126177ff`
S2 = `Exact_Dyadic_Up_Extraction/dyadic_up_extraction.tex` (20 Letter pp.)			
	1,277	46,386	`80d2feedf09875a2278ad0310edac7e63c65d483ccace42a8fb2f28528381e46`
S3 = `Exact_Geometric_Tails_Rvachev_Up/up_dyadic_extraction.tex` (21 A4 pp.)			
	1,530	54,622	`8669456e7848e9a191daf358f716affffa02747895f456543478e7a1dafd8581`
S4 = `dyadic_up_extraction_package/dyadic_up_extraction.tex` (26 A4 pp.)			
	1,801	60,526	`9840d53814848874746e7f82f6a9dc2c3d0aedfb542d1f1b8ac30d81a86251819`
S5 = `rvachev_q_extrapolation/rvachev_q_extrapolation.tex` (19 A4 pp.)			
	1,334	41,936	`c692ae4d4b2db6bc18fa1e0be43b3e214dee0c6f193ab6beb9536bbe92839108`
S6 = `rvachev_up_dyadic_extrapolation_package/rvachev_up_dyadic_extrapolation.tex` (15 A4 pp.)			
	1,143	36,098	`e100dec4adc34acdce1b71af1446e23696ef75efd6e98251cd46123045e912dc`

C.3 What each source contributed

No source was a superset of the others.

- **S1** contributed the knot-geometry explanation of the onset (x becomes a cell centre exactly at level $s(x)$, and is a knot one level earlier), the Thue–Morse derivative stencil, the q-Pochhammer closed form of the reciprocal-product coefficients, the exact ℓ^∞ noise-amplification factor, and the reconstruction of every mode coefficient. It alone used the moment-generating-function normalization $\gamma_{2\ell}/(2\ell)!$, reconciled in Section 4.
- **S2** contributed the cleanest statement and the depth $s(x)$ as the organizing integer, the terminating Taylor jet, the closed inverse for all modes, and the explicit index conversion to the repository convention.
- **S3** contributed the Prouhet-cancellation route to the local degree collapse, the Bell–Bernoulli form of the coefficients, the *odd-level onset* ($n = s(x) - 1$ already suffices when the depth is odd, with the symmetric-rectangle convention in the exceptional case $s(x) = 1$), the Lebesgue factor, the mode projector with its even-jet recovery, and a verification of 2,376 identities at 255 interior dyadic points through level 7.
- **S4** contributed the factor-count versus user-indexing discussion, head–tail self-similarity with strict positivity, the derivative-tiling proof of jet termination, the integer base-4 and common-denominator forms of the extraction row, the uniform weight bound, the argument that $d + 1$ samples are generically necessary, onset diagnostics through the annihilating recurrence, and the largest verification (2,570 window identities at 257 points, 941 fitted-versus- analytic coefficient checks, 1,542 recurrence checks).
- **S5** contributed the one-page statement, the jet transfer at a cell centre, the rational tail generating function, the universality of the q-Lagrange weights, the Romberg-style q-Richardson tableau, and the coefficient table through a_5.

- **S6** contributed the proof that *all* d modes are nonzero for depth at least 2, so that d is the exact number of geometric modes, together with the indexing bridge to the centred Fabius splines and the relation to the repository's half-base q-row.

C.4 What the consolidation added

Beyond the merge, the following results are new to the volume; none of them is in any of the six sources.

- The closed form of the defect at the knot level, Theorem 7.4: at every dyadic point of depth $s \geq 2$ the sample at $n = s-1$ differs from the law's prediction by exactly $-\varepsilon_k 2^{-\binom{s}{2}} B_s/s!$ (zero for odd s). It rests on the half-range identity $zM_+(z) = \mathrm{e}^{z/2}M(z/2) - 1$ of Theorem 7.3, a one-line consequence of Rvachev's equation, and it settles what the sources had only verified: the onset is exact at even depth (Theorem 7.6). The formula was checked in exact arithmetic at all 508 interior points of depth 2 to 8, and the retained verifier checks it through depth 7.
- The reflection identity $\mathrm{up}_n(x) + \mathrm{up}_n(1-x) = 1$ with its proof through the finite functional equation $\mathrm{up}_{n+1}(x) = \int_{2x-1}^{2x+1} \mathrm{up}_n$ (Theorem 11.1, Theorem 7.11); the sources used the symmetry of their examples without proving it.
- The identification of the Exponents volume's exact CDF law at $x = \frac{1}{2}$ with the $s = 2$ density law at $x = \frac{1}{4}$, and the attainment of its sharp C^r law at every dyadic point of depth $r + 2$ (Theorems 7.10 and 7.11).
- Rationality of the whole even jet derived from the law itself rather than from the classical arithmetic (Theorem 6.4), and a proof of the local deconvolution principle the sources stated as a heuristic (Theorem 13.1).

C.5 Conventions reconciled

Four different letters were used for the depth (r, d, m, s), four for the mode count (m, M, d, R), three for the ratio base (ρ, Q, ρ), and three normalizations for the reciprocal coefficients. The concordance in Section A maps each source's choice to this volume's $s(x)$, d, Q, a_m. Two reconciliations change formulas rather than glyphs: the mode coefficient is written here as $C_r(x)$ with the geometric factor 4^{-rn}, so that a source's $4^{-r(n+1)}$ form corresponds to absorbing one factor 4^{-r} into the coefficient; and S1's $\gamma_{2\ell}/(2\ell)!$ equals $(-1)^\ell a_\ell$, which Section 4 proves. All six sources agree on the indexing of the finite spline itself ($n+1$ factors, $k = 0, \ldots, n$); the volume keeps it and states its relation to the corpus's p_n once, in Section 2.

C.6 Status of the claims

Every theorem, proposition, lemma, and corollary carries a proof. Where a source asserted without proof, the proof is supplied or the assertion is demoted to a labelled open problem. Every rational number printed was checked against the six packages' captured verification outputs, which were treated as ground truth over the prose; each disagreement is recorded in Section B. The plan was to draft each cluster of sections and then have it reviewed by an independent adversarial pass; the drafting of the setting, mechanism and extraction sections was completed that way, but the adversarial passes and the remaining drafts were cut off by a resource limit, and the main-theorem, sharpness, examples, algorithm, consequences and register sections were written by the consolidating editor. In place of the adversarial passes, the editor's own refutation pass recomputed every printed rational with an independent exact evaluator, ran the retained verifier, resolved every cited Lean name namespace-aware against the repository, and audited every cross-reference by script; the reader should weigh the proofs accordingly.

None of the proofs is machine-checked. The formalization register in Section 14 records which statements have kernel-verified counterparts in the Lean corpus — the extraction row's weights and their moment cancellation do; the exact cell identity at $x = 1/4$ is a verified instance of the main law in CDF form — and which do not: the general dyadic-depth theorem is absent from Lean.

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